SingleRAN

UL and DL Decoupling Feature Parameter Description

Issue Draft B

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Contents

1 Change History

This chapter describes changes not included in the "Parameters", "Counters", "Glossary", and "Reference Documents" chapters. These changes include:

- Technical changes

  Changes in functions and their corresponding parameters
- Editorial changes

  Improvements or revisions to the documentation

1.1 SRAN22.1 Draft B (2026-01-30)

This issue includes the following changes.

Technical Changes

None

Editorial Changes

Deleted descriptions of the mutually exclusive relationship between UL and DL Decoupling and LNR uplink precise RB estimation in SUL and NR FDD co-deployment mode. For details, see **5.3.2.2 Mutually Exclusive Functions**.

Added descriptions of the cell requirements for configuring the **NRDUCellGigaBand**.***LnrUlRbEstCorrectCoeff*** parameter. For details, see **5.3.5 Cells**.

Added descriptions of the benefits of compatibility of UE random access control based on cell radius with SUL UEs and related configurations. For details, see **4.2.1 Benefits**, **4.4.2.1 Data Preparation**, **4.4.2.2 Using MML Commands**, **5.2.1 Benefits**, **5.4.1.1 Data Preparation**, and **5.4.1.2 Using MML Commands**.

Added the impact relationships of UL and DL Decoupling with user-rate-based dynamic network slice resource management and user-rate-based dynamic QCI resource management. For details, see **4.3.2.3 Function Impacts** and **5.3.2.3 Function Impacts**.

Revised other descriptions in this document, which do not involve changes in meanings and therefore are not detailed.

1.2 SRAN22.1 Draft A (2025-12-31)

This issue introduces the following changes to SRAN21.1 02 (2025-04-27).

Technical Changes

Change Description	Parameter Change	Base Station Model
Added support for UL and DL Decoupling with SUL and NR FDD co-deployment. For details, see: • **3.2 Deployment Mode** • **5 SUL and NR FDD Co-Deployment**	Added parameters: • **NRDUCellCollabCoop.*SulToNulMappingType*** • **NRDUCellCollabCoop.*NulCellID*** • **NRDUCellCollabCoop.*NulMcc*** • **NRDUCellCollabCoop.*NulMnc*** • **NRDUCellCollabCoop.*NulGnodebId*** • **NRDUCellCollabServ.*NulToSulMappingType*** • **NRDUCellCollabServ.*SulCellID*** • **NRDUCellCollabServ.*SulMcc*** • **NRDUCellCollabServ.*SulMnc*** • **NRDUCellCollabServ.*SulGnodebId*** Modified parameters: • Added the description of inter-gNodeB UL and DL Decoupling to the meaning of the **NRDUCellCollabServ.*NulSulMaxInterGnbDelay*** parameter. • Changed the UE type that is compatible with this parameter to "SUL UEs in SUL and NR FDD co-deployment mode" in the meaning of the **NRDUCellCollabCoop.*InterGnbSulM***	• 3900 and 5900 series base stations (macro base stations) • DBS3900 LampSite and DBS5900 LampSite

Change Description	Parameter Change	Base Station Model
	axRbPct parameter. • Changed the UE type that is compatible with this parameter to "SUL UEs in SUL and NR FDD co-deployment mode" in the meaning of the **NRDUCellCollabCoop**.***InterGnbSulMinRbPct*** parameter. • Added the description of inter-gNodeB UL and DL Decoupling to the meaning of the **NRDUCellCollabCoop**.***NulSulMaxInterGnbDelay*** parameter. • Added UL and DL Decoupling to the **Feature ID/Feature Name** field in the **NRCellRelation**.***InterGnodebSulFlag*** parameter reference. • Added UL and DL Decoupling to the **Feature ID/Feature Name** field in the **NRCellSulFrequency**.***SulFreqIndex*** parameter reference. • Added UL and DL Decoupling to the **Feature ID/Feature Name** field in the **NRCellSulFrequency**.***SsbFreqPos***	

Change Description	Parameter Change	Base Station Model
	parameter reference. ● Added UL and DL Decoupling to the **Feature ID/ Feature Name** field in the **NRCellSulFrequency**.***SulInterFreqA5RsrpThld2*** parameter reference. ● Modified the meaning of the **NRDUCellPdcch**.***SulCceAdjPolicy*** parameter and added UL and DL Decoupling to the **Feature ID/ Feature Name** field in the reference of this parameter. ● Modified the meaning of the **NRDUCellSulRelation**.***FddUlNrarfcn*** parameter and added UL and DL Decoupling to the **Feature ID/ Feature Name** field in the reference of this parameter. ● Added UL and DL Decoupling to the **Feature ID/ Feature Name** field in the **NRDUCellSulRelation**.***FddUlBandwidth*** parameter reference. ● Added the description of UL and DL Decoupling to the meaning of	

Change Description	Parameter Change	Base Station Model
	the **NRDUCellSulRelation**.***SulType*** parameter. • Revised the applicable modes of the **NRDUCellAlgoSwitch**.***UlDlDecouplingSwitch*** parameter.	
Revised the constraints affecting cell activation due to license insufficiency in independent SUL deployment mode. For details, see **4.3.1 Licenses**.	None	3900 and 5900 series base stations (macro base stations)
Added an impact relationship between UL and DL Decoupling and time-domain power saving BWP. For details, see: • **4.3.2.3 Function Impacts** • **5.3.2.3 Function Impacts**	None	• 3900 and 5900 series base stations (macro base stations) • DBS3900 LampSite and DBS5900 LampSite
Added the compatibility of UE random access control based on cell radius with SUL UEs. For details, see: • **4.1.1.3 SUL Random Access** • **4.2.2 Impacts** • **5.1.1.3 SUL Random Access** • **5.2.2 Impacts**	None	3900 and 5900 series base stations (macro base stations)
Modified the way in which the gNodeB measures random-access-related counters of NR TDD cells in SUL scenarios. For details, see: • **4.2.2 Impacts** • **5.2.2 Impacts**	None	• 3900 and 5900 series base stations (macro base stations) • DBS3900 LampSite and DBS5900 LampSite

Change Description	Parameter Change	Base Station Model
Added a mutually exclusive relationship between UL and DL Decoupling and LNR uplink precise RB estimation in SUL and NR FDD co-deployment mode. For details, see **5.3.2.2 Mutually Exclusive Functions**.	None	3900 and 5900 series base stations (macro base stations)
Removed the mutually exclusive relationship between UL and DL Decoupling and proactive avoidance of remote interference from **4.3.2.2 Mutually Exclusive Functions** and **4.4.2.1 Data Preparation**.	None	3900 and 5900 series base stations (macro base stations)
Added support for SUL CoMP to take effect for UEs with specific QCIs. For details, see: • **4.1.2.2 SUL CoMP** • **4.4.2.1 Data Preparation** • **4.4.2.2 Using MML Commands**	Added the **NRDUCellComp**.***CompConfigPol*** parameter. Modified parameter: Added the **COMP_SW** option to the **gNBQciBearer**.***ServiceDiffAlgoSwitch*** parameter.	• 3900 and 5900 series base stations (macro base stations) • DBS3900 LampSite and DBS5900 LampSite
Added impact relationships of UL and DL Decoupling with RedCap and proactive avoidance of remote interference. For details, see **4.3.2.3 Function Impacts** and **5.3.2.3 Function Impacts**.	None	• 3900 and 5900 series base stations (macro base stations) • DBS3900 LampSite and DBS5900 LampSite

Editorial Changes

Added descriptions of support for the simultaneous use with super uplink with independent SUL deployment. For details, see **4.1.3 Simultaneous Use with Super Uplink**.

Added descriptions of the impacts of UL and DL Decoupling. For details, see **4.2.2 Impacts**.

Revised other descriptions in this document, which do not involve changes in meanings and therefore are not detailed.

2 About This Document

2.1 General Statements

Purpose

Feature Parameter Description documents are intended to acquaint readers with:

- The technical principles of features and their related parameters
- The scenarios where these features are used, the benefits they provide, and the impact they have on networks and functions
- Requirements that must be met before feature activation
- Parameter configuration required for feature activation, verification of feature activation, and monitoring of feature performance

NOTE

This document only provides guidance for feature activation. Feature deployment and feature gains depend on the specifics of the network scenario where the feature is deployed. To achieve optimal gains, contact Huawei professional service engineers.

Functions mentioned in this document work properly only when enabled in the specified applicable scenarios (such as RAT and networking). If a function not mentioned in this document is enabled or a function is enabled in a scenario not specified as applicable, exceptions or other impacts may occur.

Software Interfaces

Any parameters, alarms, counters, or managed objects (MOs) described in this document apply only to the corresponding software release. For future software releases, refer to the corresponding updated product documentation.

2.2 Applicable RAT

This document applies to NR.

2.3 Features in This Document

This document describes the following NR TDD features.

Feature ID	Feature Name	Chapter/Section
FOFD-010205	UL and DL Decoupling	**4 Independent SUL Deployment** **5 SUL and NR FDD Co-Deployment**

2.4 Differences

Table 2-1 Differences between NR FDD and NR TDD

Function Name	Difference	Chapter/Section
UL and DL Decoupling	Supported only in NR TDD	**4 Independent SUL Deployment** **5 SUL and NR FDD Co-Deployment**

Table 2-2 Differences between NSA and SA

Function Name	Difference	Chapter/Section
UL and DL Decoupling	Supported in both NSA and SA networking in independent SUL deployment mode, with the following differences: • Different messages used to carry the SUL carrier information • Different SUL carrier selection and access procedures • Different mobility management for UEs for which UL and DL Decoupling takes effect Supported only in SA networking in SUL and NR FDD co-deployment mode.	**4 Independent SUL Deployment** **5 SUL and NR FDD Co-Deployment**

Table 2-3 Differences between high frequency bands and low frequency bands

Function Name	Difference	Chapter/Section
UL and DL Decoupling	Supported only in low frequency bands	**4 Independent SUL Deployment** **5 SUL and NR FDD Co-Deployment**

This document refers to frequency bands belonging to FR1 (410–7125 MHz) as low frequency bands, and those belonging to FR2 (24250–71000 MHz) as high frequency bands. For details about FR1 and FR2, see section 5.1 "General" in 3GPP TS 38.104 V17.8.0.

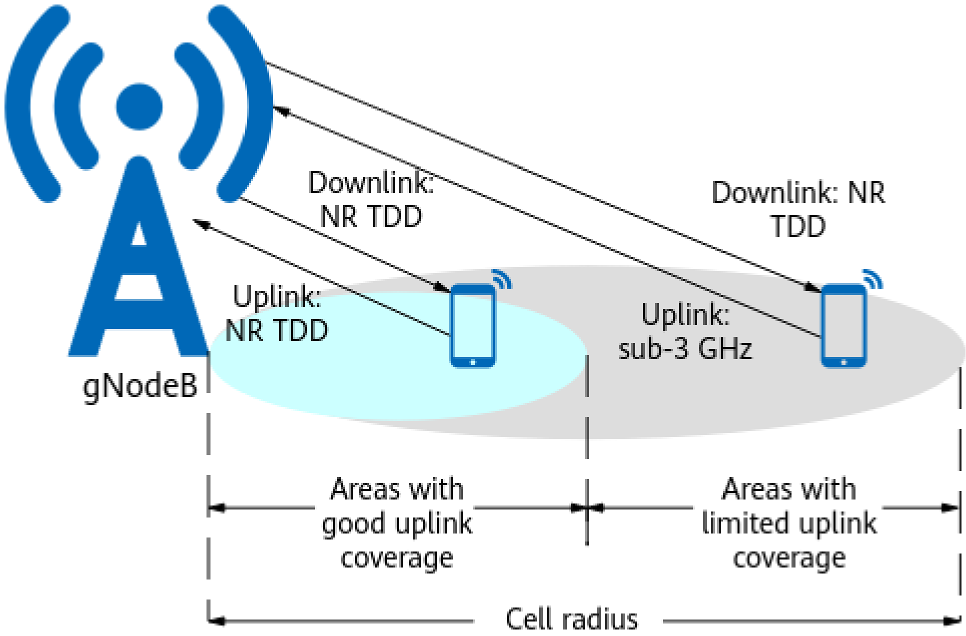
3 Overview

3.1 Background and Definition

NR TDD spectrum has the large bandwidth making it perfect for 5G enhanced Mobile Broadband (eMBB) services. Downlink coverage is better than uplink coverage on NR TDD spectrum due to the large downlink transmit power of the gNodeB and to disproportion in uplink-to-downlink slot configuration of NR. The application of technologies such as beamforming and cell-specific reference signal (CRS)-free reduces downlink interference and further increases the difference between NR TDD uplink and downlink coverage.

This feature defines new paired spectrum for areas with limited uplink coverage, with NR TDD for the downlink and a sub-3 GHz band (for example, 1.8 GHz) for the uplink, thereby improving uplink coverage. **Figure 3-1** shows how UL and DL Decoupling works. In the early stages of 5G commercial use, if no dedicated sub-3 GHz spectrum is available for 5G, then LTE FDD and NR Uplink Spectrum Sharing can be enabled to allow LTE FDD to share sub-3 GHz spectrum with NR. For details, see *LTE FDD and NR SUL Dynamic Spectrum Sharing*.

Figure 3-1 How UL and DL Decoupling works



3.2 Deployment Mode

UL and DL Decoupling can be enabled with either independent SUL deployment or SUL and NR FDD co-deployment.

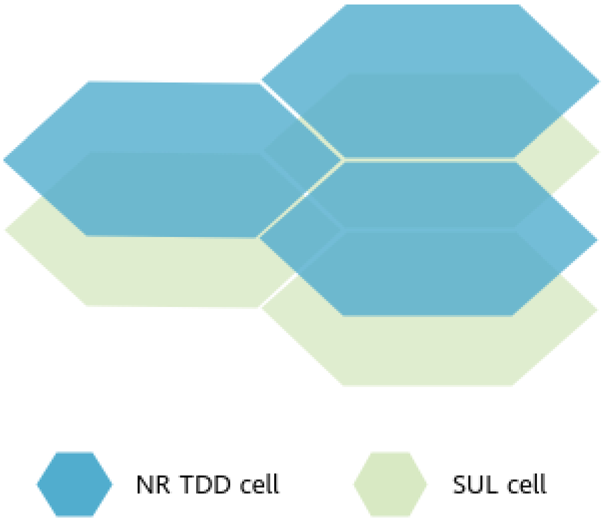
The main difference between the two deployment modes lies in the method of obtaining SUL spectrum resources.

Independent SUL Deployment

In this deployment mode, the SUL cell needs to be deployed independently, and NR TDD and SUL cells must be co-sited and cover the same area and they are always paired to implement UL and DL Decoupling, as shown in **Figure 3-2**. This deployment mode is recommended when the operator has independent SUL spectrum resources or can obtain these resources through LTE FDD and NR Uplink Spectrum Sharing.

For details about LTE FDD and NR Uplink Spectrum Sharing, see *LTE FDD and NR SUL Dynamic Spectrum Sharing*.

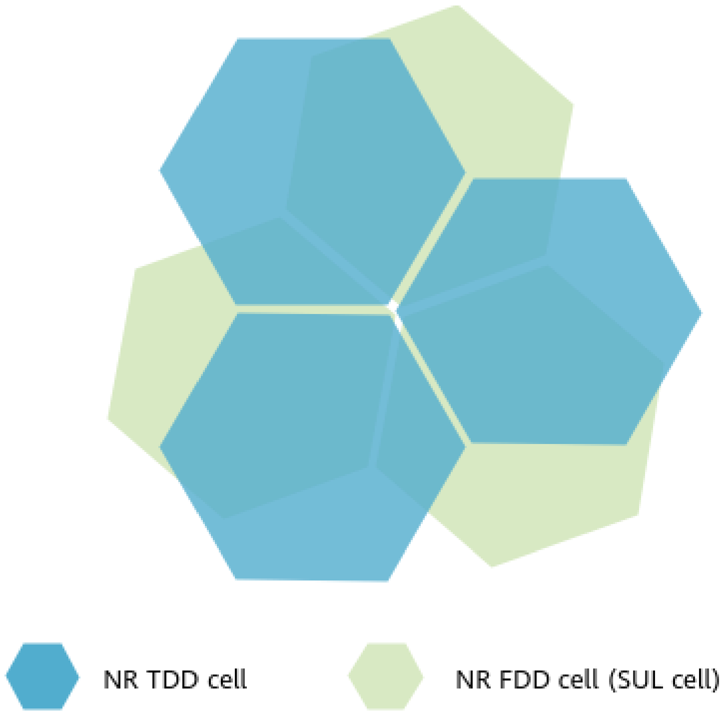
Figure 3-2 Independent SUL deployment



SUL and NR FDD Co-Deployment

In this deployment mode, the SUL link is provided by the NR FDD cell. That is, SUL and NR FDD are deployed in the same cell, as shown in **Figure 3-3**. UL and DL Decoupling is implemented based on flexible pairing between NR TDD cells and SUL cells (NR FDD cells). As such, neither the independent deployment of an SUL cell nor the co-coverage deployment of an NR TDD cell and an SUL cell (NR FDD cell) is required. In addition, an NR TDD cell and an SUL cell can be served by the same base station or different base stations. This deployment mode is recommended when the operator has spectrum resources available for both NR FDD and SUL and has established the NR FDD cell.

Figure 3-3 SUL and NR FDD co-deployment



4 Independent SUL Deployment

This chapter describes UL and DL Decoupling with independent SUL deployment. UL and DL Decoupling described in this chapter refers to UL and DL Decoupling with independent SUL deployment.

4.1 Principles

3GPP Release 15 introduces the SUL. The use of SUL effectively utilizes idle sub-3 GHz band resources, improves the uplink coverage of the NR TDD band, and enables the provisioning of 5G services in a wider area. The use of SUL also improves the service experience of cell edge users (CEUs). For details on the SUL, see section 6.9 "Supplementary Uplink" in 3GPP TS 38.300 (V15.0.0). In this feature, SUL carriers are on the sub-3 GHz band, and normal UL (NUL) carriers are on the NR TDD band.

This feature enables the decoupling between uplink and downlink spectrum by allowing for uplink data transmission on either the NUL or SUL carriers. The SUL is configured by means of an SUL cell. The NUL is implemented by an NR TDD cell. The NR TDD cell is associated with the SUL cell to implement UL and DL Decoupling. For UEs whose uplink data transmission is carried on the SUL cell, the random access, power control, scheduling, link management, and mobility management differ from the procedures for UEs whose uplink and downlink data transmission are carried on NUL.

Table 4-1 lists the supported combinations of NR TDD and sub-3 GHz bands. For details about the mapping between frequency bands and frequencies, see section 5.2 "Operating bands" in 3GPP TS 38.104 V15.5.0. In addition, both non-standalone (NSA) and standalone (SA) networking are supported.

Table 4-1 Combinations of NR TDD and sub-3 GHz bands

Networking	NR TDD Frequency Band	Frequency Band Combination
SA	n78	NR n78+NR n80 (SUL[a])
		NR n78+NR n82 (SUL)

Networking	NR TDD Frequency Band	Frequency Band Combination
		NR n78+NR n83(SUL)
		NR n78+NR n84 (SUL)
		NR n78+NR n86 (SUL)
NSA	n78	LTE band 3+[NR n78+NR n80 (SUL)]
		LTE band 20+[NR n78+NR n82 (SUL)]
		LTE band 1+[NR n78+NR n84 (SUL)]
		LTE band 28+[NR n78+NR n83 (SUL)]
		LTE band 66+[NR n78+NR n86 (SUL)]
a: SUL is short for supplementary uplink.		

The bandwidth requirement of each frequency band is described in **4.3.3 Hardware**.

4.1.1 Basic Functions

UL and DL Decoupling can be enabled by setting the **NRDUCellAlgoSwitch**.***UlDlDecouplingSwitch*** parameter to **ON** in an NR TDD cell.

If the gNodeB is shared by multiple operators, that is, the **gNBSharingMode**.***gNBMultiOpSharingMode*** parameter is set to a value other than **INDEPENDENT**, the **OP_SUL_SWITCH** option of the **NRDUCellOp**.***PolicySwitch*** parameter must be selected in the NR TDD cell for the operator that requires UL and DL Decoupling.

4.1.1.1 SUL Cell Configuration

NR TDD and SUL cells must already be configured before UL and DL Decoupling is enabled. The NR TDD and SUL cells are associated by configuring the **NRDUCellSulRelation**.***NrDuCellId*** and **NRDUCellSulRelation**.***SulNrDuCellId*** parameters.

The details about the NR TDD cell configuration and setup principle and process are elaborated in *Cell Management* in *5G RAN Feature Documentation*.

An SUL cell is configured using the **NRDUCell** MO. **Table 4-2** describes the parameters that must be configured for an SUL cell in this MO.

Table 4-2 Mandatory parameters for an SUL cell

Parameter Name	Parameter ID	Meaning
Duplex Mode	**NRDUCell**.***DuplexMode***	For an SUL cell, set this parameter to **CELL_SUL**.
Frequency Band	**NRDUCell**.***FrequencyBand***	For the frequency bands supported by an SUL cell, see **4.1 Principles**.
Uplink NARFCN	**NRDUCell**.***UlNarfcn***	For an SUL cell, only the uplink NR-ARFCN needs to be configured.
Uplink Bandwidth	**NRDUCell**.***UlBandwidth***	The uplink bandwidth varies with the operating frequency band. For details, see **4.3.3 Hardware**.
Subcarrier Spacing	**NRDUCell**.***SubcarrierSpacing***	For an SUL cell, set this parameter to **15KHZ**.
Logical Root Sequence Index	**NRDUCell**.***LogicalRootSequenceIndex***	This is used to generate the cell preamble sequence. For details, see *Cell Management* in *5G RAN Feature Documentation*.

4.1.1.2 SUL Carrier Information

To ensure the availability of the SUL carrier for uplink data transmission, gNodeBs must transmit the SUL carrier information to UEs. The SUL carrier information includes the following:

- Frame structure, system bandwidth, and NR-ARFCN
- Common channel configuration of the SUL
  - Physical random access channel (PRACH) configuration: time-frequency resource configuration, including the **NRDUCellCollabServ**.***RsrpThld*** parameter
  - Physical uplink shared channel (PUSCH) common configuration
  - Physical uplink control channel (PUCCH) common configuration
  - System information subscription configuration: When the **gNBSibConfig**.***SibTransPolicy*** parameter is set to **ON_DEMAND**, system information of a SIB type can be broadcast only after being subscribed by UEs. The system information is configured in the **gNBSibConfig** MO. For details, see *5G Networking and Signaling* in *5G RAN Feature Documentation*.

The SUL carrier information is carried in the following message:

- In SA networking:
  - System Information Block 1 (SIB1) for UEs in idle mode. For details, see section 5.2 "System information" in 3GPP TS 38.331 V15.4.0.
  - RRCReconfiguration message for UEs in RRC_CONNECTED mode. For details, see section 5.2 "System information" in 3GPP TS 38.331 V15.4.0.

- In NSA networking, the SUL carrier information is carried in the SGNB ADDITION REQUEST ACKNOWLEDGE message. For details, see section 9.1.4.2 "SGNB ADDITION REQUEST ACKNOWLEDGE" in 3GPP TS 36.423 V15.4.0.

4.1.1.3 SUL Random Access

Random access is used to establish and recover the uplink synchronization between UEs and the gNodeB.

Random access is classified into contention-based random access and non-contention-based random access.

- In contention-based random access, UEs randomly select an access preamble to be sent to the gNodeB, and the preambles selected by different UEs may conflict with each other. The gNodeB uses a contention-based resolution mechanism to handle access requests, and UEs may fail to access the network.
- In non-contention-based random access, the gNodeB allocates access preambles to UEs, and each preamble is dedicated to only one UE. Therefore, preamble conflicts will not occur. If the dedicated resources are insufficient, the gNodeB instructs UEs to initiate contention-based random access.

For details about the triggering scenarios, types, and procedures of random access, see chapter "Random Access" in *5G Networking and Signaling* in *5G RAN Feature Documentation*.

The SUL random access differs from the NUL random access in downlink beam information acquisition.

- In non-contention-based random access, the beam information is obtained by means of downlink measurement. When allocating dedicated access preambles to UEs, the gNodeB buffers the beam information. After allocating the dedicated access preamble and receiving responses from UEs, the gNodeB sends a random access response (RAR) to UEs based on the buffered beam information.
- In contention-based random access, the access preamble sequences in system broadcast messages are grouped based on the downlink beam. After completing downlink synchronization, UEs obtain the downlink beam information, and select the access preamble sequence based on the downlink beam for access. The gNodeB obtains beam information based on the access preamble sequence group, and sends an RAR to UEs. When the following conditions are met, UEs randomly select a beam. Under other circumstances, random access fails.
  - The cell downlink RSRP measured by UEs meets the cell selection criterion (also known as criterion S), where the value of $Q_{rxlevmin}$ is equal to that of **NRDUCellSelConfig.*MinimumRxLevel***. For details, see section 5.2.3.2 "Cell Selection Criterion" in 3GPP TS 38.304 V15.4.0.
  - The RSRP of the downlink beam is greater than the value of **NRDUCellPrach.*RsrpThldForSsbSelection***.

  For NUL cells, the value of **NRDUCellPrach.*RsrpThldForSsbSelection*** must be consistent with that of **NRDUCellSelConfig.*MinimumRxLevel***.

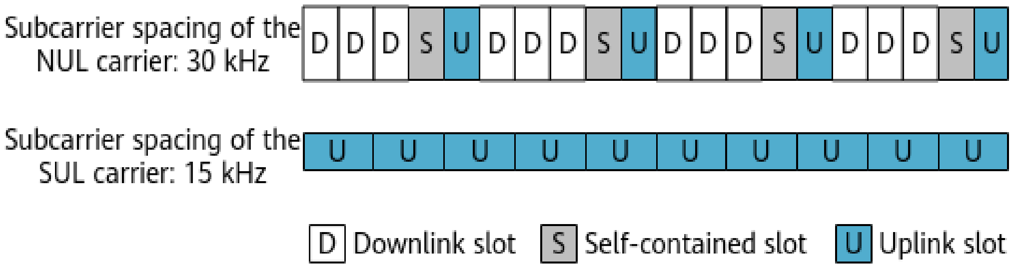
SUL UEs support the "UE random access control based on cell radius" function. When the distance between an SUL UE and an SUL cell is greater than the SUL cell radius, UEs enabled with UL and DL Decoupling are not allowed to access the cell. For details about the "UE random access control based on cell radius" function, see *Random Access Control*.

4.1.1.4 Uplink Scheduling

When UL and DL Decoupling is enabled, the downlink data transmission is carried on the NUL carrier, and the uplink data transmission can be carried on the SUL carrier. The subcarrier spacing of the NUL carrier is 30 kHz, and that of the SUL carrier is 15 kHz. The ratio of slots of the NUL carrier to slots of the SUL carrier is 2:1. Therefore, the scheduling time sequence must be considered. For the NUL carrier, the slot configuration (specified by the **NRDUCell.*SlotAssignment*** parameter) can be 4:1 or 8:2. For the SUL carrier, all slots are available for UEs in SA networking and for frequency division multiplexing (FDM) UEs in NSA networking. For the time division multiplexing (TDM) UEs in NSA networking, the resources are allocated in accordance with the TDM-Pattern. For details, see **UE Capability-based Scheduling in NSA Networking**. **Figure 4-1** uses the slot configuration of 4:1 as an example.

As uplink TDM scheduling is performed for a UE on NR TDD and SUL carriers, the slots on the SUL carrier corresponding to the symbols with SRS on the NR TDD carrier cannot be scheduled in specific scenarios. For details, see **4.1.2.3 Symbol-Level Scheduling on the SUL Carrier**.

Figure 4-1 Scheduling time sequence after UL and DL Decoupling is enabled



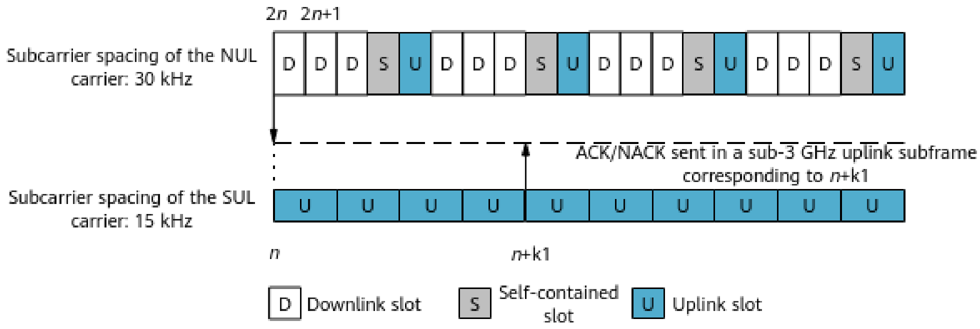
NR introduces a flexible scheduling mechanism, where k1 and k2 are used to ensure a correct scheduling time sequence between gNodeBs and UEs. For details on k1 and k2, see sections 5.2 "UE procedure for reporting channel state information (CSI)" and 5.3 "UE PDSCH processing procedure time" in 3GPP TS 38.214 V15.0.0. k1 determines the hybrid automatic repeat request (HARQ) time sequence in downlink data transmission, while k2 determines the uplink scheduling time sequence. Both k1 and k2 are automatically calculated using algorithms, gNodeBs send k1 and k2 to UEs through downlink control information (DCI). Other scheduling algorithms are the same as those used before UL and DL Decoupling is enabled.

HARQ Time Sequence

If UL and DL Decoupling is enabled, a UE transmits the ACK/NACK feedback for downlink data in slot *n*+k1.

When the SUL subcarrier spacing is 15 kHz and the UE receives downlink data in slot 2*n* or 2*n*+1 over the NR TDD band, it sends the ACK/NACK through the uplink subframe over a sub-3 GHz band that corresponds to slot *n*+k1 over SUL, as shown in **Figure 4-2**.

Figure 4-2 HARQ time sequence

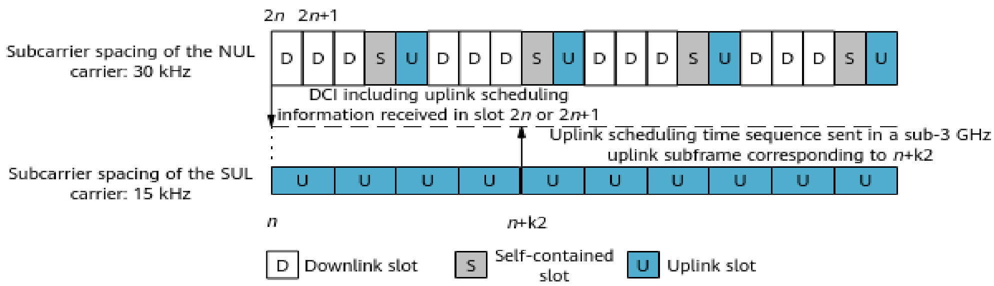


Uplink Scheduling Time Sequence

When UL and DL Decoupling is enabled, the network side sends a scheduling indication to the UE over the NR TDD band, indicating the SUL resources that can be scheduled by the UE.

When the SUL subcarrier spacing is 15 kHz and the UE receives DCI including uplink scheduling information in slot 2*n* or 2*n*+1 over the NR TDD band, it sends uplink data in the slot over the sub-3 GHz band that corresponds to slot *n*+k2 over SUL, as shown in **Figure 4-3**.

Figure 4-3 Uplink scheduling time sequence



UL and DL Decoupling allows for flexible scheduling. The resources of the SUL carrier working in a sub-3 GHz band can be scheduled in each subframe over the NR TDD band. The flexible scheduling balances the PDCCH load in each subframe over the NR TDD band. As the ratio of the slots over the NR TDD band to the slots in a sub-3 GHz band is 2:1, a subframe in the sub-3 GHz band can be scheduled in two subframes over the NR TDD band, and the PDCCHs in each of the two subframes over the NR TDD band carry only 50% of the load.

UE Capability-based Scheduling in NSA Networking

3GPP Release 15 stipulates that a UE can transmit uplink data in TDM or FDM mode in NSA networking when the master cell group (MCG) and secondary cell group (SCG) work in the same frequency band and have overlapped spectrum. When UL and DL Decoupling is enabled in NSA networking, if the LTE primary cell

(PCell) and NR SUL cell work in the same frequency band and have overlapped spectrum, uplink data can be transmitted in both TDM and FDM modes.

On the UE side:

- FDM transmission: The UE transmits uplink data in MCG LTE PCell and SCG NR SUL concurrently. In each slot, the UE can transmit uplink data on both the LTE and NR sides, and the total transmit power of the UE on both sides equals 23 dBm.
- TDM transmission: The UE transmits uplink data in MCG LTE PCell and SCG NR SUL in TDM mode. In each slot, the UE transmits uplink data on either the NR or LTE side, and the transmit power of the UE on either side can reach 23 dBm.

On the base station side:

- For the UE that uses the FDM transmission, the base station performs scheduling on the LTE and NR SUL separately.
- For the UE that uses the TDM transmission, the base station allocates slot-level resources to achieve TDM transmission of uplink data on the LTE and NR sides, and notifies the UE. When UL and DL Decoupling is enabled in NSA networking, the TDM mode is configured via the **TDM_SWITCH** option of the **NsaDcMgmtConfig.*NsaDcAlgoSwitch*** parameter on the eNodeB side. The corresponding LTE cell must be an LTE FDD cell that meets either of the following conditions:
  - When the **RACHCfg**.***PrachConfigIndexCfgInd*** parameter is set to **NOT_CFG**:
    - The **ENodeBAlgoSwitch**.***PrachTimeStagSwitch*** parameter is set to **OFF**.
    - The **RachAdjSwitch** option of the **CellAlgoSwitch**.***RachAlgoSwitch*** parameter is deselected.
  - The **RACHCfg**.***PrachConfigIndexCfgInd*** parameter is set to **CFG**, and the **RACHCfg**.***PrachConfigIndex*** parameter is set to **0**, **3**, **6**, **9**, **13**, **14**, **16**, **19**, **22**, **25**, **29**, **32**, **35**, **38**, **41**, **45**, **48**, **51**, **54**, or **57**.

The LTE and NR SUL resources on the base station side are allocated in accordance with the TDM-Pattern defined in 3GPP specifications. In this version, the TDM-Pattern of NR:LTE = 9:1 is always used in NSA networking.

The gNodeB sends the TDM-Pattern to the eNodeB through an SgNB Modification Request Acknowledge or SgNB Addition Request Acknowledge message. Upon receiving either message, the eNodeB sends the TDM-Pattern to the UE. The TDM function takes effect on both the UE and the base station sides. In NSA dual connectivity (DC), after the TDM function takes effect in an LTE FDD cell, PUCCHs in the cell always use format3, which increases the number of resource blocks (RBs) occupied by PUCCHs. As a result, the number of RBs available for PUSCHs decreases, and the overall cell throughput decreases.

NOTE

- The FDM mode has high requirements on terminal implementation and costs. Currently, mainstream chip vendors implement the TDM mode.
- For details on the principle of TDM-Pattern, see chapter 10 "Physical uplink control channel procedures" in 3GPP TS 36.213 V15.4.0. For details on how to deliver TDM-Pattern, see section 6.2 "RRC messages" in 3GPP TS 36.331 V15.4.0.

4.1.1.5 SUL Power Control

The power control principles for SUL channels are almost the same as those for uplink channels on the NUL carrier. For details, see *Power Control* in *5G RAN Feature Documentation*.

The only difference is the path loss estimation. The SUL carrier is not paired with a downlink carrier. The path loss of the SUL carrier is estimated instead based on the downlink of the NUL carrier. The path loss estimated in such a way is greater than the actual path loss. As a result, UEs use a high uplink transmit power during the random access on the SUL carrier, which increases the uplink interference. Therefore, the gNodeB performs adjustment based on the path loss difference between the SUL carrier and NUL carrier.

4.1.1.6 SUL Carrier Management

4.1.1.6.1 Uplink Carrier Selection

UEs in RRC_CONNECTED Mode Accessing the SUL Cell in NSA Networking

UEs still camp on an LTE network in NSA networking. For DC operations, the network side configures a gNodeB as the SCG for a UE and instructs the UE to initiate a random access procedure on the NR network.

When the **TDM_CHG_WITH_INTRA_CELL_HO_SW** option of the **NsaDcAlgoParam**.***NsaDcAlgoSwitch*** parameter is selected on the eNodeB, the UE always initiates a random access procedure on the NUL carrier.

1. For DC operations, the eNodeB sends inter-RAT B1 measurement configurations to the UE, instructing the UE to measure the signal quality of a certain gNodeB. The event B1 threshold delivered by the eNodeB is specified by the **NrScgFreqConfig**.***NsaDcB1ThldRsrp*** parameter.
2. After receiving an inter-RAT measurement report from the UE, the eNodeB forwards the RSRP of the NR cell to the gNodeB. The UE then initiates a random access procedure on the NUL carrier.

When the **TDM_CHG_WITH_INTRA_CELL_HO_SW** option of the **NsaDcAlgoParam**.***NsaDcAlgoSwitch*** parameter is deselected, the network side selects the NUL or SUL carrier for a UE supporting UL and DL Decoupling and notifies the UE of the selected uplink carrier by an RRCReconfiguration message.

SUL frequency band information needs to be configured to allow the LTE side to deliver the information while querying for the UE capability. This information is configured through the **NrScgFreqConfig**.***ScgDlArfcn*** and **NrScgFreqConfig**.***NrFrequencyBand*** parameters.

The uplink carrier selection procedure is as follows:

1. For DC operations, the eNodeB sends inter-RAT B1 measurement configurations to the UE, instructing the UE to measure the signal quality of a certain gNodeB. The event B1 threshold delivered by the eNodeB is specified by the **NrScgFreqConfig**.***NsaDcSulB1ThldRsrp*** parameter.
2. After receiving an inter-RAT measurement report from the UE, the eNodeB forwards the RSRPs of the NR cells to the gNodeB. The gNodeB selects an uplink carrier for the UE based on the following rules:
   - If the RSRP of an NR cell is greater than or equal to the **NRDUCellCollabServ**.***RsrpThld*** parameter value, the UE is in an area with good NR uplink coverage. The gNodeB instructs the UE to initiate a random access procedure on the NUL carrier.
   - If the RSRP of the NR cell is smaller than the **NRDUCellCollabServ**.***RsrpThld*** parameter value, the UE is in an area with weak or without NR uplink coverage. The gNodeB instructs the UE to initiate a random access procedure on the SUL carrier.
3. The gNodeB notifies the eNodeB of the selected uplink carrier, and the eNodeB sends an RRCReconfiguration message containing this information to the UE.
4. The UE initiates a random access procedure on the indicated uplink carrier.

UEs in Idle Mode Accessing the SUL Cell in SA Networking

In SA networking, when an RRC connection is set up for a UE, the UE receives system information from the gNodeB, measures the downlink RSRP over the NR TDD band, and selects an uplink carrier based on the following rules:

- If the RSRP of an NR cell is greater than or equal to the **NRDUCellCollabServ**.***RsrpThld*** parameter value, the UE is in an area with good NR uplink coverage. The UE initiates a random access procedure on the NUL carrier.
- If the RSRP of the NR cell is smaller than the **NRDUCellCollabServ**.***RsrpThld*** parameter value, the UE is in an area with weak or without NR uplink coverage. The UE initiates a random access procedure on the SUL carrier.

UEs in RRC_CONNECTED Mode Accessing the SUL Cell in SA Networking

When a UE supporting UL and DL Decoupling in SA networking is handed over to the SUL cell, the network side selects the NUL or SUL carrier for the UE and notifies the UE of the selected uplink carrier by an RRCReconfiguration message.

The uplink carrier selection procedure is as follows:

1. Before the handover, the gNodeB delivers intra-RAT measurement configurations (event A3) to the UE, instructing the UE to measure the quality of neighboring cells.
2. After receiving a measurement report from the UE, the source gNodeB forwards the RSRPs of the NR cells to the target gNodeB. The target gNodeB selects an uplink carrier for the UE based on the following rules:
   - If the RSRP of an NR cell is greater than or equal to the **NRDUCellCollabServ**.***RsrpThld*** parameter value, the UE is in an area with good NR uplink coverage. The network side instructs the UE to initiate a random access procedure on the NUL carrier.

  - If the RSRP of the NR cell is smaller than the **NRDUCellCollabServ**.***RsrpThld*** parameter value, the UE is in an area with weak or without NR uplink coverage. The network side instructs the UE to initiate a random access procedure on the SUL carrier.
3. The target gNodeB sends an RRCReconfiguration message indicating the selected uplink carrier to the UE through the source gNodeB.
4. The UE initiates a random access procedure on the indicated uplink carrier.

4.1.1.6.2 Uplink Carrier Change

When a UE enters RRC_CONNECTED mode and moves in an NR cell enabled with UL and DL Decoupling, the uplink carrier changes as the coverage area of the NUL carrier differs from that of the SUL carrier.

In this version, the uplink carrier for UEs can be selected based on either uplink quality or user experience. The criterion for selecting the uplink carrier is controlled by the **UL_CARR_SEL_FOR_USER_EXP_SW** option of the **NRDUCellCollabServ**.***SulAlgoSwitch*** parameter.

- When the **UL_CARR_SEL_FOR_USER_EXP_SW** option is selected, the uplink carrier is selected based on user experience.
  - If a UE's uplink data transmission is currently carried on the NUL carrier, the gNodeB obtains the SRS RSRP of the NUL carrier and compares it with the value of the **NRDUCellCollabServ**.***SrsRsrpThld*** parameter. If the SRS RSRP of the NUL carrier is less than the value of the **NRDUCellCollabServ**.***SrsRsrpThld*** parameter minus the hysteresis (2 dB) and if user experience provided by the SUL carrier is expected to be superior to that provided by the NUL carrier, the gNodeB instructs the UE to switch to the SUL carrier for uplink data transmission. Under other circumstances, the UE's uplink data transmission remains being carried on the NUL carrier.
  - If a UE's uplink data transmission is currently carried on the SUL carrier, the gNodeB uses the SRS switching function to enable the UE to send SRSs on the NUL carrier, obtaining the SRS RSRP of the NUL carrier.
    - If the SRS RSRP of the NUL carrier is greater than the value of the **NRDUCellCollabServ**.***SrsRsrpThld*** parameter plus the hysteresis (2 dB), the gNodeB instructs the UE to switch to the NUL carrier for uplink data transmission.
    - If the SRS RSRP of the NUL carrier is less than or equal to the value of the **NRDUCellCollabServ**.***SrsRsrpThld*** parameter minus the hysteresis (2 dB), and:
      - If user experience provided by the SUL carrier is expected to be inferior to that provided by the NUL carrier, the gNodeB instructs the UE to switch to the NUL carrier for uplink data transmission.
      - If user experience provided by the SUL carrier is expected to be superior to or similar to that provided by the NUL carrier, the UE's uplink data transmission remains being carried on the SUL carrier.
- When the **UL_CARR_SEL_FOR_USER_EXP_SW** option is deselected, the uplink carrier is selected based on uplink quality.

– If a UE's uplink data transmission is currently carried on the NUL carrier, the gNodeB obtains the SRS RSRP of the NUL carrier and compares it with the value of the **NRDUCellCollabServ**.***SrsRsrpThld*** parameter. If the SRS RSRP of the NUL carrier is less than the value of the **NRDUCellCollabServ**.***SrsRsrpThld*** minus the hysteresis (2 dB), an NUL-to-SUL carrier change is triggered. Under other circumstances, the UE's uplink data transmission remains being carried on the NUL carrier.
– If a UE's uplink data transmission is currently carried on the SUL carrier, the gNodeB uses the SRS switching function to enable the UE to send SRSs on the NUL carrier, obtaining the SRS RSRP of the NUL carrier. The gNodeB then compares the SRS RSRP of the NUL carrier with the value of the **NRDUCellCollabServ**.***SrsRsrpThld*** parameter. When the SRS RSRP of the NUL carrier is greater than the value of the **NRDUCellCollabServ**.***SrsRsrpThld*** parameter plus the hysteresis (2 dB), an SUL-to-NUL carrier change is triggered. Under other circumstances, the UE's uplink data transmission remains being carried on the SUL carrier.

4.1.1.6.3 SUL Carrier Detection and Management

In UL and DL Decoupling scenarios, when the SUL carrier of a UE cannot be used due to an exception and an NUL-to-SUL carrier change is triggered, the UE access is abnormal and the RRC connection is repeatedly reestablished.

With the SUL carrier detection and management function, the gNodeB detects an SUL carrier when a UE's uplink data transmission needs to be switched to the SUL carrier based on the SRS measurement of the NUL carrier. If the SUL carrier is normal, the UE's uplink data transmission is switched to the SUL carrier. Otherwise, the UE's uplink data transmission remains on the NUL carrier. This function is controlled by the **SUL_FAULT_TOLERANCE_SW** option of the **NRDUCellCollabServ**.***SulAlgoSwitch*** parameter.

If the gNodeB detects that the UE's SRS SINR is greater than the sum of the **NRDUCellCollabServ**.***SulSinrLowerLimit*** and **NRDUCellCollabServ**.***SulSinrLowerLimitOffset*** parameter values, and discontinuous transmission (DTX) is not used on the SUL carrier, then the SUL carrier is considered normal. Otherwise, the SUL carrier is considered abnormal.

4.1.1.7 Mobility Management

When UEs supporting UL and DL Decoupling move between NR cells, they need to reselect an SUL link during a handover or redirection.

Mobility management can be classified into mobility management in NSA networking and mobility management in SA networking.

4.1.1.7.1 Mobility Management in NSA Networking

The UE mobility management in NSA networking involves the MCG mobility management and SCG mobility management.

- MCG mobility management applies to the case where the serving eNodeB changes as the UE moves.
- SCG mobility management applies to the case where the serving eNodeB remains the same but the serving gNodeB changes as the UE moves.

MCG Mobility Management

The original eNodeB serves as the master base station (MeNB) and a gNodeB serves as the secondary base station (SgNB) for a UE. When an intra- or inter-MCG handover is triggered but the SCG remains unchanged, the original eNodeB sends an RRCReconfiguration message to the UE, instructing the UE to initiate a handover to the target eNodeB. The RRCReconfiguration message contains the non-contention-based access information of the SCG. After receiving this message, the UE initiates non-contention-based random access procedure on the SCG.

SCG Mobility Management

The eNodeB releases the connection with the original gNodeB, and adds a cell served by the target gNodeB as the SCG. For details, see *EPC-based NSA Basics*. If the target NR cell supports UL and DL Decoupling, the network side instructs the UE to initiate random access on the NUL carrier or SUL carrier.

4.1.1.7.2 Mobility Management in SA Networking

The mobility management in SA networking is classified into intra-RAT mobility management and inter-RAT mobility management.

- Intra-RAT mobility management includes intra-frequency handovers, inter-frequency handovers, inter-frequency redirections, and inter-frequency blind redirections. NR does not support intra-RAT blind handovers in the current version. This section mainly describes the differences in mobility management after UL and DL Decoupling is enabled. For details about the mobility management procedures, see *SA Mobility Management in Connected Mode* in *5G RAN Feature Documentation*.
- Inter-RAT mobility management includes inter-RAT handovers/redirections and inter-RAT blind redirections. This section mainly describes the differences in mobility management after UL and DL Decoupling is enabled. For details about the mobility management procedures, see *Interoperability Between E-UTRAN and NG-RAN in Connected Mode* in *5G RAN Feature Documentation*.

Intra-Frequency Handovers

After the UE reports an event A3, the source gNodeB performs an intra-gNodeB intra-frequency handover or inter-gNodeB intra-frequency handover. The gNodeB of the target cell selects an uplink carrier based on the downlink RSRP over the NR TDD band of the target cell reported in event A3, and informs the source gNodeB. The source gNodeB instructs the UE to initiate a handover and then to perform random access in the target gNodeB through an RRCReconfiguration message.

Inter-Frequency Handovers and Redirections

After the UE reports an event A2 or A5, the source gNodeB performs an intra-gNodeB inter-frequency handover or inter-gNodeB inter-frequency handover. The gNodeB of the target cell selects an uplink carrier based on the downlink RSRP over the NR TDD band of the target cell reported in event A5, and informs the source gNodeB. The source gNodeB instructs the UE to initiate a handover/redirection and then to perform random access in the target gNodeB through an RRCReconfiguration message.

NOTE

The frequency of the cell with the best signal quality is selected for a redirection based on the measurement results only when the UE does not support handovers. Whether the UE supports inter-frequency handovers is indicated by the handoverInterF field.

Inter-Frequency Blind Redirections

In inter-frequency blind redirections, measurement is performed only in the serving cell, not in neighboring cells.

If the measured RSRP of the serving cell meets the entering condition of event A2 for an inter-frequency blind redirection, the inter-frequency blind redirection is triggered to redirect the UE to the frequency with the highest priority. By default, the UE initiates random access in the NR TDD uplink of a cell working on the indicated frequency.

Inter-RAT Handovers and Redirections

Inter-RAT handovers include NR-to-LTE handovers and LTE-to-NR handovers.

- NR-to-LTE handovers/redirections: The NR-to-LTE handover/redirection procedure is always the same, regardless of whether SUL is enabled.
- LTE-to-NR handovers/redirections: When a source eNodeB initiates a handover/redirection request to a target gNodeB, the request message contains the downlink RSRP of the target cell. The gNodeB of the target cell selects an uplink carrier based on the downlink RSRP and informs the source eNodeB. The source eNodeB instructs the UE to initiate a handover/redirection and then to perform random access in the target gNodeB through an RRCReconfiguration message.

Inter-RAT Blind Redirections

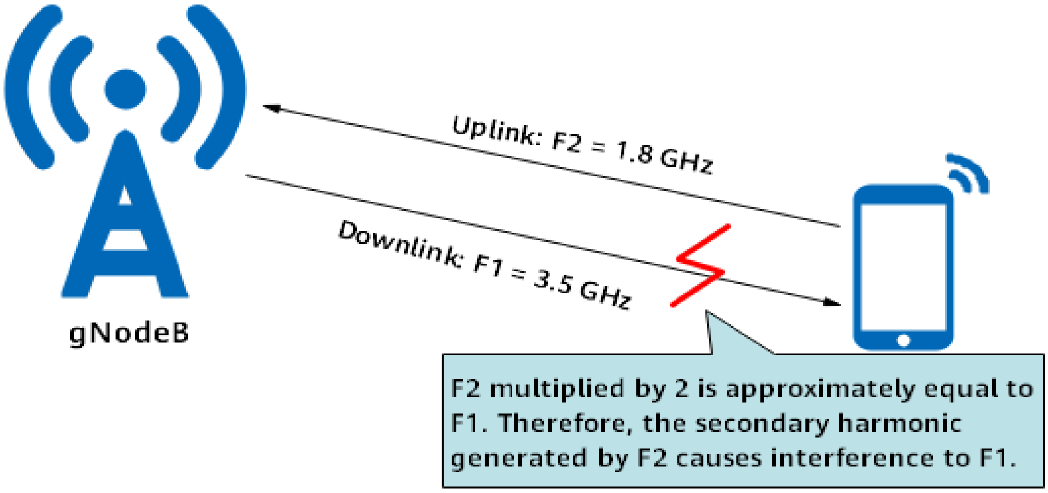
Only NR-to-LTE blind redirection is supported, and the procedure is always the same, regardless of whether SUL is enabled.

4.1.2 Enhanced Functions

4.1.2.1 Secondary Harmonic Interference Avoidance

Secondary harmonics are the self-interference signals generated by UEs during uplink and downlink data transmission. When a UE's uplink data transmission is carried on an SUL carrier and its downlink data transmission is carried on an NUL carrier, the uplink transmit frequency multiplied by two approximately equals the downlink receive frequency, and the transmission of uplink signals interferes with the reception of downlink signals, as shown in **Figure 4-4**.

Figure 4-4 Secondary harmonic

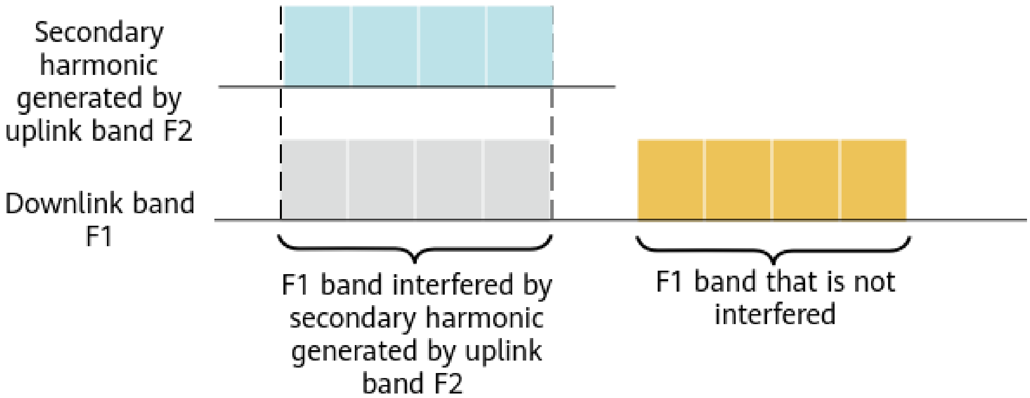


Secondary harmonic interference is generated when the following combinations of the NR TDD and sub-3 GHz bands are used:

- NR n78+NR n80 (SUL)
- NR n78+NR n86 (SUL)

If either of the preceding frequency band combinations is used, it is recommended that the secondary harmonic interference avoidance switch be turned on to enable the gNodeB to schedule downlink data for UEs for which UL and DL Decoupling takes effect in a frequency band free from interference. This reduces interference from UEs' uplink signals to downlink signals, as shown in **Figure 4-5**.

Figure 4-5 Secondary harmonic interference avoidance



The secondary harmonic interference avoidance switch is specified by the **SEC_HARMONIC_INTRF_AVOID_SW** option of the **NRDUCellCollabServ**.***SulAlgoSwitch*** parameter.

4.1.2.2 SUL CoMP

NUL and SUL carriers use different antennas, and the antenna azimuth of the NUL carrier is inconsistent with that of the SUL carrier due to engineering implementation errors. Consequently, the gains provided by UL and DL Decoupling are affected. The UL CoMP function is introduced to the SUL to compensate for the gain loss caused by inconsistent antenna azimuths (difference less than or equal to 30°). The UL CoMP on the SUL is referred to as SUL CoMP.

The SUL CoMP function is controlled by the **INTRA_GNB_SUL_COMP_SW** option of the **NRDUCellCollabServ**.***SulAlgoSwitch*** parameter. For CEUs, the gNodeB receives the UEs' uplink signals from both the serving cell and the strongest intra-gNodeB neighboring cell, and combines the received signals. This improves the demodulation capability of the gNodeB. The procedure is as follows:

1. The gNodeB selects UEs for synchronization signal and PBCH block (SSB) RSRP measurement and delivery of measurement configurations related to event A3 as follows:
   - If the **UL_2CELL_COMP_BEAR_POL_SW** option of the **NRDUCellComp**.***CompConfigPol*** parameter is deselected, the gNodeB delivers measurement configurations to all UEs in the cell.
   - If the **UL_2CELL_COMP_BEAR_POL_SW** option of the **NRDUCellComp**.***CompConfigPol*** parameter is selected, the gNodeB delivers measurement configurations to specific UEs in the cell. Specific UEs are those having at least one QCI for which the **COMP_SW** option of the **gNBQciBearer**.***ServiceDiffAlgoSwitch*** parameter is selected.

   The event A3 measurement control parameters (in the **NRCellComp** MO) are the same as those for 5G UL CoMP, including **NRCellComp**.***TimeToTrigger***, **NRCellComp**.***Hysteresis***, and **NRCellComp**.***UlCompRsrpOffset***. For details, see *CoMP* in *5G RAN Feature Documentation*.
2. UEs perform A3 measurement, and send A3 measurement reports, with NR TDD intra-frequency neighboring cells included.
3. The gNodeB determines whether the UEs are CEUs based on the received A3 measurement results, and determines the strongest intra-gNodeB neighboring NR TDD cell.
4. The gNodeB determines the strongest intra-gNodeB neighboring cell of the SUL based on the mapping between NUL and SUL carriers. The following configurations of the neighboring cell must be the same as those of the serving cell:
   - Transmit and receive mode: **NRDUCellTrp**.***TxRxMode***
   - Duplex mode: **NRDUCell**.***DuplexMode***
   - Downlink NR-ARFCN: **NRDUCell**.***DlNarfcn***
   - Uplink bandwidth: **NRDUCell**.***UlBandwidth***
   - Slot configuration: **NRDUCell**.***SlotAssignment***
   - Cyclic prefix length: **NRDUCell**.***CyclicPrefixLength***
   - Subcarrier spacing: **NRDUCell**.***SubcarrierSpacing***
   - PUSCH DMRS configuration: **NRDUCellPusch**.***UlDmrsType*** and **NRDUCellPusch**.***UlAdditionalDmrsPos***
5. The strongest intra-gNodeB neighboring cell of the SUL receives uplink SRSs from UEs. If the RSRP of the uplink SRSs sent by a UE in this neighboring cell of the SUL is greater than the sum of the RSRP of uplink SRSs in the UE's serving cell and the **NRCellComp**.***UlCompRsrpOffset*** parameter value, the gNodeB determines that the UE is an SUL CoMP UE. If not, the gNodeB determines that the UE is not an SUL CoMP UE.

6. For SUL CoMP UEs, the serving cell and the strongest intra-gNodeB neighboring cell of the SUL simultaneously receive the UE's data on the PUSCH. The received data is combined and demodulated, improving the PUSCH performance of these UEs.

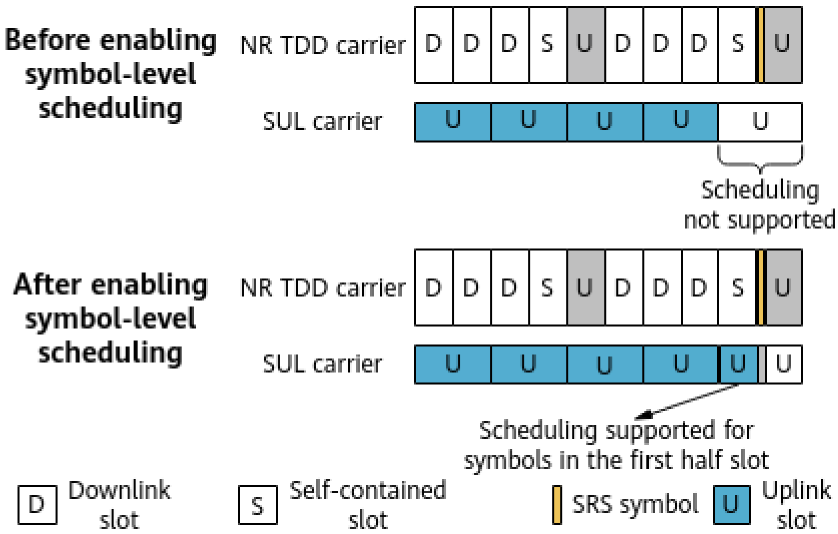
   The strongest intra-gNodeB neighboring cell of the SUL preferentially processes the data from the UEs it serves. If this cell does not have sufficient capabilities (for example, the number of PRBs used by this cell plus the number of PRBs required for SUL CoMP exceeds the cell's total number of PRBs), it does not process the SUL CoMP UE data.

For details about CoMP, see *CoMP*.

4.1.2.3 Symbol-Level Scheduling on the SUL Carrier

Symbol-level scheduling on the SUL carrier applies to scenarios where the slot configuration is 4:1 in NR TDD. When this slot configuration is in use and uplink TDM scheduling is performed for a UE on NR TDD and SUL carriers, the slots on the SUL carrier corresponding to the symbols with SRS on the NR TDD carrier cannot be scheduled. With symbol-level scheduling, the gNodeB allows a UE to transmit data over symbols in the first half slot on the SUL carrier, which corresponds to the downlink or self-contained slot on the NR TDD carrier. This improves the uplink throughput of the SUL carrier. For details, see **Figure 4-6**.

Figure 4-6 Symbol-level scheduling principle

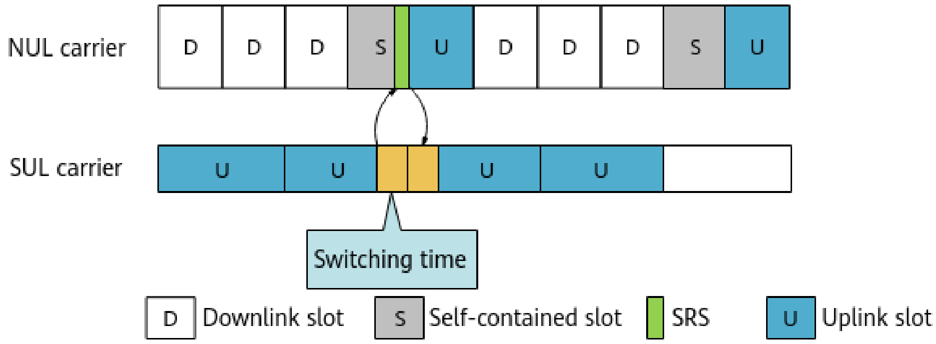


Symbol-level scheduling is controlled by the **SCHEDULE_FIRST_HALF_SLOT_SW** option of the **NRDUCellCollabServ**.***SulAlgoSwitch*** parameter.

4.1.2.4 UE Compatibility Handling

When the SUL subcarrier spacing is 15 kHz: For some UEs, the time taken to switch between the NUL and SUL carriers cannot be 0 µs. To resolve the compatibility problem of such UEs, the **NRDUCellCollabServ**.***Nul1TxToSul1TxSwitchTime*** parameter can be used to specify the time taken to switch between the NUL and SUL carriers. **Figure 4-7** shows the time for switching between NUL and SUL carriers.

Figure 4-7 Time required for switching between NUL and SUL carriers



The gNodeB reconfigures channel resources for such UEs based on the value of the **NRDUCellCollabServ**.***Nul1TxToSul1TxSwitchTime*** parameter to make UL and DL Decoupling take effect for these UEs.

4.1.3 Simultaneous Use with Super Uplink

When super uplink and UL and DL Decoupling are both enabled, super uplink takes effect in cell center areas, and UL and DL Decoupling takes effect in cell edge areas. In this case, random access and mobility management are different compared to when only super uplink is enabled. For details about super uplink, see *Super Uplink*. Super uplink can be enabled together with UL and DL Decoupling only on macro base stations, not on LampSite base stations.

Random Access

During initial RRC connection setup, a UE receives system information from the gNodeB, obtains the *rsrp-ThresholdSSB-SUL* IE in SIB1, measures the downlink RSRP of the NR TDD carrier, and determines the uplink carrier for random access. The value of *rsrp-ThresholdSSB-SUL* is specified by the **NRDUCellCollabServ**.***RsrpThld*** parameter.

- If the RSRP of an NR cell is greater than or equal to the value of the **NRDUCellCollabServ**.***RsrpThld*** parameter, the UE is in an area with good NR uplink coverage. The gNodeB instructs the UE to initiate a random access procedure on the NR TDD carrier, and super uplink takes effect.
- If the RSRP of an NR cell is less than the value of the **NRDUCellCollabServ**.***RsrpThld*** parameter, the UE is in an area with weak or no NR uplink coverage. The gNodeB instructs the UE to initiate a random access procedure on the SUL carrier, and UL and DL Decoupling takes effect.

Intra-Cell Mobility Management

When a UE moves in an NR cell, it uses super uplink in cell center areas, and UL and DL Decoupling in cell edge areas. These two functions are switched when certain conditions relating to either uplink quality or user experience are met. The **UL_CARR_SEL_FOR_USER_EXP_SW** option of the **NRDUCellCollabServ**.***SulAlgoSwitch*** parameter specifies whether uplink quality or user experience is used.

- When the **UL_CARR_SEL_FOR_USER_EXP_SW** option is selected, the uplink carrier is selected based on user experience. The gNodeB then determines whether super uplink or UL and DL Decoupling takes effect for a UE based on the selected carrier.

- When the **UL_CARR_SEL_FOR_USER_EXP_SW** option is deselected, the uplink carrier is selected based on uplink quality. The details are as follows:
  - If the SRS RSRP of a UE on the NR TDD carrier is greater than the sum of the **NRDUCellCollabServ**.***SrsRsrpThld*** parameter value and hysteresis (2 dB), super uplink takes effect for the UE. In this scenario, the UE transmits uplink data on NR TDD and SUL carriers in TDM mode.
  - If the SRS RSRP of a UE on the NR TDD carrier is less than or equal to the result of the **NRDUCellCollabServ**.***SrsRsrpThld*** parameter value minus hysteresis (2 dB), UL and DL Decoupling takes effect for the UE. In this scenario, the UE transmits uplink data on the SUL carrier only.

Inter-Cell Mobility Management

When a UE that supports super uplink moves between NR cells and a handover to the target cell for which super uplink and UL and DL Decoupling are enabled is triggered, the gNodeB selects the NR TDD or SUL carrier for the UE and notifies the UE of the uplink carrier through an RRCReconfiguration message. The specific selection procedure is as follows:

1. The gNodeB delivers intra-RAT event A3 measurement configurations to the UE, instructing it to measure the signal quality of neighboring NR cells.
2. After receiving a measurement report from the UE, the source gNodeB forwards the RSRP of the target cell to the target gNodeB, which then selects an uplink carrier for the UE based on the following rules:
   - If the RSRP of the target cell is greater than or equal to the value of the **NRDUCellCollabServ**.***RsrpThld*** parameter, the UE is in an area with good NR uplink coverage. The gNodeB instructs the UE to initiate a random access procedure on the NR TDD carrier.
   - If the RSRP of the target cell is less than the value of the **NRDUCellCollabServ**.***RsrpThld*** parameter, the UE is in an area with weak or no NR uplink coverage. The gNodeB instructs the UE to initiate a random access procedure on the SUL carrier.
3. The target gNodeB notifies the source gNodeB of the selected uplink carrier. The source gNodeB sends the uplink carrier information of the target cell to the UE through an RRCReconfiguration message.
4. The UE initiates a random access procedure on the uplink carrier indicated in the RRCReconfiguration message.

4.2 Network Analysis

4.2.1 Benefits

The basic functions of UL and DL Decoupling improve the average uplink UE throughput, and decrease the proportion of UEs whose rates fall within a low-rate range in a cell.

The benefits provided by the basic functions can be observed by using the methods described in **4.4.4 Network Monitoring**.

A lower network interference level (indicated by **N.UL.NI.Avg**), a smaller distance to the main lobe direction of the antenna, and a larger SUL bandwidth result in larger benefits. A higher network interference level, a larger distance to the main lobe direction of the antenna, and a smaller SUL bandwidth result in small benefits. UL and DL Decoupling will not provide negative gains.

SUL CoMP improves the uplink reception performance and throughput for UEs in SUL overlapping areas by performing joint reception in intra-frequency continuous networking. For details, see **4.4.4 Network Monitoring**.

UE compatibility handling allows UL and DL Decoupling to take effect for UEs for which the time taken to switch between carriers is larger than 0 µs. As a result, such UEs can obtain gains provided by UL and DL Decoupling.

After cell radius-based UE random access control is enabled, contention-based random access from UEs whose distance to the base station exceeds the cell radius is forbidden.

4.2.2 Impacts

The impacts between this function and other functions are described in **4.3.2.3 Function Impacts**.

- For UEs for which the time taken to switch between carriers equals 0 µs, their uplink UE throughput decreases when the **NRDUCellCollabServ**.***Nul1TxToSul1TxSwitchTime*** parameter is set to a value other than **MICROSECOND0**.
- The number of the RBs occupied by SIB1 may slightly increase and the downlink peak throughput slightly decreases because the gNodeB broadcasts the SUL information through SIB1.
- SUL uplink scheduling consumes CCE resources of the NR TDD cell. When CCE resources are insufficient in the NR TDD cell, uplink or downlink CCE resource application is more likely to fail. As a result, the uplink or downlink throughput of the NR TDD cell may decrease, and the downlink user-plane delay may increase.
- It is recommended that the **UL_DL_CCE_SHARING_SW** or **UL_DL_CCE_RATIO_ADAPT_SW** option of the **NRDUCellPdcch**.***PdcchAlgoSwitch*** parameter be selected to reduce the probability of CCE resource application failures.
- When UL and DL Decoupling is in effect, UEs randomly access an SUL cell. Random access–related counters of the NR TDD cell are measured, which may cause the values to fluctuate. The values of random access–related counters of the SUL cell are all 0 (except for **N.RA.Contention.Att.Max**, **N.RA.Dedicated.PreambleAssign.Num**, and **N.RA.Dedicated.PreambleAssign.Num**).
- The UE random access control based on cell radius function enables the gNodeB to detect UE access beyond the cell radius, which increases the consumption of physical resources. Since contention-based random access is prohibited for UEs whose distance to the gNodeB exceeds the cell radius, the random access success rate and RRC connection setup success rate may decrease, and the cell traffic, cell rate, and PRB usage may decrease. In addition, enabling "optimized measurement of RRC connection setup success rate based on cell radius" can prevent the RRC connection setup success rate from decreasing.

4.3 Requirements

4.3.1 Licenses

<table>
<tr><th>Feature ID</th><th>Feature Name</th><th>Model</th><th>License Control Item Name</th><th>Sales Unit</th></tr>
<tr><td>FOFD-010205</td><td>UL and DL Decoupling</td><td>NR0SULADLD00</td><td>UL and DL Decoupling (NR)</td><td>Per Cell</td></tr>
<tr><td colspan="5">For the license control item NR0SULADLD00:
● If the feature is enabled for an activated cell, the cell will not be automatically deactivated when the number of units of the license for the feature is insufficient due to license expiration or forcible license file installation. However, after the cell is reestablished, the cell cannot be activated again due to license insufficiency.
● If the feature is not enabled for an activated cell and the number of units of the license for the feature is insufficient due to license expiration or forcible license file installation:
– When the license sales number pre-check function is enabled, the NRDUCellAlgoSwitch.UlDlDecouplingSwitch parameter cannot be set to ON due to license insufficiency, preventing cell activation failures.
– When the license sales number pre-check function is disabled, the cell cannot be activated again due to license insufficiency after the feature is enabled and the parameters that cause automatic cell reset are modified.
The license sales number pre-check function is controlled by the LIC_SALE_NUM_PRE_CHECK_SW option of the gNBLicenseAlmThld.LicenseChkSw parameter.</td></tr>
</table>

NOTE

The insufficiency of certain feature licenses affects cell activation. For details, see the **Is Cell Activation Affected by License Insufficiency** column in *License Control Item Lists* in 5G RAN feature documentation.

4.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

4.3.2.1 Prerequisite Functions

None

4.3.2.2 Mutually Exclusive Functions

RAT	Function Name	Function Switch	Reference (5G RAN Feature Documentation)	Description
Low-frequency NR TDD	High-speed Railway Superior Experience	**NRDUCell**.***HighSpeedFlag*** set to **HIGH_SPEED**	*High Speed Mobility*	UL and DL Decoupling cannot be enabled in high-speed cells.
Low-frequency NR TDD	Hyper Cell	**NRDUCell**.***NrDuCellNetworkingMode*** set to **HYPER_CELL**	*Hyper Cell*	UL and DL Decoupling cannot be enabled in hyper cells.
Low-frequency NR TDD	Cell Combination	**NRDUCell**.***NrDuCellNetworkingMode*** set to **HYPER_CELL_COMBINE_MODE**	*Cell Combination*	UL and DL Decoupling cannot be enabled in combined cells.
Low-frequency NR TDD	Channel decoupling	**SUL_CHANNEL_DECOUPLE_SW** option of the **NRDUCellCollabServ**.***SulAlgoSwitch*** parameter	*Super Uplink*	Channel decoupling cannot be enabled when UL and DL Decoupling is enabled.
Low-frequency NR TDD	Super uplink (SUL and NR FDD co-deployment)	**FLEX_SUPER_UPLINK_SW** option of the **NRDUCellAlgoSwitch**.***SuperUplinkSwitch*** parameter	*Super Uplink*	Super uplink with SUL and NR FDD co-deployment cannot be enabled when UL and DL Decoupling is enabled.

RAT	Function Name	Function Switch	Reference (5G RAN Feature Documentation)	Description
Low-frequency NR TDD	SRS interference coordination based on self-contained and uplink slots	**SRS_INTRF_COORD_S_U_SLOT_SW** option of the **NRDUCellSrs**.***SrsAlgoExtSwitch*** parameter	*Channel Management*	When SRS interference coordination based on self-contained and uplink slots is enabled, the **NRDUCellCollabServ**.***Nul1TxToSul1TxSwitchTime*** parameter cannot be set to **MICROSECOND140**.
Low-frequency NR TDD	Fusion Cell	**NRDUCell**.***NrDuCellNetworkingMode*** set to **FUSION_CELL**	*Fusion Cell (Low-Frequency TDD)*	UL and DL Decoupling cannot be enabled in Fusion Cell networking.
Low-frequency NR TDD	Intelligent power control energy saving	**SMART_PWR_CTRL_ENERGY_SAVE_SW** option of the **NRDUCellAlgoSwitch**.***PowerSavingSwitch*** parameter	*Energy Conservation and Emission Reduction*	Intelligent power control energy saving cannot be enabled in SUL cells.
Low-frequency NR TDD	Pattern-free PDCCH rate matching activation rate increase	**PDCCH_NO_PAT_RM_RATIO_INC_SW** option of the **NRDUCellServExp**.***SpdTurboAlgoSwitch*** parameter	*Downlink Scheduling*	Pattern-free PDCCH rate matching activation rate increase cannot be enabled when UL and DL Decoupling is enabled.

RAT	Function Name	Function Switch	Reference (5G RAN Feature Documentation)	Description
Low-frequency NR TDD	5QI-specific UE-number-based connected mode MLB	**INTER_FREQ_CONNECTED_MLB** option of the **NRCellAlgoSwitch**.***MlbAlgoSwitch*** parameter selected, and **NRCellMlb**.***MlbTriggerMode*** set to **SPECIFIED_5QI_USER_NUM**	*Mobility Load Balancing in Connected Mode*	5QI-specific UE-number-based connected mode MLB cannot be enabled when UL and DL Decoupling is enabled.
Low-frequency NR TDD	PUSCH scheduling based on SRS interference avoidance	**SRS_INTRF_AVOID_PUSCH_SCH_SW** option of the **NRDUCellSrs**.***SrsAlgoExtSwitch*** parameter	None	PUSCH scheduling based on SRS interference avoidance cannot be enabled when UL and DL Decoupling is enabled.
Low-frequency NR TDD	Cell deployment in IAB mode	**NRDUCell**.***DeployPosition*** set to **DEPLOY_IAB**	None	UL and DL Decoupling cannot be enabled in cells deployed in IAB mode.
Low-frequency NR TDD	Cooperation between joint transmission and pattern-free PDCCH rate matching	**JT_PDCCH_NO_PATT_RM_COORD_SW** option of the **NRDUCellPdcch**.***PdcchAlgoSwitch*** parameter	None	Cooperation between joint transmission and pattern-free PDCCH rate matching cannot be enabled when UL and DL Decoupling is enabled.

RAT	Function Name	Function Switch	Reference (5G RAN Feature Documentation)	Description
Low-frequency NR TDD	Uplink assistance	**UL_ASSISTANCE_SW** option of the **NRDUCellCollabServ**.***MultiCellMimoSwitch*** parameter	None	Uplink assistance cannot be enabled when UL and DL Decoupling is enabled.

4.3.2.3 Function Impacts

RAT	Function Name	Function Switch	Reference	Description
LTE FDD LTE TDD	Downlink FDD+TDD CA	**InterFddTddCaSwitch** option of the **CaMgtCfg**.***CellCaAlgoSwitch*** parameter	*Carrier Aggregation*	When an 8:2 slot configuration and a 3-ms frame offset are configured for the NR TDD band, and when UL and DL Decoupling is enabled together with LTE FDD and NR Uplink Spectrum Sharing, a 3-ms frame offset must be configured for LTE FDD. If LTE TDD is deployed in the same frequency band as NR TDD, downlink FDD+TDD CA cannot be enabled.
Low-frequency NR TDD	Uplink fallback to LTE	**UL_FALLBACK_TO_LTE_SWITCH** option of the **NRCellNsaDcConfig**.***NsaDcAlgoSwitch*** parameter	*EPC-based NSA Basics*	UL and DL Decoupling provides larger uplink coverage gains than the uplink fallback to LTE function in the NSA Networking based on EPC feature. Therefore, if the uplink fallback to LTE function is used together with UL and DL Decoupling, the uplink fallback to LTE function does not take effect.

RAT	Function Name	Function Switch	Reference	Description
LTE FDD LTE TDD Low-frequency NR TDD	TDM power control	• LTE: **NSA_DC_ENH_UL_POWER_CONTROL_SW** and **TDM_SWITCH** options of the **NsaDcMgmtConfig**.***NsaDcAlgoSwitch*** parameter • NR: **NSA_DC_ENH_UL_POWER_CONTROL_SW** option of the **NRCellNsaDcConfig**.***NsaDcAlgoSwitch*** parameter	*EPC-based NSA Performance Enhancement*	If UL and DL Decoupling has taken effect, TDM power control and secondary intermodulation interference avoidance in the time domain do not take effect. The TDM-Pattern corresponding to UL and DL Decoupling is used.

RAT	Function Name	Function Switch	Reference	Description
LTE FDD LTE TDD Low-frequency NR TDD	Secondary intermodulation interference avoidance in the time domain	• LTE: **INTERFERENCE_AVOID_SW** and **TDM_SWITCH** options of the **NsaDcMgmtConfig**.***NsaDcAlgoSwitch*** parameter • NR: **CROSS_MDLT_INTRF_AVOID_SW** option of the **NRDUCellAlgoSwitch**.***NsaDcAlgoSwitch*** parameter	*EPC-based NSA Basics*	

RAT	Function Name	Function Switch	Reference	Description
LTE FDD LTE TDD Low-frequency NR TDD	Uplink data transmission path selection	• LTE **NSA_DC_UL_PATH_SELECTION_SW** option of the **NsaDcMgmtConfig**.***NsaDcAlgoSwitch*** parameter • NR **NSA_DC_UL_PATH_SELECTION_SW** option of the **NRCellNsaDcConfig**.***NsaDcAlgoSwitch*** parameter	*EPC-based NSA Performance Enhancement*	Uplink data transmission path selection does not take effect in an NR cell enabled with UL and DL Decoupling.

RAT	Function Name	Function Switch	Reference	Description
LTE FDD LTE TDD Low-frequency NR TDD	Network-coordinated dynamic UE power sharing	• LTE **NSA_DC_COOPERATION_DPS_SW** option of the **NsaDcMgmtConfig**.***NsaDcAlgoSwitch*** parameter • NR **NSA_DC_COOPERATION_DPS_SW** option of the **NRCellNsaDcConfig**.***NsaDcAlgoSwitch*** parameter	*EPC-based NSA Performance Enhancement*	If UL and DL Decoupling has taken effect and the base station performs uplink scheduling in TDM mode, network-coordinated dynamic UE power sharing does not take effect.
Low-frequency NR TDD	Power saving BWP	**BWP2_SWITCH** option of the **NRDUCellUePwrSaving**.***BwpPwrSavingSw*** parameter	*UE Power Saving*	UEs working in BWP2 for power saving do not support SUL CoMP.
Low-frequency NR TDD	Low latency and high reliability	**HIGH_RELIABILITY_BASIC_SW** option of the **NRDUCellAlgoSwitch**.***HighReliabilitySwitch*** parameter	*URLLC*	The low latency and high reliability function does not take effect for UEs using UL and DL Decoupling.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency NR TDD	Spectral-efficiency-based connected mode MLB	**INTER_FREQ_CONNECTED_MLB_SW** option of the **NRCellAlgoSwitch**.***MlbAlgoSwitch*** parameter, and **NRCellMlb**.***MlbTriggerMode*** set to **SPEC_EFF_BASED_USER_NUM**	*Mobility Load Balancing in Connected Mode*	The gNodeB does not select UEs for which UL and DL Decoupling has taken effect as the UEs to be handed over.
Low-frequency NR TDD	Intra-band CA (TDD)	**INTRA_BAND_CA_SW** option of the **NRDUCellAlgoSwitch**.***CaAlgoSwitch*** parameter	*Carrier Aggregation*	For UEs that support both UL and DL Decoupling and intra-band CA, either UL and DL Decoupling or intra-band CA can take effect at a time.
Low-frequency NR TDD	Intra-FR inter-band CA	**INTRA_FR_INTER_BAND_CA_SW** option of the **NRDUCellAlgoSwitch**.***CaAlgoSwitch*** parameter	*Carrier Aggregation*	For UEs that support both UL and DL Decoupling and intra-FR inter-band CA, either UL and DL Decoupling or intra-FR inter-band CA can take effect at a time.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency NR TDD	NR DU cell resource management between network slices	**RB_DYNAMIC_CONTROL_SW** option of the **NRDUCellAlgoSwitch**.***NetworkSliceAlgoSwitch*** parameter	*Network Slicing*	When both the NR DU cell resource management between network slices function and UL and DL Decoupling are enabled, and the **RB_DYNAMIC_CONTROL_SW** option of the **NRDUCellAlgoSwitch.*NetworkSliceAlgoSwitch*** parameter is selected for the SUL cell, this option must also be selected for the corresponding TDD cell, and the proportion of RB resources configured for the TDD cell takes effect in the SUL cell.
Low-frequency NR TDD	RFSP-based RB resource control	**RB_DYNAMIC_CONTROL_SW** option of the **NRDUCellCommonSw**.***RfspAlgoSwitch*** parameter	*Resource Control in NSA Networking*	Assume that both RFSP-based RB resource control and UL and DL Decoupling are enabled. If RFSP-based RB resource control is enabled in the SUL cell, it must also be enabled in the associated TDD cell, and the proportion of RB resources configured for the TDD cell takes effect in the SUL cell.
Low-frequency NR TDD	Uplink multi-frequency coordination	**MULTI_FREQ_UL_COORD_SW** option of the **NRDUCellFeatureSw**.***MultiCarrierSwitch*** parameter	*Multi-Frequency Smart Selection*	The gNodeB does not select UEs for which UL and DL Decoupling has taken effect.
Low-frequency NR TDD	Multi-frequency beam coordination	**MULTI_FREQ_BEAM_COORD_SW** option of the **NRDUCellFeatureSw**.***MultiCarrierSwitch*** parameter	*Multi-Frequency Smart Selection*	The gNodeB does not select UEs for which UL and DL Decoupling has taken effect.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency NR TDD	PDCCH skipping	**PDCCH_SKIP_SW** option of the **NRDUCellUePwrSaving**.***SchPwrSavingSw*** parameter	*UE Power Saving*	When both SUL CoMP and PDCCH skipping are enabled, PDCCH skipping does not take effect for SUL UEs.
Low-frequency NR TDD	Downlink massive CA	**NRDUCellCarrMgmt**.***CaDlMaxCcNum*** set to **DL4CC** or **DL5CC**.	*CA Experience Improvement*	UL and DL Decoupling and downlink massive CA cannot take effect at the same time.
Low-frequency NR TDD	Inter-FR CA	**INTER_FR_FR1TOFR2_CA_SW** option of the **NRDUCellAlgoSwitch**.***CaAlgoSwitch*** parameter	*Carrier Aggregation*	Inter-FR CA and UL and DL Decoupling cannot take effect at the same time.
Low-frequency NR TDD	3GPP R15-compliant uplink slot aggregation	**VONR_UL_SLOT_AGGREGATION_SW** option of the **NRDUCellUlSch**.***VoiceUlSchSwitch*** parameter	*VoNR*	3GPP R15-compliant uplink slot aggregation does not take effect on UEs with UL and DL Decoupling in effect.
Low-frequency NR TDD	3GPP R16-compliant uplink slot aggregation	**VOICE_R16_UL_SLOT_AGG_SW** option of the **NRDUCellUlSch**.***VoiceUlSchSwitch*** parameter	*VoNR*	3GPP R16-compliant uplink slot aggregation does not take effect on UEs with UL and DL Decoupling in effect.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency NR TDD	QCI-based resource control	**RB_DYNAMIC_CONTROL_SW** option of the **NRDUCellFeatureSw**.***QciBasedDiffServExpSw*** parameter	*QCI-based Resource Control*	When QCI-based resource control (controlled by the **RB_DYNAMIC_CONTROL_SW** option of the **NRDUCellFeatureSw**.***QciBasedDiffServExpSw*** parameter) is enabled together with UL and DL Decoupling and is enabled in the SUL cell, the proportion of RB resources configured for the TDD cell takes effect in the SUL cell.
Low-frequency NR TDD	Time-domain power saving BWP (low-frequency TDD)	**TDM_BWP2_SWITCH** option of the **NRDUCellUePwrSaving**.***BwpPwrSavingSw*** parameter	*UE Power Saving*	UEs on the NUL carrier can be switched to BWP2 while UEs on the SUL carrier cannot be switched to BWP2. The scheduling delay of BWP2 UEs on the NUL carrier increases when the SUL carrier is activated.
Low-frequency NR TDD	RedCap Introduction	**REDCAP_SW** option of the **NRDUCellFeatureSw**.***UeCapbBasedFunSw*** parameter	*RedCap*	UL and DL Decoupling does not take effect on RedCap UEs when proactive avoidance of remote interference is enabled.
Low-frequency NR TDD	Proactive avoidance of remote interference	**SRS_RIM_INTRF_AVOID_SW** and **SRS_INTRF_COORD_S_SLOT_SW** options of the **NRDUCellSrs**.***SrsAlgoExtSwitch*** parameter	*Remote Interference Management (Low-Frequency TDD)*	

RAT	Function Name	Function Switch	Reference	Description
Low-frequency NR TDD	User-rate-based dynamic network slice resource management	**DYNAMIC_NETWORKSLICE_SW** option of the **NRDUCellCommonSw**.***ServiceDiffAlgoExtSwitch*** parameter	*Network Slicing*	When user-rate-based dynamic network slice resource management is enabled for an SUL cell with UL and DL Decoupling in effect, you are advised to also enable it for the associated TDD cell. Otherwise, the gNodeB may perform inaccurate dynamic resource adjustments.
Low-frequency NR TDD	User-rate-based dynamic QCI resource management	**DYNAMIC_QCI_GROUP_SW** option of the **NRDUCellFeatureSw**.***QciBasedDiffServExpSw*** parameter	*QCI-based Resource Control*	When user-rate-based dynamic QCI resource management is enabled for an SUL cell with UL and DL Decoupling in effect, you are advised to also enable it for the associated TDD cell. Otherwise, the gNodeB may perform inaccurate dynamic resource adjustments.

4.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910, and 5900 series base stations must be configured with the BBU5900, BBU5910, or BBU5900A.

Boards

The gNodeB supports the following main control boards and baseband processing units:

- Main control boards: UMPTe, UMPTg, UMPTga, and UMPTha
- Baseband processing units for macro base stations: All baseband processing units that support SUL cell specifications are compatible. For details about these boards, see **Product Specifications** > **BBU Technical Specifications** > **Technical Specifications of Baseband Processing Units** > **NR TDD Technical Specifications of Baseband Processing Units** in *3900 & 5900 Series Base Station Product Documentation*. If the value of **Frequency Band Type of a Cell** corresponding to a board includes SUL, the board supports SUL cell specifications.

The constraints on the gNodeB baseband processing units are as follows:

- The SUL cells served by a UBBPfw1 board must be configured with the same bandwidth and number of receive antennas.
- The paired NR TDD cell and SUL cell must be deployed on the same baseband processing unit.
- The UBBPfw1 does not support SUL CoMP.

Constraints on the eNodeB baseband processing units: When TDM is enabled in NSA networking, the LBBPc, LBBPd1, LBBPd2, LBBPd3, and LBBPd5 do not support this feature.

RF Modules

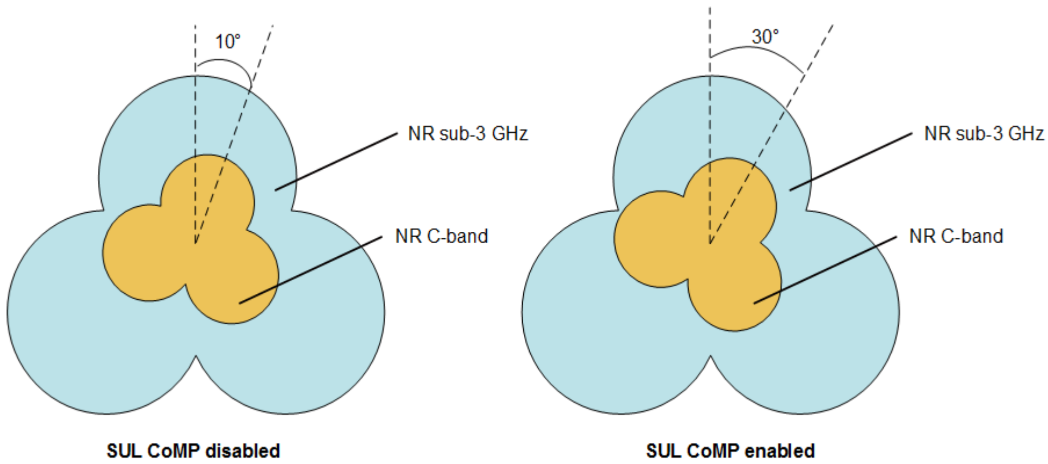
Frequency Band	Supported Module	TX/RX Mode	Bandwidth Requirement
n78 (3.5 GHz/3.7 GHz)	Modules supporting 32T32R and 64T64R	32T32R and 64T64R	40 MHz, 50 MHz, 60 MHz, 70 MHz, 80 MHz, 90 MHz, 100 MHz
n80 (1.8 GHz)	RRU3952m, RRU3953, RRU3953w, RRU3959, RRU3959a, RRU3959w, RRU3962, RRU3971, RRU3971a, 5000 series RRUs[a]	2R and 4R. Splitting is not supported.	15 MHz, 20 MHz
n82 (800 MHz)	RRU3965, 5000 series RRUs[a]	2R and 4R. Splitting is not supported.	10 MHz
n83 (700 MHz)	RRU3262, 5000 series RRUs[a]	2R and 4R. Splitting is not supported.	10 MHz
n84 (2.1 GHz)	RRU3952, RRU3952m, RRU3953, RRU3958, RRU3959, RRU3959a, RRU3962, RRU3971, 5000 series RRUs[a]	2R and 4R. Splitting is not supported.	15 MHz, 20 MHz
n86 (AWS)	RRU3971, 5000 series RRUs[a]	2R and 4R. Splitting is not supported.	15 MHz, 20 MHz

Frequency Band	Supported Module	TX/RX Mode	Bandwidth Requirement
a: 5000 series RRUs refer to RRUs named RRU5*xxx*, for example, RRU5909.			

4.3.4 Networking

- The RF modules for the NUL carrier and SUL carrier must be deployed on the same site and cover the same geographical area.
- The power control and link management of the SUL carrier depend on the downlink measurement of the NUL carrier, because the SUL carrier is not paired with a downlink carrier. When the SUL carrier coverage is identical with the coverage of the corresponding NUL carrier, the optimal decoupling effect can be achieved. Before deploying this feature, ensure that the difference between the antenna azimuth of the SUL carrier and that of the corresponding NUL carrier is less than or equal to 10°. When SUL CoMP is enabled, the maximum difference between the antenna azimuth of the SUL carrier and that of the corresponding NUL carrier can reach 30°, as shown in **Figure 4-8**. For details, see **4.1.2.2 SUL CoMP**.

Figure 4-8 Azimuth requirements of the antennas working in NR sub-3 GHz and NR TDD bands



4.3.5 Cells

- UL and DL Decoupling with independent SUL deployment can be enabled only when the **NRDUCellSulRelation**.***SulType*** parameter is set to **FIX_SUL**.

4.3.6 Others

UEs must support the combinations of NR TDD and sub-3 GHz bands. If BandCombinationList in the UECapabilityInformation message contains the combinations of NR TDD and sub-3 GHz bands, the concerned UE supports these combinations.

4.4 Operation and Maintenance

4.4.1 Precautions

About Feature Deployment

When enabling this feature, ensure that the setting of the **NRDUCellUlTaConfig**.***UlTimeAlignmentTimer*** parameter for the SUL cell is consistent with that for the NUL cell.

About TA Offset Configuration

A timing advance (TA) offset is introduced to NR TDD, which is specified by the **NRDUCell**.***TaOffset*** parameter. NR SUL uses the same TA offset as NR TDD, to ensure that the uplink time of NR SUL is aligned with that of NR TDD. If the SUL spectrum is obtained through LTE FDD and NR Uplink Spectrum Sharing, the uplink time of LTE FDD needs to align with that of NR SUL. The frame offset must be configured on the LTE side to make the TA offsets aligned. For details about the configuration requirements, see *LTE FDD and NR SUL Dynamic Spectrum Sharing*.

About the MOs Configured for SUL Cells

Since SUL cells have only uplinks, the configuration of SUL cells involves only certain MOs and parameters. In addition to the parameters required for the deployment of UL and DL Decoupling (see **4.4.2.1 Data Preparation**), the following lists the MOs and parameters that take effect for SUL cells.

In the following MOs, all parameters take effect for SUL cells:

- **NRDUCellPrach**
- **NRDUCellQciBearer**
- **NRDUCellUlPcConfig**
- **NRDUCellCollabServ**
- **NRDUCellSulRelation**
- **NRDUCellUlTaConfig**
- **NRDUCellUeCoop**
- **NRDUCellSelConfig**
- **NRDUCellSpctCloud**

In the MOs described in the tables below, only some parameters take effect for SUL cells.

Table 4-3 Parameters in an **NRDUCellPucch** MO that take effect

Parameter Name	Parameter ID
Format3 CSI-dedicated RB Number	**NRDUCellPucch**.***CsiDedicatedRbNum***

Parameter Name	Parameter ID
CSI Report Period	**NRDUCellPucch**.***CsiReportPeriod***
CSI Resource Algo Switch	**NRDUCellPucch**.***CsiResoureAlgoSwitch***
Format1 RB Number	**NRDUCellPucch**.***Format1RbNum***
Format3 RB Number	**NRDUCellPucch**.***Format3RbNum***
Format4 CSI-dedicated RB Number	**NRDUCellPucch**.***Format4CsiDedicatedRbNum***
Format4 RB Number	**NRDUCellPucch**.***Format4RbNum***
SR Period	**NRDUCellPucch**.***SrPeriod***
SR Resource Algo Switch	**NRDUCellPucch**.***SrResoureAlgoSwitch***

Table 4-4 Parameters in an **NRDUCellPusch** MO that take effect

Parameter Name	Parameter ID
Uplink Target IBLER	**NRDUCellPusch**.***UlTargetIbler***
SINR Threshold for Waveform Selection	**NRDUCellPusch**.***SinrThldforWaveformSel***
Uplink Additional DMRS Position	**NRDUCellPusch**.***UlAdditionalDmrsPos***
Uplink DMRS Type	**NRDUCellPusch**.***UlDmrsType***
Uplink PUSCH Algorithm Switch	**NRDUCellPusch**.***UlPuschAlgoSwitch***
Initial Uplink SINR Adjustment	**NRDUCellPusch**.***InitUlSinrAdjust***
Uplink Preallocation Switch	**NRDUCellPusch**.***UlPreallocationSwitch***
Uplink Smart Preallocation Duration	**NRDUCellPusch**.***UlSmartPreallocationDur***

Table 4-5 Parameters in an **NRDUCellTrp** MO that take effect

Parameter Name	Parameter ID
NR DU Cell TRP ID	**NRDUCellTrp**.***NrDuCellTrpId***
NR DU Cell ID	**NRDUCellTrp**.***NrDuCellId***
Transmit and Receive Mode	**NRDUCellTrp**.***TxRxMode***

Parameter Name	Parameter ID
Baseband Equipment ID	**NRDUCellTrp**.***BasebandEqmId***
Baseband Resource Mutual Aid Switch	**NRDUCellTrp**.***BbResMutualAidSw***
CPRI Compression	**NRDUCellTrp**.***CpriCompression***
Frequency Range and Duplex Mode	**NRDUCellTrp**.***FrAndDuplexMode***

Table 4-6 Parameters in an **NRDUCellSrs** MO that take effect

Parameter Name	Parameter ID
SRS Period	**NRDUCellSrs**.***SrsPeriod***
SRS Algorithm Switch	**NRDUCellSrs**.***SrsAlgoSwitch***

Table 4-7 Parameters in an **NRDUCellCoverage** MO that take effect

Parameter Name	Parameter ID
NR DU Cell TRP ID	**NRDUCellCoverage**.***NrDuCellTrpId***
NR DU Cell Coverage ID	**NRDUCellCoverage**.***NrDuCellCoverageId***
Sector Equipment ID	**NRDUCellCoverage**.***SectorEqmId***

Table 4-8 Parameters in an **NRDUCell** MO that take effect

Parameter Name	Parameter ID
Frequency Band	**NRDUCell**.***FrequencyBand***
Uplink Bandwidth	**NRDUCell**.***UlBandwidth***
Cell Radius	**NRDUCell**.***CellRadius***
Subcarrier Spacing	**NRDUCell**.***SubcarrierSpacing***
Cyclic Prefix Length	**NRDUCell**.***CyclicPrefixLength***
Cell ID	**NRDUCell**.***CellId***
Physical Cell ID	**NRDUCell**.***PhysicalCellId***
Uplink NARFCN	**NRDUCell**.***UlNarfcn***
NR DU Cell Networking Mode	**NRDUCell**.***NrDuCellNetworkingMode***
Logical Root Sequence Index	**NRDUCell**.***LogicalRootSequenceIndex***

Parameter Name	Parameter ID
PRACH Frequency Start Position	**NRDUCell**.***PrachFreqStartPosition***

Table 4-9 Parameter in an **NRDUCellUePwrSaving** MO that takes effect

Parameter Name	Parameter ID	Option
NR DU Cell DRX Algorithm Switch	**NRDUCellUePwrSaving**.***NrDuCellDrxAlgoSwitch***	BASIC_DRX_SW

Table 4-10 Parameters in an **NRDUCellAlgoSwitch** MO that take effect

Parameter Name	Parameter ID	Option
Adaptive Edge Experience Enhance Switch	**NRDUCellAlgoSwitch**.***AdaptiveEdgeExpEnhSwitch***	UL_WAVEFORM_ADAPT_SW
		UL_IBLER_ADAPT_SW
Spectrum Cloudification Switch	**NRDUCellAlgoSwitch**.***SpectrumCloudSwitch***	LTE_NR_UL_SPECTRUM_SHARING_SW
NSA DC Algorithm Switch	**NRDUCellAlgoSwitch**.***NsaDcAlgoSwitch***	CROSS_MDLT_INTRF_AVOID_SW
		HARMONIC_INTRF_AVOID_SW
Uplink Interference Random Schedule Switch	**NRDUCellAlgoSwitch**.***UlIntrfRandomSchSwitch***	None
UL and DL Decoupling Switch	**NRDUCellAlgoSwitch**.***UlDlDecouplingSwitch***	None
Super Uplink Switch	**NRDUCellAlgoSwitch**.***SuperUplinkSwitch***	SUPER_UL_TDM_SCH_SW
UE Link Abnormal Detect Switch	**NRDUCellAlgoSwitch**.***UeLinkAbnormalDetectSwitch***	None
CoMP Switch	**NRDUCellAlgoSwitch**.***CompSwitch***	INTRA_GNB_UL_COMP_SW
		INTRA_GNB_OVERLAPPING_MEAS_SW
Power Saving Switch	**NRDUCellAlgoSwitch**.***PowerSavingSwitch***	TIMING_CARRIER_SHUTDOWN_SW

Parameter Name	Parameter ID	Option
		LNR_SMART_CARRIER_SHUTDOWN_SW
BWP Configuration Policy Switch	**NRDUCellAlgoSwitch**.***BwpConfigPolicySwitch***	INIT_BWP_FULL_BW_SW
Uplink 256QAM Switch	**NRDUCellAlgoSwitch**.***Ul256QamSwitch***	UL_256QAM_OFF
		UL_256QAM_FIXED
		UL_256QAM_ADAPT

About SUL CoMP

To compare the throughput before and after SUL CoMP is enabled, you are advised to select the **INTRA_GNB_OVERLAPPING_MEAS_SW** option of the **NRDUCellAlgoSwitch.*CompSwitch*** parameter for the NR TDD cell one week before SUL CoMP is enabled so that statistics of throughput-related counters for UEs in overlapping areas can be collected in advance. For specific counters, see **4.4.4 Network Monitoring**.

To observe the gains of SUL CoMP, you are advised to set the **NRCellComp**.***OverlappingRsrpOffset*** and **NRCellComp**.***UlCompRsrpOffset*** parameters to the same value. The reason is that UEs in overlapping areas related to the specific counters are determined based on event A3 in overlapping areas. That is, the SSB RSRP needs to meet the event A3 reporting condition in which the A3 offset is specified by the **NRCellComp**.***OverlappingRsrpOffset*** parameter while other variables are the same as those for SUL CoMP event A3.

4.4.2 Data Configuration

4.4.2.1 Data Preparation

Table 4-11 Parameters used for configuration on the gNodeB side

Parameter Name	Parameter ID	Setting Notes
UL and DL Decoupling Switch	**NRDUCellAlgoSwitch**.***UlDlDecouplingSwitch***	Set this parameter to **ON** in the NR TDD cell when this feature is required.
Policy Switch	**NRDUCellOp**.***PolicySwitch***	Select the **OP_SUL_SWITCH** option of this parameter for the operator that requires UL and DL Decoupling if the gNodeB is shared by multiple operators.

Parameter Name	Parameter ID	Setting Notes
Duplex Mode	**NRDUCell**.***DuplexMode***	Set this parameter to **CELL_SUL** for SUL carriers.
Frequency Band	**NRDUCell**.***FrequencyBand***	Set this parameter based on the operator's network plan. For details, see section 5.2 "Operating bands" in 3GPP TS 38.104 V15.0.0.
Uplink NARFCN	**NRDUCell**.***UlNarfcn***	Set this parameter based on the operator's network plan. For details, see section 5.4.2.3 "Channel raster entries for each operating band" in 3GPP TS 38.104 V15.0.0.
Uplink Bandwidth	**NRDUCell**.***UlBandwidth***	Set this parameter based on the operator's network plan.
Subcarrier Spacing	**NRDUCell**.***SubcarrierSpacing***	Set this parameter to **15KHZ** for the SUL carrier.
Slot Assignment	**NRDUCell**.***SlotAssignment***	Set this parameter based on the operator's network plan.
Logical Root Sequence Index	**NRDUCell**.***LogicalRootSequenceIndex***	Set this parameter based on the operator's network plan.
NR DU Cell ID	**NRDUCellSulRelation**.***NrDuCellId***	Set this parameter based on the operator's network plan.
SUL NR DU Cell ID	**NRDUCellSulRelation**.***SulNrDuCellId***	Set this parameter based on the operator's network plan.
RSRP Threshold	**NRDUCellCollabServ**.***RsrpThld***	It is recommended that this parameter use its default value during initial activation. ● When the NR uplink load is high, set this parameter to a large value to enable more UEs to be carried on the SUL carrier. ● When the NR SUL load is high, set this parameter to a small value to enable more UEs to be carried on the NUL carrier.

Parameter Name	Parameter ID	Setting Notes
SRS RSRP Threshold	**NRDUCellCollabServ**.***SrsRsrpThld***	It is recommended that this parameter use its default value during initial activation. ● When the NR uplink load is high, set this parameter to a large value to enable more UEs to be carried on the SUL carrier. ● When the NR SUL load is high, set this parameter to a small value to enable more UEs to be carried on the NUL carrier. ● When standard interface capability selection for UL and DL Decoupling is enabled by selecting the **UL_DL_DECOUPLE_STD_CAP_SEL_SW** option of the **NRCellFeatureSwitch**.***MultiFreqAlgoSwitch*** parameter, it is recommended that **SRS RSRP Threshold** be set to a small value so that super uplink switches to UL and DL Decoupling later or UL and DL Decoupling switches to super uplink earlier.

Parameter Name	Parameter ID	Setting Notes
SUL Algorithm Switch	**NRDUCellCollabServ**.***SulAlgoSwitch***	● Select the **SEC_HARMONIC_INTRF_AVOID_SW** option to enable secondary harmonic interference avoidance. ● Select the **UL_CARR_SEL_FOR_USER_EXP_SW** option to enable user-experience-based uplink carrier selection. It is recommended that this option be deselected in NSA networking, and be selected in SA networking. ● Select the **INTRA_GNB_SUL_COMP_SW** option to enable SUL CoMP. It is recommended that this option be selected when the antenna azimuth difference is greater than 10°. ● Select the **SUL_FAULT_TOLERANCE_SW** option to enable SUL carrier detection and management.
RSRP Threshold for SSB Selection	**NRDUCellPrach**.***RsrpThldForSsbSelection***	It is recommended that this parameter be set to **29** in NR TDD cells. The value of this parameter is equal to the sum of **NRDUCellSelConfig**.***MinimumRxLevel*** and 157.

Parameter Name	Parameter ID	Setting Notes
NUL 1TX to SUL 1TX Switching Time	**NRDUCellCollabServ**.***Nul1TxToSul1TxSwitchTime***	It is recommended that this parameter be set based on UE conditions. ● A smaller value of this parameter results in more symbols available for the SUL carrier and higher throughput of the SUL carrier. However, UL and DL Decoupling cannot take effect for some UEs for which the time taken to switch between carriers is larger than the value of this parameter. ● A larger value of this parameter results in more UEs for which UL and DL Decoupling takes effect. However, it also results in fewer symbols available for the SUL carrier and lower throughput of the SUL carrier.
SUL SINR Lower Limit	**NRDUCellCollabServ**.***SulSinrLowerLimit***	The default value is recommended.
SUL SINR Lower Limit Offset	**NRDUCellCollabServ**.***SulSinrLowerLimitOffset***	The default value is recommended.
CoMP Configuration Policy	**NRDUCellComp**.***CompConfigPol***	Select the **UL_2CELL_COMP_BEAR_POL_SW** option to enable SUL CoMP to take effect for UEs with specific QCIs.
Service Differentiation Algorithm Switch	**gNBQciBearer**.***ServiceDiffAlgoSwitch***	Select the **COMP_SW** option for the QCI of UEs requiring SUL CoMP for a specific QCI.

Parameter Name	Parameter ID	Setting Notes
PRACH Algorithm Extension Switch	**NRDUCellPrach**.***PrachAlgoExtSwitch***	If accurate control of cell coverage is required, select the **ACCESS_BYND_CELL_RAD_PROHIB_SW** option to enable UE random access control based on cell radius. It is recommended that this option be deselected in other scenarios.

Table 4-12 describes the parameters used for configuration on the eNodeB side, which need to be configured only in NSA networking.

Table 4-12 Parameters used for configuration on the eNodeB side

Parameter Name	Parameter ID	Setting Notes
NSA DC Algorithm Switch	**NsaDcMgmtConfig**.***NsaDcAlgoSwitch***	It is recommended that the **TDM_SWITCH** option be selected in NSA networking.
NSA DC Algorithm Switch	**NsaDcAlgoParam**.***NsaDcAlgoSwitch***	In NSA networking, if a UE always initiates a random access procedure on the NUL carrier, select the **TDM_CHG_WITH_INTRA_CELL_HO_SW** option.
Indication of PRACH Configuration Index	**RACHCfg**.***PrachConfigIndexCfgInd***	The default value is recommended.
PRACH Configuration Index	**RACHCfg**.***PrachConfigIndex***	When the **RACHCfg**.***PrachConfigIndexCfgInd*** parameter is set to **CFG**, set this parameter to **0**, **3**, **6**, **9**, **13**, **14**, **16**, **19**, **22**, **25**, **29**, **32**, **35**, **38**, **41**, **45**, **48**, **51**, **54**, or **57**.
SCG DL ARFCN	**NrScgFreqConfig**.***ScgDlArfcn***	Set this parameter to **0** when SUL is required.
NR Frequency Band	**NrScgFreqConfig**.***NrFrequencyBand***	Set this parameter based on the SUL frequency band.

Parameter Name	Parameter ID	Setting Notes
NSA DC SUL B1 Event RSRP Threshold	**NrScgFreqConfig**.***NsaDcSulB1ThldRsrp***	The default value is recommended.

4.4.2.2 Using MML Commands

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

An NR TDD cell needs to be configured before running the following commands. For details about how to configure the NR TDD cell, see *Cell Management*. In addition, refer to **4.3 Requirements** and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

- On the gNodeB side

```
//(Required when SUL spectrum is obtained from an independent frequency band) Adding the RF equipment and sector equipment for the SUL cell
ADD RRUCHAIN: RCN=0, TT=CHAIN, BM=COLD, AT=LOCALPORT, HSRN=0, HSN=2, HPN=2, CR=AUTO, USERDEFRATENEGOSW=OFF;
ADD RRU: CN=0, SRN=60, SN=0, TP=TRUNK, RCN=0, PS=0, RT=MRRU, RS=LN, RN="5G06_60", RXNUM=4, TXNUM=4, MNTMODE=NORMAL, RFDCPWROFFALMDETECTSW=OFF, RFTXSIGNDETECTSW=OFF;
ADD SECTOR: SECTORID=1, SECNAME="Sector_0", USERLABEL="for cell", ANTNUM=4, ANT1CN=0, ANT1SRN=60, ANT1SN=0, ANT1N=R0A, ANT2CN=0, ANT2SRN=60, ANT2SN=0, ANT2N=R0B, ANT3CN=0, ANT3SRN=60, ANT3SN=0, ANT3N=R0C, ANT4CN=0, ANT4SRN=60, ANT4SN=0, ANT4N=R0D, CREATESECTOREQM=FALSE;
ADD SECTOREQM: SECTOREQMID=1, SECTORID=1, ANTCFGMODE=ANTENNAPORT, ANTNUM=4, ANT1CN=0, ANT1SRN=60, ANT1SN=0, ANT1N=R0A, ANTTYPE1=RX_MODE, ANT2CN=0, ANT2SRN=60, ANT2SN=0, ANT2N=R0B, ANTTYPE2=RX_MODE, ANT3CN=0, ANT3SRN=60, ANT3SN=0, ANT3N=R0C, ANTTYPE3=RX_MODE, ANT4CN=0, ANT4SRN=60, ANT4SN=0, ANT4N=R0D, ANTTYPE4=RX_MODE;
```

  - When UL and DL Decoupling is enabled with super uplink, if no SUL cell has been added, run the following command.

```
//Adding an SUL cell
ADD NRDUCELL: NrDuCellId=1, NrDuCellName="1", DuplexMode=CELL_SUL, FrequencyBand=N80, UlNarfcn=344500, UlBandwidth=CELL_BW_15M, SubcarrierSpacing=15KHZ, LogicalRootSequenceIndex=0;
ADD NRDUCELLTRP: NrDuCellTrpId=1, NrDuCellId=1, TxRxMode=4R, CpriCompression=NO_COMPRESSION,FrAndDuplexMode=SUL;
ADD NRDUCELLCOVERAGE: NrDuCellTrpId=1, NrDuCellCoverageId=1, SectorEqmId=1;
```

  - When UL and DL Decoupling is enabled with super uplink, if the NUL cell has not been associated with the SUL cell, run the following command.

```
//Adding an SUL relationship for the NR DU cell
ADD NRDUCELLSULRELATION: NrDuCellId=0, SulIndex=0, SulType=FIX_SUL, SulNrDuCellId=1;
```

```
//Setting the thresholds for uplink carrier selection
MOD NRDUCELLCOLLABSERV: NrDuCellId=0, Nul1TxToSul1TxSwitchTime=MICROSECOND0, RsrpThld=-110, SrsRsrpThld=-111;
//Enabling UL and DL Decoupling for the NR TDD cell
MOD NRDUCELLALGOSWITCH: NrDuCellId=0, UlDlDecouplingSwitch=ON;
```

```
//(Optional) Enabling UE random access control based on cell radius
MOD NRDUCELLPRACH: NrDuCellId=0, PrachAlgoExtSwitch=ACCESS_BYND_CELL_RAD_PROHIB_SW-1;
```

   – When UL and DL Decoupling is enabled with super uplink, if the operator SUL switch has not been turned on, run the following command.

```
//(Optional) Turning on the operator SUL switch for the NR TDD cell in multi-operator sharing
scenarios
MOD NRDUCELLOP: NrDuCellId=0, OperatorId=0, PolicySwitch=OP_SUL_SWITCH-1;
```

- On the eNodeB side

```
//Modifying the RACH configuration information and setting the PrachConfigIndexCfgInd to NOT_CFG
MOD RACHCFG: LocalCellId=0, PrachConfigIndexCfgInd=NOT_CFG;
//Configuring the SUL frequency band in NSA networking
ADD NRSCGFREQCONFIG: PccDlEarfcn=1300, ScgDlArfcn=0, ScgDlArfcnPriority=7,
NrFrequencyBand=N80-1, NsaDcSulB1ThldRsrp=-115;
//Turning on the TDM switch in NSA networking
MOD NSADCMGMTCONFIG: LocalCellId=0, NsaDcAlgoSwitch=TDM_SWITCH-1;
//Turning on the switch for triggering an intra-cell handover due to a TDM status change in NSA
networking
MOD NSADCALGOPARAM: NsaDcAlgoSwitch=TDM_CHG_WITH_INTRA_CELL_HO_SW-1;
```

Optimization Command Examples

```
//Setting the RSRP threshold for SSB selection upon cell access
MOD NRDUCELLPRACH: NrDuCellId=0, RsrpThldForSsbSelection=29;
//Turning on the secondary harmonic interference avoidance switch, user-experience-based uplink carrier
selection switch, SUL CoMP switch, and SUL carrier detection and management switch
MOD NRDUCELLCOLLABSERV: NrDuCellId=0, Nul1TxToSul1TxSwitchTime=MICROSECOND0,
SulAlgoSwitch=SEC_HARMONIC_INTRF_AVOID_SW-1&UL_CARR_SEL_FOR_USER_EXP_SW-1&INTRA_GNB_SU
L_COMP_SW-1&SUL_FAULT_TOLERANCE_SW-1, SulSinrLowerLimit=-80, SulSinrLowerLimitOffset=10;
//Setting the CoMP configuration policy
MOD NRDUCELLCOMP: NrDuCellId=0, CompConfigPol=UL_2CELL_COMP_BEAR_POL_SW-1;
//Selecting the COMP_SW option of the Service Differentiation Algorithm Switch parameter for QCI 9
MOD GNBQCIBEARER: Qci=9, ServiceDiffAlgoSwitch=COMP_SW-1;
```

Deactivation Command Examples

```
//Turning off the UL and DL Decoupling switch
MOD NRDUCELLALGOSWITCH: NrDuCellId=0, UlDlDecouplingSwitch=OFF;
//Deselecting the COMP_SW option of the Service Differentiation Algorithm Switch parameter for QCI 9
MOD GNBQCIBEARER: Qci=9, ServiceDiffAlgoSwitch=COMP_SW-0;
//Restoring the CoMP configuration policy
MOD NRDUCELLCOMP: NrDuCellId=0, CompConfigPol=UL_2CELL_COMP_BEAR_POL_SW-0;

//On the eNodeB side
//Turning off the TDM switch
MOD NSADCMGMTCONFIG: LocalCellId=0, NsaDcAlgoSwitch=TDM_SWITCH-0;
```

4.4.2.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

4.4.3 Activation Verification

Basic Functions of UL and DL Decoupling

Use either of the following methods to check whether the basic functions of UL and DL Decoupling have taken effect:

- Using MML commands

  Run the **DSP NRDUCELL** command on the gNodeB for the SUL cell. If the value of **NR DU Cell State** is **Normal**, the basic functions of UL and DL Decoupling have taken effect.
- Observing counters

  When the basic functions of UL and DL Decoupling are enabled and used by UEs, observe the values of the counters listed in the table below. If any of them has a non-zero value, the basic functions of UL and DL Decoupling have taken effect. If the counter values are always zero, the basic functions of UL and DL Decoupling have not taken effect.

Counter ID	Counter Name
1911816547	**N.User.Decouple.Avg**
1911816832	**N.Decoupling.PUCCH.IntraCell.NonSultoSul.Att**
1911816833	**N.Decoupling.PUCCH.IntraCell.NonSultoSul.Succ**
1911816834	**N.Decoupling.PUCCH.IntraCell.SultoNonSul.Att**
1911816835	**N.Decoupling.PUCCH.IntraCell.SultoNonSul.Succ**

SUL CoMP

Use either of the following methods to check whether SUL CoMP has taken effect:

- Observing counters

  If the value of the **N.IntragNB.UL.CoMP.SchUser.Sum** counter is not zero, this function has taken effect.
- Tracing signaling

  On the MAE-Access, choose **Monitor** > **Signaling Trace** > **Signaling Trace Management** > **NR** > **Cell Performance Monitoring** > **UL CoMP (Cell) Monitoring**. Check the number of SUL CoMP UEs and the number of RBs used by the UEs. If the values are not zero, SUL CoMP has taken effect.

4.4.4 Network Monitoring

Basic Functions of UL and DL Decoupling

The basic functions of UL and DL Decoupling improve the average uplink UE throughput, and decrease the proportion of UEs whose rates fall within a low-throughput range in a cell.

- The average uplink throughput of UEs in a cell is measured by **User Uplink Average Throughput (DU)**.
- The proportion of UEs whose throughput falls into a low-throughput range can be obtained by dividing the number of samples in the low-throughput range by the total number of samples in a cell. A low-throughput range is relative. For example, range 0 is a low-throughput range compared with range 1. The table below lists the counters used to observe the number of UEs in each throughput range.

Counter ID	Counter Name
1911816708	**N.Thp.UL.Samp.Index0**
1911816709	**N.Thp.UL.Samp.Index1**
1911816710	**N.Thp.UL.Samp.Index2**
1911816711	**N.Thp.UL.Samp.Index3**
1911816712	**N.Thp.UL.Samp.Index4**
1911816713	**N.Thp.UL.Samp.Index5**
1911816714	**N.Thp.UL.Samp.Index6**
1911816715	**N.Thp.UL.Samp.Index7**
1911816716	**N.Thp.UL.Samp.Index8**
1911816717	**N.Thp.UL.Samp.Index9**
1911816937	**N.Thp.UL.Samp.Index10**
1911816941	**N.Thp.UL.Samp.Index11**
1911816938	**N.Thp.UL.Samp.Index12**
1911816939	**N.Thp.UL.Samp.Index13**
1911816940	**N.Thp.UL.Samp.Index14**

SUL CoMP

SUL CoMP increases the uplink throughput of UEs in overlapping areas excluding small-packet throughput without decreasing the average uplink UE throughput and the uplink UE throughput excluding small-packet throughput. You can evaluate the gains of SUL CoMP by comparing the values of the following counters before and after this function is enabled. As the counter values depend on parameter settings, set related parameters before the evaluation by referring to **4.4.1 Precautions**.

- Uplink throughput of UEs in overlapping areas excluding small-packet throughput = (**N.ThpVol.UL.CellOverlap** – **N.ThpVol.UL.SmallPkt.CellOverlap**)/ **N.ThpTime.UL.RmvSmallPkt.CellOverlap**
- Uplink UE throughput excluding small-packet throughput = (**N.ThpVol.UL** – **N.ThpVol.UE.UL.SmallPkt**)/**N.ThpTime.UE.UL.RmvSmallPkt**
- Average uplink UE throughput = **N.ThpVol.UL**/**N.ThpTime.UL**

5 SUL and NR FDD Co-Deployment

This chapter describes UL and DL Decoupling with SUL and NR FDD co-deployment. UL and DL Decoupling described in this chapter refers to UL and DL Decoupling with SUL and NR FDD co-deployment.

5.1 Principles

In SUL and NR FDD co-deployment mode, the SUL function is provided by NR FDD cells. That is, UL and DL Decoupling is implemented using NR TDD cells and NR FDD cells. UL and DL Decoupling supports one-to-one mapping between intra-gNodeB and inter-gNodeB NR TDD cells and SUL cells (NR FDD cells). Unless otherwise specified, the descriptions in the following sections apply to both intra-gNodeB and inter-gNodeB deployment.

Table 5-1 lists the combinations of NR TDD, NR FDD, and SUL frequency bands supported by UL and DL Decoupling. For details about the mapping between frequency bands and frequencies, see 3GPP TS 38.104. The bandwidth requirement of each frequency band is described in **RF Modules** in **5.3.3 Hardware**.

Table 5-1 Combinations of NR TDD, NR FDD, and SUL frequency bands supported by UL and DL Decoupling

NR TDD Frequency Band	SUL Frequency Band	NR FDD Frequency Band	Applicable Base Station Model
n78	n80	n3	Macro base station
	n81	n8	Macro base station
	n82	n20	Macro base station
	n83	n28	Macro base station

NR TDD Frequency Band	SUL Frequency Band	NR FDD Frequency Band	Applicable Base Station Model
	n84	n1	Macro and LampSite base stations
n41	n80	n3	Macro and LampSite base stations
	n81	n8	Macro base station
	n83	n28	Macro base station
	n84	n1	Macro base station
n40	n83	n28	Macro base station

NOTE

The frequency band combinations n41+n81 and n40+n83 are only for trial use.

Trial functions are functions that are not yet ready for full commercial release for certain reasons. For example, the industry chain (terminals/CN) may not be sufficiently compatible. However, these functions can still be used for testing purposes or commercial network trials. Anyone who desires to use the trial functions shall contact Huawei and enter into a memorandum of understanding (MoU) with Huawei prior to an official application of such trial functions. Trial functions are not for sale in the current version but customers may try them for free.

Customers acknowledge and undertake that trial functions may have a certain degree of risk due to absence of commercial testing. Before using them, customers shall fully understand not only the expected benefits of such trial functions but also the possible impact they may exert on the network. In addition, customers acknowledge and undertake that since trial functions are free, Huawei is not liable for any trial function malfunctions or any losses incurred by using the trial functions. Huawei does not promise that problems with trial functions will be resolved in the current version. Huawei reserves the rights to convert trial functions into commercial functions in later R/C versions. If trial functions are converted into commercial functions in a later version, customers shall pay a licensing fee to obtain the relevant licenses prior to using the said commercial functions. If a customer fails to purchase such a license, the trial function(s) will be invalidated automatically when the product is upgraded.

5.1.1 Basic Functions

UL and DL Decoupling is enabled by setting the **NRDUCellAlgoSwitch**.***UlDlDecouplingSwitch*** parameter to **ON** in both NR TDD and NR FDD cells and setting the following parameters and options to specific values.

For an NR TDD cell:

- **NRDUCellSulRelation**.***SulType*** is set to **DYN_SUL**.
- **NRDUCellCollabServ**.***NulToSulMappingType*** is set to **1TO1**.
- The NCGI-related parameters (including **NRDUCellCollabServ**.***SulMcc***, **NRDUCellCollabServ**.***SulMnc***, **NRDUCellCollabServ**.***SulGnodebId***, and **NRDUCellCollabServ**.***SulCellID***) are configured for the NR FDD cell in SUL and NR FDD co-deployment mode. NCGI stands for NR cell global identifier.

For an NR FDD cell:

- **NRDUCellCollabCoop**.***SulToNulMappingType*** is set to **1TO1**.
- The NCGI-related parameters (including **NRDUCellCollabCoop**.***NulMcc***, **NRDUCellCollabCoop**.***NulMnc***, **NRDUCellCollabCoop**.***NulGnodebId***, and **NRDUCellCollabCoop**.***NulCellID***) are configured for the NUL cell bound in 1:1 mode.

If the gNodeB is shared by multiple operators, that is, the **gNBSharingMode**.***gNBMultiOpSharingMode*** parameter is set to a value other than **INDEPENDENT**, the **OP_SUL_SWITCH** option of the **NRDUCellOp**.***PolicySwitch*** parameter must be selected in the NR TDD cell for the operator that requires UL and DL Decoupling.

5.1.1.1 Cell Configuration

Before enabling UL and DL Decoupling, parameters must be configured for NR TDD and NR FDD cells.

NR TDD Cell Configuration

In addition to the basic configurations of the NR TDD cell, the following configurations must be made:

- Parameters in an **NRDUCellSulRelation** MO (as listed in **Table 5-2**) are set to configure the relationship between NR TDD and SUL cells.
- Parameters in an **NRDUCellCollabServ** MO (as listed in **Table 5-3**) are set to configure the NR TDD cell as the serving NR DU cell corresponding to a coordination feature.

Table 5-2 Parameters for configuring SUL relationship specific for the NR TDD cell

Parameter Name	Parameter ID	Setting Notes
NR DU Cell ID	**NRDUCellSulRelation**.***NrDuCellId***	This parameter indicates the ID of the NR TDD cell.
SUL Index	**NRDUCellSulRelation**.***SulIndex***	This parameter identifies a unique SUL instance within the NR TDD cell. Currently, only one SUL carrier is supported.

Parameter Name	Parameter ID	Setting Notes
SUL Type	**NRDUCellSulRelation**.***SulType***	This parameter indicates the SUL type when UL and DL Decoupling is enabled. Set this parameter to **DYN_SUL** for UL and DL Decoupling with SUL and NR FDD co-deployment.
FDD Frequency Band	**NRDUCellSulRelation**.***FddFrequencyBand***	This parameter indicates the operating band of the FDD cell on the SUL carrier in UL and DL Decoupling with SUL and NR FDD co-deployment. This parameter must be set to the same value as the **NRDUCell**.***FrequencyBand*** parameter for the NR FDD cell.
FDD UL NR-ARFCN	**NRDUCellSulRelation**.***FddUlNrarfcn***	This parameter indicates the uplink NR-ARFCN of the FDD cell on the SUL carrier in UL and DL Decoupling with SUL and NR FDD co-deployment. This parameter is mandatory when the **NRDUCellSulRelation**.***SulType*** parameter is set to **DYN_SUL**. This parameter must be set to the same value as the **NRDUCell**.***UlNarfcn*** parameter for the NR FDD cell.
FDD UL Bandwidth	**NRDUCellSulRelation**.***FddUlBandwidth***	This parameter indicates the uplink bandwidth of the FDD cell on the SUL carrier in UL and DL Decoupling with SUL and NR FDD co-deployment. This parameter must be set to the same value as the **NRDUCell**.***UlBandwidth*** parameter for the NR FDD cell.

Parameter Name	Parameter ID	Setting Notes
FDD UL BWP0 Configuration Mode	**NRDUCellSulRelation**.***FddUlBwp0ConfigMode***	This parameter indicates whether FDD uplink BWP0 configuration is automatic or manual for UL and DL Decoupling with SUL and NR FDD co-deployment. ● If this parameter is set to **AUTO**, FDD uplink BWP0 parameters are configured automatically. Automatic configuration applies to NR FDD cells with a standard bandwidth of 20 MHz or lower. It also applies to NR FDD cells with a large bandwidth greater than 20 MHz where spectrum use is exclusive (that is, spectrum is not shared between NR FDD and LTE FDD). ● If this parameter is set to **MANUAL**, FDD uplink BWP0 parameters are configured manually. Manual configuration applies to NR FDD cells with a large bandwidth greater than 20 MHz where spectrum is shared with LTE FDD. It also applies in scenarios where FDD uplink BWP0 must be specially configured.
FDD UL BWP0 Bandwidth	**NRDUCellSulRelation**.***FddUlBwp0Bandwidth***	This parameter indicates the number of RBs across the uplink BWP0 bandwidth in UL and DL Decoupling with SUL and NR FDD co-deployment.
FDD UL BWP0 Start PRB	**NRDUCellSulRelation**.***FddUlBwp0StartPrb***	This parameter indicates the start frequency-domain position of SUL BWP0 in UL and DL Decoupling with SUL and NR FDD co-deployment. PRB 0 is used as the reference point.

Parameter Name	Parameter ID	Setting Notes
FDD UL Subcarrier Spacing	**NRDUCellSulRelation**.***FddSubcarrierSpacing***	This parameter indicates the subcarrier spacing of the FDD cell on the SUL carrier in UL and DL Decoupling with SUL and NR FDD co-deployment. This parameter is mandatory when the **NRDUCellSulRelation**.***SulType*** parameter is set to **DYN_SUL**. This parameter must be set to the same value as the **NRDUCell**.***SubcarrierSpacing*** parameter for the NR FDD cell.
FDD Cyclic Prefix Length	**NRDUCellSulRelation**.***FddCyclicPrefixLength***	This parameter indicates the cyclic prefix (CP) length of the FDD cell on the SUL carrier in UL and DL Decoupling with SUL and NR FDD co-deployment. Currently, only normal CP is supported. This parameter must be set to the same value as the **NRDUCell**.***CyclicPrefixLength*** parameter for the NR FDD cell.
FDD LTE NR SPCTSHR Flag	**NRDUCellSulRelation**.***FddLteNrSpctShrFlag***	This parameter indicates whether LTE FDD and NR spectrum is shared in the NR FDD cell for UL and DL Decoupling with SUL and NR FDD co-deployment.

Table 5-3 Parameters for configuring the serving NR DU cell corresponding to a coordination feature

Parameter Name	Parameter ID	Setting Notes
NUL To SUL Mapping Type	**NRDUCellCollabServ**.***NulToSulMappingType***	This parameter must be set to **1TO1**.
SUL MCC	**NRDUCellCollabServ**.***SulMcc***	This parameter indicates the mobile country code (MCC) of the SUL cell bound to the TDD cell in UL and DL Decoupling with SUL and NR FDD co-deployment.

Parameter Name	Parameter ID	Setting Notes
SUL MNC	**NRDUCellCollabServ**.***SulMnc***	This parameter indicates the mobile network code (MNC) of the SUL cell bound to the TDD cell in UL and DL Decoupling with SUL and NR FDD co-deployment.
SUL gNodeB ID	**NRDUCellCollabServ**.***SulGnodebId***	This parameter indicates the ID of the base station serving the SUL cell bound to the TDD cell in UL and DL Decoupling with SUL and NR FDD co-deployment.
SUL Cell ID	**NRDUCellCollabServ**.***SulCellID***	This parameter indicates the ID of the SUL cell bound to the TDD cell in UL and DL Decoupling with SUL and NR FDD co-deployment.

NR FDD Cell Configuration

When UL and DL Decoupling is enabled, basic configurations of the NR FDD cell (specified by the **NRDUCell** MO) must be consistent with the parameter settings in the **NRDUCellSulRelation** MO.

Parameters in an **NRDUCellCollabCoop** MO (as listed in **Table 5-4**) are set to configure the NR FDD cell as the cooperating NR DU cell corresponding to a coordination feature.

Table 5-4 Parameters for configuring the cooperating NR DU cell corresponding to a coordination feature

Parameter Name	Parameter ID	Setting Notes
SUL To Nul Mapping Type	**NRDUCellCollabCoop**.***SulToNulMappingType***	This parameter must be set to **1TO1**.
NUL MCC	**NRDUCellCollabCoop**.***NulMcc***	This parameter indicates the MCC of the NUL cell bound to the FDD cell in UL and DL Decoupling with SUL and NR FDD co-deployment.
NUL MNC	**NRDUCellCollabCoop**.***NulMnc***	This parameter indicates the MNC of the NUL cell bound to the FDD cell in UL and DL Decoupling with SUL and NR FDD co-deployment.

Parameter Name	Parameter ID	Setting Notes
NUL gNodeB ID	**NRDUCellCollabCoop**.***NulGnodebId***	This parameter indicates the ID of the base station serving the NUL cell bound to the FDD cell in UL and DL Decoupling with SUL and NR FDD co-deployment.
NUL Cell ID	**NRDUCellCollabCoop**.***NulCellID***	This parameter indicates the ID of the NUL cell bound to the FDD cell in UL and DL Decoupling with SUL and NR FDD co-deployment.

Inter-gNodeB Deployment Requirements

If the NR TDD and NR FDD cells are served by different gNodeBs, the delay level must be configured for these cells.

The delay levels of NR TDD and NR FDD cells must be the same. Otherwise, UL and DL Decoupling cannot be enabled. The maximum allowed delay for an NR TDD cell is specified by the **NRDUCellCollabServ**.***NulSulMaxInterGnbDelay*** parameter, and that for an NR FDD cell is specified by the **NRDUCellCollabCoop**.***NulSulMaxInterGnbDelay*** parameter.

The delay level needs to be determined based on the inter-gNodeB one-way delay. The inter-gNodeB one-way delay (unit: 0.1 microsecond) equals max(**gNodeB DU eXn User Plane One-Way Delay**, **gNodeB DU eXn User Plane Two-Way Delay – gNodeB DU eXn User Plane One-Way Delay**).

The procedure for querying the values of **gNodeB DU eXn User Plane One-Way Delay** and **gNodeB DU eXn User Plane Two-Way Delay** is as follows:

- Before inter-gNodeB UL and DL Decoupling is enabled, you are advised to manually configure an IP performance monitoring (IP PM) session with the activation direction being uplink to measure the eXn-U transmission link delay. Then, you can run the **DSP IPPMSESSION** command to query the delay.
- After inter-gNodeB UL and DL Decoupling is enabled, the gNodeB automatically creates an IP PM session to measure the eXn-U transmission link delay. In this case, you can run the **DSP GNBDUEXNUPINFO** command to query the delay.

After the inter-gNodeB one-way delay is obtained, the delay level needs to be configured based on the following principles:

- If the inter-gNodeB one-way delay is less than or equal to 0.5 ms, **NRDUCellCollabServ**.***NulSulMaxInterGnbDelay*** and **NRDUCellCollabCoop**.***NulSulMaxInterGnbDelay*** need to be set to **DELAY_0DOT5MS**.

NOTE

Flow control against multi-carrier operations at the gNodeB now applies to the procedure for enabling UL and DL Decoupling. The flow control level is specified by the **gNodeBAlgo**.***MultiCarrierFlowCtrlLevel*** parameter. Multiple flow control levels correspond to one CPU load threshold. When the CPU usage reaches the relevant threshold, UL and DL Decoupling will no longer take effect for newly admitted UEs.

In terms of intra-gNodeB UL and DL Decoupling, the **NRDUCellCollabServ**.***NulSulMaxInterGnbDelay*** and **NRDUCellCollabCoop**.***NulSulMaxInterGnbDelay*** parameters specifying the delay levels must be set to **DELAY_0DOT5MS** for the NR TDD and NR FDD cells, respectively.

5.1.1.2 SUL Carrier Information

SUL carrier information for UL and DL Decoupling with SUL and NR FDD co-deployment is the same as that for UL and DL Decoupling with independent SUL deployment. For details, see **4.1.1.2 SUL Carrier Information**.

5.1.1.3 SUL Random Access

SUL carrier information for UL and DL Decoupling with SUL and NR FDD co-deployment is the same as that for UL and DL Decoupling with independent SUL deployment. For details, see **4.1.1.3 SUL Random Access**.

SUL UEs support the "UE random access control based on cell radius" function. When the distance between an SUL UE and an NR FDD cell is greater than the NR FDD cell radius, UEs enabled with UL and DL Decoupling are not allowed to access the cell. For details about the "UE random access control based on cell radius" function, see *Random Access Control*.

5.1.1.4 Carrier Management

When UL and DL Decoupling is enabled, NR TDD and SUL cells are bound through configuration. Instead, UEs are paired freely based on the NR TDD cell configuration information and measurement events.

SUL Carrier Configuration

The SUL carrier configuration procedure is triggered for one of the following UE types in an NR TDD cell:

- UEs performing a random access procedure
- UEs performing an incoming handover
- UEs performing an incoming RRC connection reestablishment
- UEs performing RRC connection resumption
- Activated UEs that support super uplink but are not configured with UL and DL Decoupling and whose uplink traffic volume meets the triggering condition

The procedure is as follows:

1. If the SRS RSRP in the NR TDD cell is less than the sum of the **NRDUCellCollabServ**.***SrsRsrpThld*** parameter value and hysteresis (2 dB), the gNodeB delivers measurement configurations related to event A5 for SUL

configuration. Event A5 indicates that the signal quality of the serving cell is lower than threshold 1 and the signal quality of a neighboring cell is higher than threshold 2. Table 5-5 describes related parameters.

2. The UE reports the NR FDD cells that meet the threshold for event A5.
3. The **gNodeBAlgo**.***MultiCarrierMeasWaitTimer*** parameter specifies the length of the A5 measurement waiting timer. Assume that the gNodeB has delivered the A5 measurement configurations. If the measurement reports of the FDD frequencies corresponding to the SUL frequency band bound to TDD frequencies are reported before the timer expires, the gNodeB stops the timer immediately and adds SCells and SUL carriers based on these measurement reports. Otherwise, the gNodeB will not add SCells or SUL carriers until the timer expires, and such addition will be performed only based on the received measurement reports.
4. The gNodeB selects the NR FDD cell bound in 1:1 mode (if any) as the SUL cell based on the event A5 measurement result.

   For the LampSite base station, the SSB SINR of the NR FDD cell carried in the event A5 measurement result also needs to be considered. The NR FDD cell can serve as an SUL cell only when its SSB SINR is greater than the sum of the **NRCellSulFrequency**.***SulSsbSinrThld*** and **NRCellSulFrequency**.***SulSsbSinrOffset*** parameter values.

Inter-gNodeB UL and DL Decoupling and intra-gNodeB UL and DL Decoupling cannot both take effect.

Table 5-5 Parameters related to event A5 for SUL configuration

Variable	Meaning	Value
Hys	Hysteresis for event A5	1 dB
TimeToTrigger	Time-to-trigger for event A5	320 ms
A5 Thresh1	Threshold 1 for event A5	–30 dBm
A5 Thresh2	Threshold 2 for event A5	Specified by the **NRCellSulFrequency**.***SulInterFreqA5RsrpThld2*** parameter
Ofn	Frequency offset for connected mode	0 dB
Ocn	Cell individual offset	Specified by the **NRCellRelation**.***CellIndividualOffset*** parameter

SUL Carrier Removal

The gNodeB removes the SUL carrier of a UE when any of the following conditions is met:

- There are consecutive DTXs for the UE on the SUL carrier.

  If the gNodeB detects that the SRS SINR of the UE on the SUL carrier is greater than or equal to the value of the **NRDUCellCollabServ**.***SulDtxSinrLowerLimit*** parameter, it determines whether the SUL carrier is abnormal based on the DTX result. If abnormal, the gNodeB removes the SUL carrier. If not, the DTX result is still not conclusive enough to rule out the SUL carrier as abnormal.

- The SRS SINR meets the SUL carrier removal threshold.

  If the gNodeB detects that the SRS SINR of the UE on the SUL carrier is less than the value of the **NRDUCellCollabServ**.***SulSinrLowerLimit*** parameter, it determines that the SUL carrier provides limited coverage or is abnormal and triggers SUL carrier removal.

- The TA difference does not meet the restriction condition.

  If the gNodeB finds that the TA difference between SUL and TDD carriers is too large through calculation, it triggers SUL carrier removal.

- (Only for the LampSite base station) The SSB SINR of the NR FDD cell where the SUL carrier is located does not meet the restriction condition.

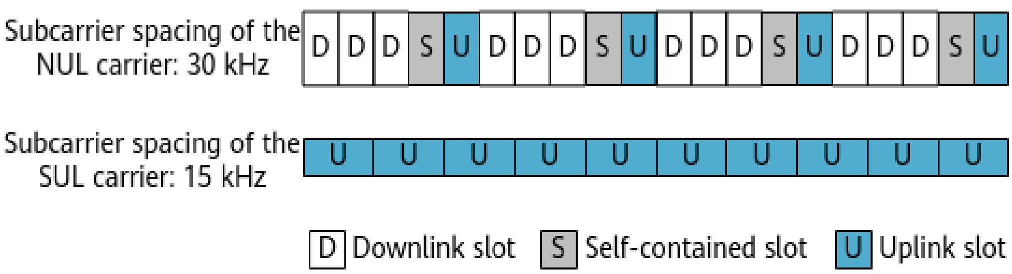
  If the gNodeB detects that the SSB SINR of the NR FDD cell corresponding to the SUL carrier serving the UE is less than the result of the value of the **NRCellSulFrequency**.***SulSsbSinrThld*** parameter minus that of the **NRCellSulFrequency**.***SulSsbSinrOffset*** parameter, it determines that the UE is in the overlapping area of the NR FDD cells. In this case, strong neighboring cell interference will exist, triggering SUL carrier removal.

5.1.1.5 Uplink Scheduling

When UL and DL Decoupling is enabled, the downlink data transmission is carried on the NUL carrier, and the uplink data transmission can be carried on the SUL carrier. The subcarrier spacing of the NUL carrier is 30 kHz, and that of the SUL carrier is 15 kHz. The ratio of slots of the NUL carrier to slots of the SUL carrier is 2:1. Therefore, the scheduling time sequence must be considered. For the NUL carrier, the timeslot allocation (configured by the **NRDUCell**.***SlotAssignment*** parameter) can be 4:1 or 8:2. For the SUL carrier, all slots are available for all UEs in SA networking. **Figure 5-1** uses the slot configuration of 4:1 as an example.

As uplink TDM scheduling is performed for a UE on NR TDD and SUL carriers, the slots on the SUL carrier corresponding to the symbols with SRS on the NR TDD carrier cannot be scheduled in specific scenarios. For details, see **5.1.2.2 Symbol-Level Scheduling on the SUL Carrier**.

Figure 5-1 Scheduling time sequence when the SUL subcarrier spacing is 15 kHz after UL and DL Decoupling is enabled



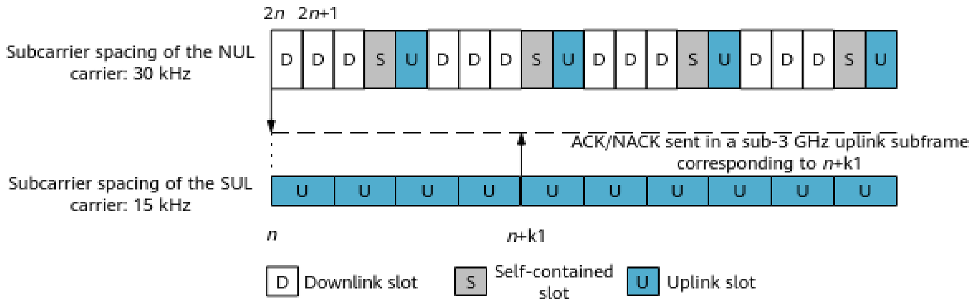
NR introduces a flexible scheduling mechanism, where k1 and k2 are used to ensure a correct scheduling time sequence between gNodeBs and UEs. For details on k1 and k2, see sections 5.2 "UE procedure for reporting channel state information (CSI)" and 5.3 "UE PDSCH processing procedure time" in 3GPP TS 38.214 V15.0.0. k1 determines the HARQ time sequence in downlink data transmission, and k2 determines the uplink scheduling time sequence. k1 and k2 are automatically calculated using algorithms. gNodeBs send k1 and k2 to UEs through downlink control information (DCI). Other scheduling algorithms are the same as those used before UL and DL Decoupling is enabled.

HARQ Time Sequence

If UL and DL Decoupling is enabled, a UE transmits the ACK/NACK feedback for downlink data in slot *n*+k1.

When the SUL subcarrier spacing is 15 kHz and the UE receives downlink data in slot 2*n* or 2*n*+1 over the NR TDD band, it sends the ACK/NACK through the uplink subframe over a sub-3 GHz band that corresponds to slot *n*+k1 over SUL, as shown in **Figure 5-2**.

Figure 5-2 HARQ time sequence when the SUL subcarrier spacing is 15 kHz

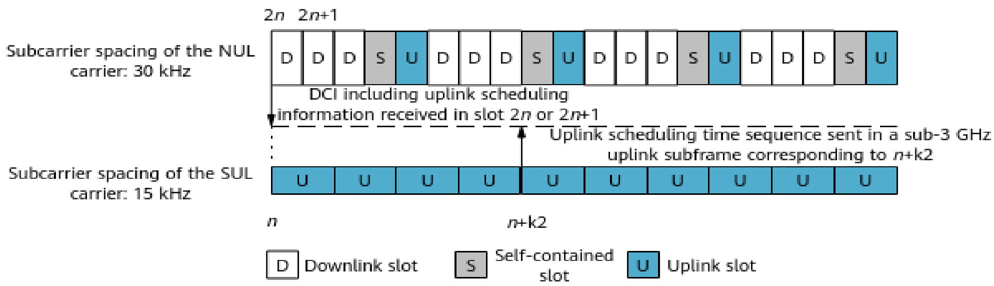


Uplink Scheduling Time Sequence

When UL and DL Decoupling is enabled, the network side sends a scheduling indication to the UE over the NR TDD band, indicating the SUL resources that can be scheduled by the UE.

When the SUL subcarrier spacing is 15 kHz and the UE receives DCI including uplink scheduling information in slot 2*n* or 2*n*+1 over the NR TDD band, it sends uplink data in the slot over the sub-3 GHz band that corresponds to slot *n*+k2 over SUL, as shown in **Figure 5-3**.

Figure 5-3 Uplink scheduling time sequence when the SUL subcarrier spacing is 15 kHz



UL and DL Decoupling allows for flexible scheduling. The resources of the SUL carrier working in a sub-3 GHz band can be scheduled in each subframe over the NR TDD band. The flexible scheduling balances the PDCCH load in each subframe over the NR TDD band. As the ratio of the slots over the NR TDD band to the slots in a sub-3 GHz band is 2:1, a subframe in the sub-3 GHz band can be scheduled in two subframes over the NR TDD band, and the PDCCHs in each of the two subframes over the NR TDD band carry only 50% of the load.

5.1.1.6 SUL Power Control

When UL and DL Decoupling is enabled, power control is separately performed on the NR TDD carrier and SUL carrier. The NR FDD cell sends the TPC commands of the SUL carrier to the NR TDD cell and the gNodeB serving the NR TDD cell delivers these power control commands to the UEs.

5.1.1.7 SUL Carrier Management

5.1.1.7.1 Uplink Carrier Selection

UEs in Idle Mode Accessing the SUL Cell in SA Networking

In SA networking, when an RRC connection is set up for a UE, the UE receives system information from the gNodeB, measures the downlink RSRP over the NR TDD band, and selects an uplink carrier based on the following rules:

- If the RSRP of an NR cell is greater than or equal to the **NRDUCellCollabServ**.***RsrpThld*** parameter value, the UE is in an area with good NR uplink coverage. The UE initiates a random access procedure on the NUL carrier.
- If the RSRP of the NR cell is smaller than the **NRDUCellCollabServ**.***RsrpThld*** parameter value, the UE is in an area with weak or without NR uplink coverage. The UE initiates a random access procedure on the SUL carrier.

UEs in RRC_CONNECTED Mode Accessing the SUL Cell in SA Networking

When a UE supporting UL and DL Decoupling in SA networking is handed over to the SUL cell, the network side selects the NUL carrier for the UE and notifies the UE of the selected uplink carrier by an RRCReconfiguration message.

5.1.1.7.2 SUL Carrier Detection and Management

In UL and DL Decoupling scenarios, when the SUL carrier of a UE cannot be used due to an exception and an NUL-to-SUL carrier change is triggered, the UE access is abnormal and the RRC connection is repeatedly reestablished.

With the SUL carrier detection and management function, the gNodeB detects an SUL carrier when a UE's uplink data transmission needs to be switched to the SUL carrier based on the SRS measurement of the NUL carrier. If the SUL carrier is normal, the UE's uplink data transmission is switched to the SUL carrier. Otherwise, the UE's uplink data transmission remains on the NUL carrier. This function is controlled by the **SUL_FAULT_TOLERANCE_SW** option of the **NRDUCellCollabServ**.***SulAlgoSwitch*** parameter.

If the gNodeB detects that the UE's SRS SINR is greater than the sum of the **NRDUCellCollabServ**.***SulSinrLowerLimit*** and **NRDUCellCollabServ**.***SulSinrLowerLimitOffset*** parameter values, and discontinuous transmission (DTX) is not used on the SUL carrier, then the SUL carrier is considered normal. Otherwise, the SUL carrier is considered abnormal.

5.1.1.8 Mobility Management

When UEs supporting UL and DL Decoupling move between NR cells, they need to reselect an SUL link during a handover or redirection. Mobility management is supported only in SA networking.

The mobility management in SA networking is classified into intra-RAT mobility management and inter-RAT mobility management.

- Intra-RAT mobility management includes intra-frequency handovers, inter-frequency handovers, inter-frequency redirections, and inter-frequency blind redirections. NR does not support intra-RAT blind handovers in the current version. This section mainly describes the differences in mobility management after UL and DL Decoupling is enabled. For details about the mobility management procedures, see *SA Mobility Management in Connected Mode* in *5G RAN Feature Documentation*.
- Inter-RAT mobility management includes inter-RAT handovers/redirections and inter-RAT blind redirections. This section mainly describes the differences in mobility management after UL and DL Decoupling is enabled. For details about the mobility management procedures, see *Interoperability Between E-UTRAN and NG-RAN in Connected Mode* in *5G RAN Feature Documentation*.

Intra-Frequency Handovers

After the UE reports an event A3, the source gNodeB performs an intra-gNodeB intra-frequency handover or inter-gNodeB intra-frequency handover. The gNodeB of the target cell selects an uplink carrier based on the downlink RSRP over the NR TDD band of the target cell reported in event A3, and informs the source gNodeB. The source gNodeB instructs the UE to initiate a handover and then to perform random access in the target gNodeB through an RRCReconfiguration message.

Inter-Frequency Handovers and Redirections

After the UE reports an event A2 or A5, the source gNodeB performs an intra-gNodeB inter-frequency handover or inter-gNodeB inter-frequency handover. The gNodeB of the target cell selects an uplink carrier based on the downlink RSRP over the NR TDD band of the target cell reported in event A5, and informs the source gNodeB. The source gNodeB instructs the UE to initiate a handover/redirection and then to perform random access in the target gNodeB through an RRCReconfiguration message.

NOTE

The frequency of the cell with the best signal quality is selected for a redirection based on the measurement results only when the UE does not support handovers. Whether the UE supports inter-frequency handovers is indicated by the handoverInterF field.

Inter-Frequency Blind Redirections

In inter-frequency blind redirections, measurement is performed only in the serving cell, not in neighboring cells.

If the measured RSRP of the serving cell meets the entering condition of event A2 for an inter-frequency blind redirection, the inter-frequency blind redirection is triggered to redirect the UE to the frequency with the highest priority. By default, the UE initiates random access in the NR TDD uplink of a cell working on the indicated frequency.

Inter-RAT Handovers and Redirections

Inter-RAT handovers include NR-to-LTE handovers and LTE-to-NR handovers.

- NR-to-LTE handovers/redirections: The NR-to-LTE handover/redirection procedure is always the same, regardless of whether SUL is enabled.
- LTE-to-NR handovers/redirections: When a source eNodeB initiates a handover/redirection request to a target gNodeB, the request message contains the downlink RSRP of the target cell. The gNodeB of the target cell selects an uplink carrier based on the downlink RSRP and informs the source eNodeB. The source eNodeB instructs the UE to initiate a handover/redirection and then to perform random access in the target gNodeB through an RRCReconfiguration message.

Inter-RAT Blind Redirections

Only NR-to-LTE blind redirection is supported, and the procedure is always the same, regardless of whether SUL is enabled.

5.1.2 Enhanced Functions

5.1.2.1 SUL CoMP

When UL and DL Decoupling is enabled, SUL CoMP is used to increase uplink benefits in the overlapping area of SUL cells. This function only applies to intra-gNodeB UL and DL Decoupling.

SUL CoMP is controlled by the **INTRA_GNB_SUL_COMP_SW** option of the **NRDUCellCollabServ**.***SulAlgoSwitch*** parameter configured for the TDD cell. In addition, UL CoMP must be enabled in the cells where NR FDD and SUL are co-deployed. UL CoMP is controlled by the **INTRA_GNB_UL_COMP_SW** option of the **NRDUCellAlgoSwitch**.***CompSwitch*** parameter.

SUL CoMP is triggered when the SUL carrier is configured or changed. The procedure is as follows:

1. The gNodeB delivers measurement configurations related to event A5. For details about parameters related to event A5, see **5.1.1.4 Carrier Management**.
2. UEs perform event A5 measurement, and send A5 measurement reports, which include NR FDD neighboring cells.
3. The gNodeB selects candidate UEs for SUL CoMP as follows:

  - If the **UL_2CELL_COMP_BEAR_POL_SW** option of the **NRDUCellComp**.***CompConfigPol*** parameter is deselected, all UEs in the cells can be selected as candidate UEs for SUL CoMP.
  - If the **UL_2CELL_COMP_BEAR_POL_SW** option of the **NRDUCellComp**.***CompConfigPol*** parameter is selected, the gNodeB selects specific UEs as candidate UEs for SUL CoMP. Specific UEs are those having at least one QCI for which the **COMP_SW** option of the **gNBQciBearer**.***ServiceDiffAlgoSwitch*** parameter is selected.
4. The gNodeB selects a serving SUL cell based on **5.1.1.4 Carrier Management**. At the same time, the gNodeB selects candidate SUL CoMP cooperating cells from NR FDD cells that overlap the serving SUL cell. The RSRP difference between each candidate SUL CoMP cooperating cell and the serving SUL cell must be greater than the value of **NRCellComp**.***UlCompRsrpOffset***.
5. The gNodeB measures the uplink SRSs of a UE in each candidate SUL CoMP cooperating cell. Among these cells, the cell in which the RSRP of uplink SRSs sent by the UE is greater than the sum of the RSRP of uplink SRSs in the UE's serving cell and the value of **NRCellComp**.***UlCompRsrpOffset*** is selected as the SUL CoMP cooperating cell. At the same time, the gNodeB determines that SUL CoMP takes effect for the UE.
6. For UEs for which SUL CoMP takes effect, the serving SUL cell and the SUL CoMP cooperating cell simultaneously receive the UEs' data on PUSCHs. The received data is combined and demodulated, improving the PUSCH performance of these UEs.

   The SUL CoMP cooperating cell preferentially processes the data from the UEs it serves. If this cell does not have sufficient capabilities (for example, the number of PRBs used by this cell plus the number of PRBs required for uplink joint reception exceeds the cell's total number of PRBs), it does not process the UL CoMP UE data.

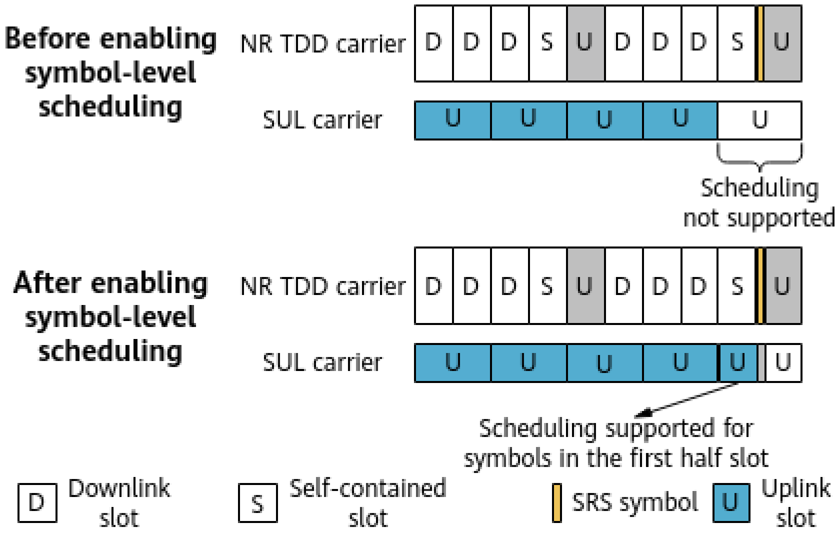
When the SUL CoMP function in UL and DL Decoupling is enabled with an LTE FDD and NR spectrum sharing function (LTE FDD and NR Flash Dynamic Spectrum Sharing/Hybrid DSS Based on Asymmetric Bandwidth), SUL CoMP can take effect only when both the serving SUL cell and the SUL CoMP cooperating cell are in the LTE FDD and NR spectrum sharing state.

For details about CoMP, see *CoMP*.

5.1.2.2 Symbol-Level Scheduling on the SUL Carrier

Symbol-level scheduling on the SUL carrier applies to scenarios where the slot configuration is 4:1 in NR TDD. When this slot configuration is in use and uplink TDM scheduling is performed for a UE on NR TDD and SUL carriers, the slots on the SUL carrier corresponding to the symbols with SRS on the NR TDD carrier cannot be scheduled. With symbol-level scheduling, the gNodeB allows a UE to transmit data over symbols in the first half slot on the SUL carrier, which corresponds to the downlink or self-contained slot on the NR TDD carrier. This improves the uplink throughput of the SUL carrier. For details, see **Figure 5-4**.

Figure 5-4 Symbol-level scheduling principle



Symbol-level scheduling is controlled by the **SCHEDULE_FIRST_HALF_SLOT_SW** option of the **NRDUCellCollabServ**.***SulAlgoSwitch*** parameter.

5.1.2.3 SUL CCE Adjustment Policy

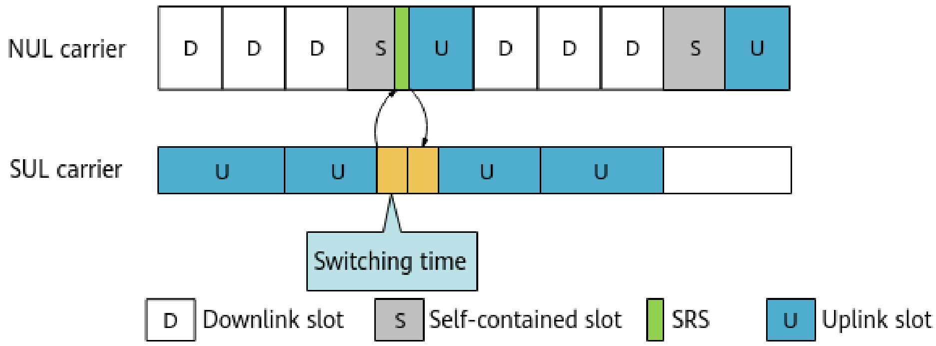
SUL scheduling consumes CCE resources of the NR TDD cell. The **NRDUCellPdcch**.***SulCceAdjPolicy*** parameter is used to specify the SUL CCE adjustment policy, specifically, the percentage of SUL CCEs in uplink CCEs.

- If this parameter is set to a value other than **DEFAULT**, the percentage of SUL CCEs in uplink CCEs is fixed at the configured value.
- If this parameter is set to **DEFAULT**, the SUL CCE adjustment policy does not take effect.

5.1.2.4 UE Compatibility Handling

When the SUL subcarrier spacing is 15 kHz: For some UEs, the time taken to switch between the NUL and SUL carriers cannot be 0 µs. To resolve the compatibility problem of such UEs, the **NRDUCellCollabServ**.***Nul1TxToSul1TxSwitchTime*** parameter has been added to specify the time taken to switch between the NUL and SUL carriers. **Figure 5-5** shows the time for switching between NUL and SUL carriers.

Figure 5-5 Time required for switching between NUL and SUL carriers



The gNodeB reconfigures channel resources for such UEs based on the value of the **NRDUCellCollabServ**.***Nul1TxToSul1TxSwitchTime*** parameter to make UL and DL Decoupling take effect for these UEs.

5.1.3 Simultaneous Use with Super Uplink

When super uplink and UL and DL Decoupling are both enabled, super uplink takes effect in cell center areas, and UL and DL Decoupling takes effect in cell edge areas. In this case, random access and mobility management are different compared to when only super uplink is enabled. For details about super uplink, see *Super Uplink*. Super uplink can be enabled together with UL and DL Decoupling only on macro base stations, not on LampSite base stations.

With UL and DL Decoupling and super uplink both enabled for an NR FDD cell, if two NR TDD cells with different slot configurations need to be paired with an NR FDD cell to implement the functions at the same time, the **SUL_WITH_MULTI_SLOT_CFG_NUL_SW** option of the **NRDUCellCollabCoop**.***SulAlgoSwitch*** parameter must be selected for the NR FDD cell.

Random Access

During initial RRC connection setup, a UE receives system information from the gNodeB, obtains the *rsrp-ThresholdSSB-SUL* IE in SIB1, measures the downlink RSRP of the NR TDD carrier, and determines the uplink carrier for random access. The value of *rsrp-ThresholdSSB-SUL* is specified by the **NRDUCellCollabServ**.***RsrpThld*** parameter.

- If the RSRP of an NR cell is greater than or equal to the value of the **NRDUCellCollabServ**.***RsrpThld*** parameter, the UE is in an area with good NR uplink coverage. The gNodeB instructs the UE to initiate a random access procedure on the NR TDD carrier, and super uplink takes effect.
- If the RSRP of an NR cell is less than the value of the **NRDUCellCollabServ**.***RsrpThld*** parameter, the UE is in an area with weak or no NR uplink coverage. The gNodeB instructs the UE to initiate a random access procedure on the SUL carrier, and UL and DL Decoupling takes effect.

Intra-Cell Mobility Management

When a UE moves in an NR cell, it uses super uplink in cell center areas, and UL and DL Decoupling in cell edge areas. Switching between these functions is performed based on uplink quality.

- If the SRS RSRP of a UE on the NR TDD carrier is greater than the sum of the **NRDUCellCollabServ**.***SrsRsrpThld*** parameter value and hysteresis (2 dB), super uplink takes effect for the UE. In this scenario, the UE transmits uplink data on NR TDD and SUL carriers in TDM mode.
- If the SRS RSRP of a UE on the NR TDD carrier is less than or equal to the result of the **NRDUCellCollabServ**.***SrsRsrpThld*** parameter value minus hysteresis (2 dB), UL and DL Decoupling takes effect for the UE. In this scenario, the UE transmits uplink data on the SUL carrier only.

Inter-Cell Mobility Management

When a UE that supports super uplink moves between NR cells and a handover to the target cell for which super uplink and UL and DL Decoupling are enabled is triggered, the gNodeB instructs the UE to initiate a random access procedure on a TDD carrier through an RRCReconfiguration message. The procedure is as follows:

1. The target gNodeB notifies the source gNodeB of the selected uplink carrier. The source gNodeB sends a UE an RRCReconfiguration message, indicating that the uplink carrier on which random access to the target cell is to be performed is a TDD carrier.
2. The UE initiates a random access procedure on the uplink carrier indicated in the RRCReconfiguration message.

5.2 Network Analysis

5.2.1 Benefits

The basic functions of UL and DL Decoupling improve the average uplink UE throughput, and decrease the proportion of UEs whose rates fall within a low-throughput range in a cell.

The benefits provided by the basic functions can be observed by using the methods described in **5.4.3 Network Monitoring**.

A lower network interference level (indicated by **N.UL.NI.Avg**), a smaller distance to the main lobe direction of the antenna, and a larger SUL bandwidth result in larger benefits. A higher network interference level, a larger distance to the main lobe direction of the antenna, and a smaller SUL bandwidth result in small benefits. UL and DL Decoupling will not provide negative gains.

SUL CoMP improves the uplink reception performance and throughput for UEs in SUL overlapping areas by performing joint reception in intra-frequency continuous networking. For details, see **5.4.3 Network Monitoring**.

UE compatibility handling allows UL and DL Decoupling to take effect for UEs for which the time taken to switch between carriers is larger than 0 µs. As a result, such UEs can obtain gains provided by UL and DL Decoupling.

After cell radius-based UE random access control is enabled, contention-based random access from UEs whose distance to the base station exceeds the cell radius is forbidden.

5.2.2 Impacts

The impacts between this function and other functions are described in **5.3.2.3 Function Impacts**.

Basic functions:

- When UL and DL Decoupling is enabled, only one RRC connection is established between a UE and the network. The "RRC Connected User License" is consumed only in the NR TDD cell. However, each UE occupies a set of hardware resources in both the NR TDD cell and NR FDD cell.

Therefore, the actual number of RRC_CONNECTED UEs supported on the network is halved in extreme scenarios (where all UEs on the network support SUL).

- When UL and DL Decoupling is enabled, uplink resources of the NR FDD cell can be fully utilized. Therefore, the uplink PRB usage of the NR FDD cell increases. When the NR FDD cell load is high, the uplink throughput of NR FDD UEs in the cell decreases.
- When UL and DL Decoupling is enabled, event A5 measurement configurations are delivered in the NR TDD cell. Since event A5 measurement configurations require measurement gaps, uplink and downlink throughput of the UE in the NR TDD cell decreases.
- The number of the RBs occupied by SIB1 may slightly increase and the downlink peak throughput slightly decreases because the gNodeB broadcasts the SUL information through SIB1.
- Uplink scheduling consumes CCE resources of the NR TDD cell after UL and DL Decoupling is enabled. When CCE resources are insufficient in the NR TDD cell, uplink or downlink CCE resource application is more likely to fail. As a result, the uplink or downlink throughput of the NR TDD cell may decrease, and the downlink user-plane delay may increase.
- It is recommended that the **UL_DL_CCE_SHARING_SW** or **UL_DL_CCE_RATIO_ADAPT_SW** option of the **NRDUCellPdcch**.***PdcchAlgoSwitch*** parameter be selected to reduce the probability of CCE resource application failures.
- In an NR TDD cell enabled with uplink MU-MIMO, if scheduling time alignment for SUL is enabled (that is, the **UL_SCH_ALIGN_SW** option of the **NRDUCellPusch**.***PuschPerformanceSwitch*** parameter is selected), the average uplink cell throughput and the **N.CCE.UL.Avail.Equivalent** and **N.CCE.ULDCI.Avail.Equivalent** counter values increase, as does the average delay of non-SUL UEs by up to 1%.
- The uplink throughput of UEs that switch between carriers in 0 µs decreases when UE compatibility handling is enabled with the **NRDUCellCollabServ**.***Nul1TxToSul1TxSwitchTime*** parameter set to a value other than 0 µs.
- When flow control against multi-carrier operations at the gNodeB is triggered and the CPU usage reaches the load threshold corresponding to the flow control level, SUL will no longer take effect for newly admitted UEs. As a result, the average uplink UE throughput (**User Uplink Average Throughput (DU)**) and average downlink UE throughput (**User Downlink Average Throughput (DU)**) decrease. When the **gNodeBAlgo**.***MultiCarrierFlowCtrlLevel*** parameter is set to **HIGH**, the cell service drop rate (**Service Call Drop Rate (CU, Inactive)**) increases as well.
- When inter-gNodeB UL and DL Decoupling is enabled and the NR FDD cell load is high, SUL UEs may not benefit from SUL.
- When inter-gNodeB UL and DL Decoupling is enabled, more cell configuration information is exchanged over the eXn interface between gNodeBs. Therefore, the values of counters related to the received and transmitted traffic volumes in the **SCTP Link Measurement** function subset increase.
- When UL and DL Decoupling is in effect, UEs randomly access an SUL cell. Random access–related counters of the NR TDD cell are measured, which may cause the values to fluctuate. The values of random access–related counters

of the SUL cell co-deployed with the NR FDD cell are not measured (except for **N.RA.Contention.Att.Max**, **N.RA.Dedicated.PreambleAssign.Num**, and **N.RA.Dedicated.PreambleAssign.Num**).

- The UE random access control based on cell radius function enables the gNodeB to detect UE access beyond the cell radius, which increases the consumption of physical resources. Since contention-based random access is prohibited for UEs whose distance to the gNodeB exceeds the cell radius, the random access success rate and RRC connection setup success rate may decrease, and the cell traffic, cell rate, and PRB usage may decrease. In addition, enabling "optimized measurement of RRC connection setup success rate based on cell radius" can prevent the RRC connection setup success rate from decreasing.

Enhanced functions:

- Enabling the symbol-level scheduling function affects the uplink throughput of the NR FDD cell.
- When the SUL CCE adjustment policy function is enabled and uplink-to-downlink CCE ratio adaptation (controlled by the **UL_DL_CCE_RATIO_ADAPT_SW** option of the **NRDUCellPdcch**.***PdcchAlgoSwitch*** parameter) is enabled:
  - If the proportion of SUL UEs is low, the uplink user-perceived rates of these UEs increase while those of other UEs in the NR TDD cell are not affected.
  - If the proportion of SUL UEs is high, the probability of uplink CCE resource application failures increases and scheduling opportunities for other UEs in the NR TDD cell decrease.
- When the SUL CCE adjustment policy function is enabled and uplink-to-downlink CCE ratio adaptation is disabled, there is no network impact.

The simultaneous use with super uplink has the following impacts:

If either of the following conditions is met, UL and DL Decoupling is disabled for UEs in a cell, and super uplink is disabled for UEs in the bound neighboring cell.

- UL and DL Decoupling is disabled in the NR TDD cell, or any of the **NRDUCellCollabServ**.***SulMcc***, **NRDUCellCollabServ**.***SulMnc***, **NRDUCellCollabServ**.***SulGnodebId***, and **NRDUCellCollabServ**.***SulCellID*** parameters is modified for the NR FDD cell bound to the NR TDD cell.
- UL and DL Decoupling is disabled in the NR FDD cell, or any of the **NRDUCellCollabCoop**.***NulMcc***, **NRDUCellCollabCoop**.***NulMnc***, **NRDUCellCollabCoop**.***NulGnodebId***, and **NRDUCellCollabCoop**.***NulCellID*** parameters is modified for the NR TDD cell bound to the NR FDD cell.

5.3 Requirements

5.3.1 Licenses

The following feature license is required by UL and DL Decoupling with SUL and NR FDD co-deployment. The number of required license units is calculated based on the NR TDD cells but not NR FDD cells.

Feature ID	Feature Name	Model	License Control Item Name	Sales Unit
FOFD-010205	UL and DL Decoupling	NR0SULADLD00	UL and DL Decoupling (NR)	Per Cell
For the license control item NR0SULADLD00: ● If the feature is enabled for an activated cell, the cell will not be automatically deactivated when the number of units of the license for the feature is insufficient due to license expiration or forcible license file installation. However, after the cell is reestablished, the cell cannot be activated again due to license insufficiency. ● If the feature is not enabled for an activated cell and the number of units of the license for the feature is insufficient due to license expiration or forcible license file installation: – When the license sales number pre-check function is enabled, the **NRDUCellAlgoSwitch**.***UlDlDecouplingSwitch*** parameter cannot be set to **ON** due to license insufficiency, preventing cell activation failures. – When the license sales number pre-check function is disabled, the cell cannot be activated again due to license insufficiency after the feature is enabled and the parameters that cause automatic cell reset are modified. The license sales number pre-check function is controlled by the **LIC_SALE_NUM_PRE_CHECK_SW** option of the **gNBLicenseAlmThld**.***LicenseChkSw*** parameter.				

NOTE

The insufficiency of certain feature licenses affects cell activation. For details, see the **Is Cell Activation Affected by License Insufficiency** column in *License Control Item Lists* in 5G RAN feature documentation.

5.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

5.3.2.1 Prerequisite Functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency NR TDD	Intra-base-station UL CoMP	**INTRA_GNB_UL_COMP_SW** option of the **NRDUCellAlgoSwitch**.***CompSwitch*** parameter	*CoMP*	SUL CoMP requires that intra-base-station UL CoMP be enabled in NR FDD cells when UL and DL Decoupling with SUL and NR FDD co-deployment is enabled.
Low-frequency NR TDD	PDCCH uplink and downlink CCE sharing	**UL_DL_CCE_SHARING_SW** option of the **NRDUCellPdcch**.***PdcchAlgoSwitch*** parameter	*Channel Management*	Only the SUL CCE adjustment policy function requires that the switch for uplink and downlink CCE sharing be turned on.

5.3.2.2 Mutually Exclusive Functions

RAT	Function Name	Function Switch	Reference (5G RAN Feature Documentation)	Description
Low-frequency NR TDD	High-speed Railway Superior Experience	**NRDUCell**.***HighSpeedFlag*** set to **HIGH_SPEED**	*High Speed Mobility*	UL and DL Decoupling cannot be enabled in high-speed cells.
Low-frequency NR TDD	Hyper Cell	**NRDUCell**.***NrDuCellNetworkingMode*** set to **HYPER_CELL**	*Hyper Cell*	UL and DL Decoupling cannot be enabled in hyper cells.
Low-frequency NR TDD	Cell Combination	**NRDUCell**.***NrDuCellNetworkingMode*** set to **HYPER_CELL_COMBINE_MODE**	*Cell Combination*	UL and DL Decoupling cannot be enabled in combined cells.

RAT	Function Name	Function Switch	Reference (5G RAN Feature Documentation)	Description
Low-frequency NR TDD	Channel decoupling	**SUL_CHANNEL_DECOUPLE_SW** option of the **NRDUCellCollabServ**.***SulAlgoSwitch*** parameter	*Super Uplink*	Channel decoupling cannot be enabled when UL and DL Decoupling is enabled.
Low-frequency NR TDD	SRS interference coordination based on self-contained and uplink slots	**SRS_INTRF_COORD_S_U_SLOT_SW** option of the **NRDUCellSrs**.***SrsAlgoExtSwitch*** parameter	*Channel Management*	When SRS interference coordination based on self-contained and uplink slots is enabled, the **NRDUCellCollabServ**.***Nul1TxToSul1TxSwitchTime*** parameter cannot be set to **MICROSECOND140**.
Low-frequency NR TDD	Fusion Cell	**NRDUCell**.***NrDuCellNetworkingMode*** set to **FUSION_CELL**	*Fusion Cell (Low-Frequency TDD)*	UL and DL Decoupling cannot be enabled in Fusion Cell networking.
Low-frequency NR TDD	Pattern-free PDCCH rate matching activation rate increase	**PDCCH_NO_PAT_RM_RATIO_INC_SW** option of the **NRDUCellServExp**.***SpdTurboAlgoSwitch*** parameter	*Downlink Scheduling*	Pattern-free PDCCH rate matching activation rate increase cannot be enabled when UL and DL Decoupling is enabled.

RAT	Function Name	Function Switch	Reference (5G RAN Feature Documentation)	Description
Low-frequency NR TDD	5QI-specific UE-number-based connected mode MLB	**INTER_FREQ_CONNECTED_MLB** option of the **NRCellAlgoSwitch**.***MlbAlgoSwitch*** parameter selected, and **NRCellMlb**.***MlbTriggerMode*** set to **SPECIFIED_5QI_USER_NUM**	*Mobility Load Balancing in Connected Mode*	5QI-specific UE-number-based connected mode MLB cannot be enabled when UL and DL Decoupling is enabled.
Low-frequency NR TDD	PUSCH scheduling based on SRS interference avoidance	**SRS_INTRF_AVOID_PUSCH_SCH_SW** option of the **NRDUCellSrs**.***SrsAlgoExtSwitch*** parameter	None	PUSCH scheduling based on SRS interference avoidance cannot be enabled when UL and DL Decoupling is enabled.
Low-frequency NR TDD	Cell deployment in IAB mode	**NRDUCell**.***DeployPosition*** set to **DEPLOY_IAB**	None	UL and DL Decoupling cannot be enabled in cells deployed in IAB mode.
Low-frequency NR TDD	Cooperation between joint transmission and pattern-free PDCCH rate matching	**JT_PDCCH_NO_PATT_RM_COORD_SW** option of the **NRDUCellPdcch**.***PdcchAlgoSwitch*** parameter	None	"UL and DL Decoupling" and "cooperation between joint transmission and pattern-free PDCCH rate matching" cannot be both enabled.

RAT	Function Name	Function Switch	Reference (5G RAN Feature Documentation)	Description
Low-frequency NR TDD	Uplink assistance	**UL_ASSISTANCE_SW** option of the **NRDUCellCollabServ**.***MultiCellMimoSwitch*** parameter	None	UL and DL Decoupling and uplink assistance cannot be both enabled.
NR FDD	Hybrid DSS Based on Asymmetric Bandwidth	**LTE_NR_FDD_SPCT_SHR_SW** option of the **NRDUCellAlgoSwitch**.***SpectrumCloudSwitch*** parameter	*LTE FDD and NR Hybrid Dynamic Spectrum Sharing*	When UL and DL Decoupling with SUL and NR FDD co-deployment is enabled, Hybrid DSS Based on Asymmetric Bandwidth cannot be enabled.
NR FDD	LTE FDD and NR Flash Dynamic Spectrum Sharing	**LTE_NR_FDD_SPCT_SHR_SW** option of the **NRDUCellAlgoSwitch**.***SpectrumCloudSwitch*** parameter	*LTE FDD and NR Dynamic Spectrum Sharing*	When UL and DL Decoupling with SUL and NR FDD co-deployment is enabled, LTE FDD and NR Flash Dynamic Spectrum Sharing cannot be enabled.

5.3.2.3 Function Impacts

RAT	Function Name	Function Switch	Reference	Description
LTE FDD LTE TDD Low-frequency NR TDD	Uplink data transmission path selection	• LTE **NSA_DC_UL_PATH_SELECTION_SW** option of the **NsaDcMgmtConfig**.***NsaDcAlgoSwitch*** parameter • NR **NSA_DC_UL_PATH_SELECTION_SW** option of the **NRCellNsaDcConfig**.***NsaDcAlgoSwitch*** parameter	*EPC-based NSA Performance Enhancement*	Uplink data transmission path selection does not take effect in an NR cell enabled with UL and DL Decoupling.

RAT	Function Name	Function Switch	Reference	Description
LTE FDD LTE TDD Low-frequency NR TDD	Network-coordinated dynamic UE power sharing	• LTE **NSA_DC_COOPERATION_DPS_SW** option of the **NsaDcMgmtConfig**.***NsaDcAlgoSwitch*** parameter • NR **NSA_DC_COOPERATION_DPS_SW** option of the **NRCellNsaDcConfig**.***NsaDcAlgoSwitch*** parameter	*EPC-based NSA Performance Enhancement*	If UL and DL Decoupling has taken effect and the base station performs uplink scheduling in TDM mode, network-coordinated dynamic UE power sharing does not take effect.
Low-frequency NR TDD	Power saving BWP	**BWP2_SWITCH** option of the **NRDUCellUePwrSaving**.***BwpPwrSavingSw*** parameter	*UE Power Saving*	UEs working in BWP2 for power saving do not support SUL CoMP.
Low-frequency NR TDD	Low latency and high reliability	**HIGH_RELIABILITY_BASIC_SW** option of the **NRDUCellAlgoSwitch**.***HighReliabilitySwitch*** parameter	*URLLC*	The low latency and high reliability function does not take effect for UEs using UL and DL Decoupling.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency NR TDD	Spectral-efficiency-based connected mode MLB	**INTER_FREQ_CONNECTED_MLB_SW** option of the **NRCellAlgoSwitch**.***MlbAlgoSwitch*** parameter, and **NRCellMlb**.***MlbTriggerMode*** set to **SPEC_EFF_BASED_USER_NUM**	*Mobility Load Balancing in Connected Mode*	The gNodeB does not select UEs for which UL and DL Decoupling has taken effect as the UEs to be handed over.
Low-frequency NR TDD	Intra-band CA (TDD)	**INTRA_BAND_CA_SW** option of the **NRDUCellAlgoSwitch**.***CaAlgoSwitch*** parameter	*Carrier Aggregation*	For UEs that support both UL and DL Decoupling and intra-band CA, either UL and DL Decoupling or intra-band CA can take effect at a time.
Low-frequency NR TDD	Intra-FR inter-band CA	**INTRA_FR_INTER_BAND_CA_SW** option of the **NRDUCellAlgoSwitch**.***CaAlgoSwitch*** parameter	*Carrier Aggregation*	For UEs that support both UL and DL Decoupling and intra-FR inter-band CA, either UL and DL Decoupling or intra-FR inter-band CA can take effect at a time.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency NR TDD	NR DU cell resource management between network slices	**RB_DYNAMIC_CONTROL_SW** option of the **NRDUCellAlgoSwitch**.***NetworkSliceAlgoSwitch*** parameter	*Network Slicing*	When both the NR DU cell resource management between network slices function and UL and DL Decoupling are enabled, and the **RB_DYNAMIC_CONTROL_SW** option of the **NRDUCellAlgoSwitch**.***NetworkSliceAlgoSwitch*** parameter is selected for the SUL cell, this option must also be selected for the corresponding TDD cell, and the proportion of RB resources configured for the TDD cell takes effect in the SUL cell.
Low-frequency NR TDD	RFSP-based RB resource control	**RB_DYNAMIC_CONTROL_SW** option of the **NRDUCellCommonSw**.***RfspAlgoSwitch*** parameter	*Resource Control in NSA Networking*	Assume that both RFSP-based RB resource control and UL and DL Decoupling are enabled. If RFSP-based RB resource control is enabled in the SUL cell, it must also be enabled in the associated TDD cell, and the proportion of RB resources configured for the TDD cell takes effect in the SUL cell.
Low-frequency NR TDD	Uplink multi-frequency coordination	**MULTI_FREQ_UL_COORD_SW** option of the **NRDUCellFeatureSw**.***MultiCarrierSwitch*** parameter	*Multi-Frequency Smart Selection*	The gNodeB does not select UEs for which UL and DL Decoupling has taken effect.
Low-frequency NR TDD	Multi-frequency beam coordination	**MULTI_FREQ_BEAM_COORD_SW** option of the **NRDUCellFeatureSw**.***MultiCarrierSwitch*** parameter	*Multi-Frequency Smart Selection*	The gNodeB does not select UEs for which UL and DL Decoupling has taken effect.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency NR TDD	PDCCH skipping	**PDCCH_SKIP_SW** option of the **NRDUCellUePwrSaving**.***SchPwrSavingSw*** parameter	*UE Power Saving*	When both SUL CoMP and PDCCH skipping are enabled, PDCCH skipping does not take effect for SUL UEs.
Low-frequency NR TDD	Downlink massive CA	**NRDUCellCarrMgmt**.***CaDlMaxCcNum*** set to **DL4CC** or **DL5CC**.	*CA Experience Improvement*	UL and DL Decoupling and downlink massive CA cannot take effect at the same time.
Low-frequency NR TDD	Inter-FR CA	**INTER_FR_FR1TOFR2_CA_SW** option of the **NRDUCellAlgoSwitch**.***CaAlgoSwitch*** parameter	*Carrier Aggregation*	Inter-FR CA and UL and DL Decoupling cannot take effect at the same time.
Low-frequency NR TDD	3GPP R15-compliant uplink slot aggregation	**VONR_UL_SLOT_AGGREGATION_SW** option of the **NRDUCellUlSch**.***VoiceUlSchSwitch*** parameter	*VoNR*	3GPP R15-compliant uplink slot aggregation does not take effect on UEs with UL and DL Decoupling in effect.
Low-frequency NR TDD	3GPP R16-compliant uplink slot aggregation	**VOICE_R16_UL_SLOT_AGG_SW** option of the **NRDUCellUlSch**.***VoiceUlSchSwitch*** parameter	*VoNR*	3GPP R16-compliant uplink slot aggregation does not take effect on UEs with UL and DL Decoupling in effect.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency NR TDD	QCI-based resource control	**RB_DYNAMIC_CONTROL_SW** option of the **NRDUCellFeatureSw**.***QciBasedDiffServExpSw*** parameter	*QCI-based Resource Control*	When QCI-based resource control (controlled by the **RB_DYNAMIC_CONTROL_SW** option of the **NRDUCellFeatureSw**.***QciBasedDiffServExpSw*** parameter) is enabled together with UL and DL Decoupling and is enabled in the SUL cell, the proportion of RB resources configured for the TDD cell takes effect in the SUL cell.
Low-frequency NR TDD	Timing carrier shutdown	**TIMING_CARRIER_SHUTDOWN_SW** option of the **NRDUCellAlgoSwitch**.***PowerSavingSwitch*** parameter	*Energy Conservation and Emission Reduction*	After the NR FDD carrier is shut down, SUL scheduling cannot be performed for SUL UEs.
Low-frequency NR TDD	Intelligent carrier shutdown	**INTRA_GNB_MULTI_CARR_SD_SW** or **INTER_GNB_MULTI_CARR_SD_SW** option of the **NRDUCellAlgoSwitch**.***PowerSavingSwitch*** parameter	*Energy Conservation and Emission Reduction*	After the NR FDD carrier is shut down, SUL scheduling cannot be performed for SUL UEs.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency NR TDD	Fast carrier shutdown	For capacity-layer cells: • **INTRA_GNB_MULTI_CARR_SD_SW** option of the **NRDUCellAlgoSwitch**.***PowerSavingSwitch*** parameter selected (intra-base-station multi-carrier scenarios) or **INTER_GNB_MULTI_CARR_SD_SW** option of the **NRDUCellAlgoSwitch**.***PowerSavingSwitch*** parameter selected (inter-base-station multi-carrier scenarios) • **FAST_CARRIER_SHUTDOWN_SW** option of the **NRDUCel**	*Energy Conservation and Emission Reduction*	After the NR FDD carrier is shut down, SUL scheduling cannot be performed for SUL UEs.

RAT	Function Name	Function Switch	Reference	Description
		lAlgoSwitch.***PowerSavingExtSwitch*** parameter For basic-layer cells: **FAST_CARR_SHUT_SERV_ASSUR_SW** option of the **NRDUCellAlgoSwitch**.***PowerSavingExtSwitch*** parameter selected		
Low-frequency NR TDD	Rate-matching-pattern-configuration-based PDCCH rate matching	**PDCCH_RATEMATCH_SW** option of the **NRDUCellPdsch**.***RateMatchSwitch*** parameter	*Downlink Scheduling*	UL and DL Decoupling cannot take effect when rate-matching-pattern-configuration-based PDCCH rate matching is enabled and the **NRDUCellPdcch**.***OccupiedRbNum*** parameter is set to **1** or **2**.
Low-frequency NR TDD	WUS	**DRX_ADAPTATION_SW** option of the **NRDUCellUePwrSaving.** ***NrDuCellDrxAlgoSwitch*** parameter	*UE Power Saving*	WUS does not take effect on a UE with inter-gNodeB SUL in effect. If WUS has taken effect on a UE before inter-gNodeB SUL, the UE exits WUS before inter-gNodeB SUL can take effect.
Low-frequency NR TDD	Time-domain power saving BWP (low-frequency TDD)	**TDM_BWP2_SWITCH** option of the **NRDUCellUePwrSaving**. ***BwpPwrSavingSw*** parameter	*UE Power Saving*	UEs on the NUL carrier can be switched to BWP2 while UEs on the SUL carrier cannot be switched to BWP2. The scheduling delay of BWP2 UEs on the NUL carrier increases when the SUL carrier is activated.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency NR TDD	Secondary harmonic interference avoidance	**SEC_HARMONIC_INTRF_AVOID_SW** option of the **NRDUCellCollabServ**.***SulAlgoSwitch*** parameter	None	Secondary harmonic interference avoidance does not take effect for UEs enabled with UL and DL Decoupling with SUL and NR FDD co-deployment.
Low-frequency NR TDD	RedCap Introduction	**REDCAP_SW** option of the **NRDUCellFeatureSw**.***UeCapbBasedFunSw*** parameter	*RedCap*	UL and DL Decoupling does not take effect on RedCap UEs when proactive avoidance of remote interference is enabled.
Low-frequency NR TDD	Proactive avoidance of remote interference	**SRS_RIM_INTRF_AVOID_SW** and **SRS_INTRF_COORD_S_SLOT_SW** options of the **NRDUCellSrs**.***SrsAlgoExtSwitch*** parameter	*Remote Interference Management (Low-Frequency TDD)*	
Low-frequency NR TDD	User-rate-based dynamic network slice resource management	**DYNAMIC_NETWORKSLICE_SW** option of the **NRDUCellCommonSw**.***ServiceDiffAlgoExtSwitch*** parameter	*Network Slicing*	When user-rate-based dynamic network slice resource management is enabled for an SUL cell with UL and DL Decoupling in effect, you are advised to also enable it for the associated TDD cell. Otherwise, the gNodeB may perform inaccurate dynamic resource adjustments.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency NR TDD	User-rate-based dynamic QCI resource management	**DYNAMIC_QCI_GROUP_SW** option of the **NRDUCellFeatureSw**.***QciBasedDiffServExpSw*** parameter	*QCI-based Resource Control*	When user-rate-based dynamic QCI resource management is enabled for an SUL cell with UL and DL Decoupling in effect, you are advised to also enable it for the associated TDD cell. Otherwise, the gNodeB may perform inaccurate dynamic resource adjustments.

5.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

- 3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910, and 5900 series base stations must be configured with the BBU5900, BBU5910, or BBU5900A.
- DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

Boards

All NR-capable main control boards support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

Requirements on baseband processing units:

- Macro base stations:
  - For NR TDD cells, all NR TDD-capable baseband processing units can be used.
  - For NR FDD cells:
    - Baseband processing units that support 32T32R/64T64R and those that support 2T2R/4T4R/8T8R cannot be both used.
    - NR TDD-capable baseband processing units and NR FDD-capable baseband processing units cannot be both used.
    - Baseband processing units that can serve a maximum of 24 cells can be used.
- LampSite base stations:

- For NR TDD cells, baseband processing units must support distributed massive MIMO. For details, see *Distributed Antenna Solutions (Low-Frequency TDD)*.
- For NR FDD cells, all NR FDD-capable baseband processing units can be used.

RF Modules

Table 5-6 lists the RF modules supported by macro base stations.

Table 5-6 RF modules supported by macro base stations

Frequency Band	Supported Module	Bandwidth Requirement
n78 (3.5/3.7 GHz)	Modules supporting 32T32R and 64T64R	Same as the bandwidths supported by the RF modules. For details, see related technical specifications in *3900 & 5900 Series Base Station Product Documentation*.
n40 (2.3 GHz)	Modules supporting 32T32R and 64T64R	
n41 (2.6 GHz)	Modules supporting 32T32R and 64T64R	
n3/n80 (1.8 GHz)	Modules supporting NR FDD	10 MHz, 15 MHz, and 20 MHz
n20/n82 (800 MHz)	Modules supporting NR FDD	5 MHz, 10 MHz, 15 MHz, and 20 MHz
n28/n83 (700 MHz)	Modules supporting NR FDD	10 MHz, 15 MHz, 20 MHz, and 30 MHz
n1/n84 (2.1 GHz)	Modules supporting NR FDD	10 MHz, 15 MHz, 20 MHz, 30 MHz, 40 MHz, and 50 MHz
n8/n81 (900 MHz)	Modules supporting NR FDD	10 MHz, 15 MHz, and 20 MHz

Table 5-7 lists the RF modules supported by LampSite base stations.

Table 5-7 RF modules supported by LampSite base stations

Frequency Band	Supported Module	TX/RX Mode	Bandwidth Requirement
n78 (3.5/3.7 GHz)	pRRU56*xx* series, pRRU596*x* series[a], pRRU5973, and pRRU5973H	4T4R	20 MHz, 40 MHz, 60 MHz, 80 MHz, and 100 MHz

Frequency Band	Supported Module	TX/RX Mode	Bandwidth Requirement
n41 (2.6 GHz)	pRRU596*x* series	4T4R	60 MHz, 80 MHz, and 100 MHz
n3/n80 (1.8 GHz)	pRRU56*xx* series and pRRU596*x* series	2R	10 MHz, 15 MHz, and 20 MHz
n1/n84 (2.1 GHz)	pRRU56*xx* series and pRRU596*x* series	2R	10 MHz, 15 MHz, 20 MHz, 30 MHz, 40 MHz, and 50 MHz
a: *x* in pRRU56*xx* series and pRRU596*x* series can be a digit or a combination of a digit and a letter (for example, pRRU5961).			

5.3.4 Networking

NR FDD and NR TDD cells can be deployed in the same BBU or different BBUs. In the case that these cells are deployed in the same BBU, they can be configured on the same baseband processing unit or different baseband processing units.

The networking requirements of inter-BBU UL and DL Decoupling are as follows:

- Paired NR TDD and NR FDD sites must be routed to each other over an inter-BBU eXn link. For details about eXn interface configurations, see *eXn Self-Management*.
- The actual maximum one-way transmission delay between sites cannot exceed the transmission delay configured in the gNodeB. For details, see **5.1.1 Basic Functions**. Otherwise, the inter-gNodeB SUL relationship cannot be established.

  When the transmission between sites is interrupted or the transmission delay exceeds the configured delay due to network congestion, the gNodeB imposes a 10-minute penalty on the peer gNodeB. Within 10 minutes, no inter-gNodeB SUL relationship can be established for any cell served by the gNodeB under penalty. For cells for which the inter-gNodeB SUL relationship has been established, the SUL carriers configured for the involved UEs are automatically deleted. After 10 minutes, if the inter-gNodeB transmission recovers and the transmission delay is less than the configured delay, the inter-gNodeB SUL relationship can be reestablished for all cells served by the gNodeB under penalty.
- The distance between NR TDD and NR FDD sites cannot exceed 1000 m. If the distance is beyond this range, performance of UL and DL Decoupling cannot be ensured. The inter-site distance refers to the distance between the antennas of the NR TDD cell where UEs camp and the NR FDD cell corresponding to the SUL carrier.
- The time synchronization precision of NR TDD and NR FDD sites must be within ±1.5 μs. If the time synchronization precision does not meet the requirement, UL and DL Decoupling does not take effect.

- The following base stations are compatible with this function:
  - Both NR FDD and NR TDD cells are served by macro base stations.
  - Both NR FDD and NR TDD cells are served by LampSite base stations.
  - The NR TDD cell is served by a macro base station, and the NR FDD cell is served by a LampSite base station.
  - The NR TDD cell is served by a LampSite base station, and the NR FDD cell is served by a macro base station.

5.3.5 Cells

- UL and DL Decoupling with SUL and NR FDD co-deployment can be enabled only when the **NRDUCellSulRelation**.***SulType*** parameter is set to **DYN_SUL**.
- When UL and DL Decoupling with SUL and NR FDD co-deployment is enabled, the **NrDuCell**.***CellRadius*** parameter must be set to a value less than or equal to **7205** and the **NRDUCellPrach**.***PrachConfigurationIndex*** parameter must be set to a value in the range of 18 to 21, **12**, **14**, **16**, or **65535** for the NR FDD cell.
- If **NRDUCellSulRelation**.***SulType*** is set to **DYN_SUL**, **NRDUCellGigaBand**.***LnrUlRbEstCorrectCoeff*** must be set to **0**.

5.3.6 Others

In SUL and NR FDD co-deployment mode, UEs support NR FDD and A5 inter-frequency measurements.

Requirements on UEs complying with 3GPP R15:

- UEs must support the combinations of NR TDD and SUL bands. If BandCombinationList in the UECapabilityInformation message contains the combinations of NR TDD and SUL bands, UEs support these combinations.

Requirements on UEs complying with 3GPP R16:

- UEs must support the combinations of NR TDD and SUL bands. If the UECapabilityInformation message contains all of the following, UL and DL Decoupling defined in 3GPP R16 is supported.
  - BandCombination-UplinkTxSwitch-r16 containing the combinations of NR TDD and SUL bands
  - BandCombination-UplinkTxSwitch-r16 containing supportedBandPairListNR-r16
- UEs must support the reporting of the corresponding uplink switching time. If the supportedBandPairListNR-r16 message contains uplinkTxSwitchingPeriod-r16, the corresponding uplink switching time is reported.

Requirements on UEs complying with 3GPP R18 in addition to those on UEs complying with 3GPP R16:

- UEs must support the combinations of NR TDD and SUL bands. If the UECapabilityInformation message contains all of the following, UL and DL Decoupling defined in 3GPP R18 is supported.
  - BandCombination-UplinkTxSwitch-v1800 or BandCombination-UplinkTxSwitch-v1830 containing the combinations of NR TDD and SUL bands

  - BandCombination-UplinkTxSwitch-v1800 containing supportedBandPairListNR-r18
  - switchingPeriodFor2T-r18, which indicates that UEs support the combinations of NR 2Tx and SUL 2Tx bands
  - switchingPeriodFor1T-r18, which indicates that UEs support the combinations of NR 2Tx and SUL 1Tx bands, NR 1Tx and SUL 2Tx bands, or NR 1Tx and SUL 1Tx bands
- UEs must support the reporting of the corresponding uplink switching time. If the supportedBandPairListNR-r18 message contains uplinkTxSwitchingPeriodForBandPair-r18, the corresponding uplink switching time is reported.

5.4 Operation and Maintenance

5.4.1 Data Configuration

5.4.1.1 Data Preparation

The NR TDD carrier and SUL carrier use TDM scheduling and share the same TAG. Therefore, it is required that the subframe and uplink time of the NR TDD cell be aligned with those of the NR FDD cell. Before activating the NR TDD and NR FDD cells, ensure that the frame offset and TA offset of the NR TDD cell are the same as those of the NR FDD cell.

- The frame offset is specified by the **gNodeBParam**.***FrameOffset*** or **gNBFreqBandConfig**.***FrameOffset*** parameter. If both parameters are configured, the value of the **gNBFreqBandConfig**.***FrameOffset*** parameter is used.
- The TA offset is specified by the **NRDUCell**.***TaOffset*** parameter.

If the configurations of the NR TDD cell and NR FDD cell need to be modified, you are advised to retain the configurations of the NR TDD cell, modify the TA offset of the NR FDD cell, and then modify the frame offset of the NR FDD cell. If the frame offset and TA offset of the NR TDD cell cannot be the same as those of the NR FDD cell, contact Huawei engineers to confirm the values.

Table 5-8 and **Table 5-9** describe the parameters used for function activation in NR TDD and NR FDD cells, respectively. This section does not describe parameters related to cell establishment. For details about these parameters, see *Cell Management*.

Table 5-8 Parameters used for function activation in the NR TDD cell

Parameter Name	Parameter ID	Setting Notes
UL and DL Decoupling Switch	**NRDUCellAlgoSwitch**.***UlDlDecouplingSwitch***	Set this parameter to **ON** in the NR TDD cell to enable this feature.

Parameter Name	Parameter ID	Setting Notes
NR DU Cell ID	**NRDUCellSulRelation**.*NrDuCellId*	Set this parameter based on the network plan.
SUL Type	**NRDUCellSulRelation**.*SulType*	Set this parameter to **DYN_SUL** for SUL and NR FDD co-deployment.
FDD Frequency Band	**NRDUCellSulRelation**.*FddFrequencyBand*	Set this parameter to the same value as the **NRDUCell**.*FrequencyBand* parameter for the NR FDD cell.
FDD UL NR-ARFCN	**NRDUCellSulRelation**.*FddUlNrarfcn*	Set this parameter to the same value as the **NRDUCell**.*UlNarfcn* parameter for the NR FDD cell.
FDD UL Bandwidth	**NRDUCellSulRelation**.*FddUlBandwidth*	Set this parameter to the same value as the **NRDUCell**.*UlBandwidth* parameter for the NR FDD cell.
FDD UL BWP0 Configuration Mode	**NRDUCellSulRelation**.*FddUlBwp0ConfigMode*	Set this parameter based on the network plan.
FDD UL BWP0 Bandwidth	**NRDUCellSulRelation**.*FddUlBwp0Bandwidth*	This parameter needs to be specified only when the **NRDUCellSulRelation**.*FddUlBwp0ConfigMode* parameter is set to **MANUAL**. Set this parameter based on the BWP0 bandwidth of the NR FDD cell queried by using the **DSP NRDUCELLBWP** command.
FDD UL BWP0 Start PRB	**NRDUCellSulRelation**.*FddUlBwp0StartPrb*	This parameter needs to be specified only when the **NRDUCellSulRelation**.*FddUlBwp0ConfigMode* parameter is set to **MANUAL**. Set this parameter based on the start PRB of the BWP0 bandwidth of the NR FDD cell queried by using the **DSP NRDUCELLBWP** command.

Parameter Name	Parameter ID	Setting Notes
FDD UL Subcarrier Spacing	**NRDUCellSulRelation**.***FddSubcarrierSpacing***	Set this parameter to the same value as the **NRDUCell**.***SubcarrierSpacing*** parameter for the NR FDD cell.
FDD Cyclic Prefix Length	**NRDUCellSulRelation**.***FddCyclicPrefixLength***	Set this parameter to the same value as the **NRDUCell**.***CyclicPrefixLength*** parameter for the NR FDD cell.
SUL Carrier Change SINR Threshold	**NRDUCellSulRelation**.***SulCarrChangeSinrThld***	The default value is recommended.
SUL Inter-freq A5 RSRP Threshold 2	**NRCellSulFrequency**.***SulInterFreqA5RsrpThld2***	The default value is recommended.
NUL To SUL Mapping Type	**NRDUCellCollabServ**.***NulToSulMappingType***	Set this parameter to **1TO1**.
SUL MCC	**NRDUCellCollabServ**.***SulMcc***	Set this parameter based on the network plan.
SUL MNC	**NRDUCellCollabServ**.***SulMnc***	Set this parameter based on the network plan.
SUL gNodeB ID	**NRDUCellCollabServ**.***SulGnodebId***	Set this parameter based on the network plan.
SUL Cell ID	**NRDUCellCollabServ**.***SulCellID***	Set this parameter based on the network plan.
NUL-SUL Maximum Inter-gNodeB Delay	**NRDUCellCollabServ**.***NulSulMaxInterGnbDelay***	The value of the **NRDUCellCollabServ**.***NulSulMaxInterGnbDelay*** parameter of the NR TDD cell must be the same as that of the **NRDUCellCollabCoop**.***NulSulMaxInterGnbDelay*** parameter of the NR FDD cell. Otherwise, UL and DL Decoupling cannot take effect for the corresponding NR TDD and NR FDD cells.
Inter-gNodeB SUL Flag	**NRCellRelation**.***InterGnodebSulFlag***	In inter-gNodeB scenarios, set this parameter to **TRUE**.

Parameter Name	Parameter ID	Setting Notes
Inter-gNodeB Cell Config Mode	**NRCellAnr**.***InterGnbCellCfgMode***	The default value is recommended. If this parameter is set to **AUTO**, this parameter must be set to **AUTO** for the NR FDD cell.
SUL Algorithm Switch	**NRDUCellCollabServ**.***SulAlgoSwitch***	● The **SCHEDULE_FIRST_HALF_SLOT_SW** option must be selected to enable symbol-level scheduling on the SUL carrier. The **SCHEDULE_FIRST_HALF_SLOT_SW** option of the **NRDUCellCollabCoop**.***SulAlgoSwitch*** parameter must also be selected for the corresponding NR FDD cell. Otherwise, UL and DL Decoupling cannot take effect for the corresponding NR TDD and NR FDD cells. ● The **SUL_UE_POWER_SAVING_SW** option must be selected to enable SUL UE power saving.
Multiple Frequency Uplink PRB Usage Threshold	**NRDUCellSulRelation**.***MultiFreqUlPrbUsageThld***	The default value is recommended.
Specified UE Cooperation Switch	**NRDUCellUeCoop**.***SpecUeCooperationSwitch***	Select the **SPEC_UE_IDENTIFY_SW** option.

Parameter Name	Parameter ID	Setting Notes
Multi-Carrier Measurement Wait Timer	**gNodeBAlgo**.***MultiCarrierMeasWaitTimer***	You are advised to use the recommended value. The smaller the value of this parameter, the shorter the measurement waiting time. In scenarios with a large number of candidate frequencies, however, if the value is too small, SCells may not be configured because no measurement reports are received, or better SCells may not be configured because no measurement reports on better candidates are received. As a result, the user-perceived rates of CA UEs decrease. A larger value of this parameter indicates a longer measurement waiting time and results in a higher probability of configuring better SCells but a longer time taken for SCell configuration.
SUL SSB SINR Threshold	**NRCellSulFrequency**.***SulSsbSinrThld***	Set this parameter based on the network plan. The smaller the value of this parameter, the higher the probability that SUL carriers are added and the higher the proportion of times SUL takes effect. However, interference between SUL carriers increases and the throughput of SUL UEs decreases. The larger the value of this parameter, the lower the probability that SUL carriers are added and the lower the proportion of times SUL takes effect. However, interference between SUL carriers decreases and the throughput of SUL UEs increases.

Parameter Name	Parameter ID	Setting Notes
SUL SSB SINR Offset	**NRCellSulFrequency**.***SulSsbSinrOffset***	Set this parameter based on the network plan. A smaller value of this parameter results in a shorter interval between SUL carrier addition and removal and a higher probability of ping-pong SUL carrier addition and removal. A larger value of this parameter results in a lower probability of ping-pong SUL carrier addition and removal. However, SUL carriers are not promptly removed, interference between SUL carriers increases, and the throughput of SUL UEs decreases.
NUL 1TX to SUL 1TX Switching Time	**NRDUCellCollabServ**.***Nul1TxToSul1TxSwitchTime***	The default value is recommended.
SRS RSRP Threshold	**NRDUCellCollabServ**.***SrsRsrpThld***	The default value is recommended. When standard interface capability selection for UL and DL Decoupling is enabled by selecting the **UL_DL_DECOUPLE_STD_CAP_SEL_SW** option of the **NRCellFeatureSwitch**.***MultiFreqAlgoSwitch*** parameter, it is recommended that **SRS RSRP Threshold** be set to a small value so that super uplink switches to UL and DL Decoupling later or UL and DL Decoupling switches to super uplink earlier.

Parameter Name	Parameter ID	Setting Notes
RSRP Threshold	**NRDUCellCollabServ**.***RsrpThld***	The default value is recommended during initial activation. ● When the NR uplink load is high, set this parameter to a large value to enable additional UEs to be carried on the NR SUL carrier. ● When the NR SUL load is high, set this parameter to a small value to enable additional UEs to be carried on the NR uplink carrier.
SUL Cov-based HO RSRP Threshold Offset	**NRCellMeasConfig**.***SulCovHoRsrpThldOffset***	Set this parameter based on the network plan. ● A larger value of this parameter results in a higher probability of triggering event A2 related to coverage-based handover and a lower probability of triggering event A1 related to coverage-based handover. ● A smaller value of this parameter results in the opposite effects.

Parameter Name	Parameter ID	Setting Notes
SUL CCE Adjustment Policy	**NRDUCellPdcch**.***SulCceAdjPolicy***	Set this parameter based on the network plan. ● If this parameter is set to **DEFAULT**, network performance is not affected. ● A smaller value of this parameter results in a smaller number of CCE resources available to the SUL carrier, a smaller probability of UEs for which UL and DL Decoupling has taken effect being scheduled on the SUL carrier, and a larger probability of other UEs in the NR TDD cell being scheduled. ● A larger value of this parameter results in increases in the probability of UEs for which UL and DL Decoupling has taken effect being scheduled on the SUL carrier and the user-perceived rate of these UEs. When there are a large number of such UEs, the probability of other UEs in the NR TDD cell being scheduled decreases.
PDCCH Algorithm Switch	**NRDUCellPdcch**.***PdcchAlgoSwitch***	Select the **UL_DL_CCE_SHARING_SW** option.
Multi-carrier Flow Control Level	**gNodeBAlgo**.***MultiCarrierFlowCtrlLevel***	Use the recommended value.
Multi-Frequency Algorithm Switch	**NRCellFeatureSwitch**.***MultiFreqAlgoSwitch***	Select the **UL_DL_DECOUPLE_STD_CAP_SEL_SW** option to enable standard interface capability selection for UL and DL Decoupling.

Parameter Name	Parameter ID	Setting Notes
Blacklist Control Extension Switch	**gNBUeCompatSw**.***BlacklistCtrlExtSwitch***	Select the **R16_140_UL_DL_DECOUPLE_SW_OFF** option to enable the function of disabling R16-compliant 140 µs-based UL and DL Decoupling.
PRACH Algorithm Extension Switch	**NRDUCellPrach**.***PrachAlgoExtSwitch***	If accurate control of cell coverage is required, select the **ACCESS_BYND_CELL_RAD_PROHIB_SW** option to enable UE random access control based on cell radius. It is recommended that this option be deselected in other scenarios.

Table 5-9 Parameters used for function activation in the NR FDD cell

Parameter Name	Parameter ID	Setting Notes
UL and DL Decoupling Switch	**NRDUCellAlgoSwitch**.***UlDlDecouplingSwitch***	Set this parameter to **ON** in the NR FDD cell to enable this feature.
SUL To Nul Mapping Type	**NRDUCellCollabCoop**.***SulToNulMappingType***	Set this parameter based on the network plan.
NUL MCC	**NRDUCellCollabCoop**.***NulMcc***	Set this parameter based on the network plan.
NUL MNC	**NRDUCellCollabCoop**.***NulMnc***	Set this parameter based on the network plan.
NUL gNodeB ID	**NRDUCellCollabCoop**.***NulGnodebId***	Set this parameter based on the network plan.
NUL Cell ID	**NRDUCellCollabCoop**.***NulCellID***	Set this parameter based on the network plan.

Parameter Name	Parameter ID	Setting Notes
NUL-SUL Maximum Inter-gNodeB Delay	**NRDUCellCollabCoop**.***NulSulMaxInterGnb-Delay***	The value of the **NRDUCellCollabCoop**.***NulSulMaxInterGnbDelay*** parameter of the NR FDD cell must be the same as that of the **NRDUCellCollabServ**.***NulSulMaxInterGnbDelay*** parameter of the NR TDD cell. Otherwise, UL and DL Decoupling cannot take effect for the corresponding NR FDD and NR TDD cells.
Inter-gNodeB SUL Flag	**NRCellRelation**.***InterGnodebSulFlag***	In inter-gNodeB scenarios, set this parameter to **TRUE**.
Inter-gNodeB Cell Config Mode	**NRCellAnr**.***InterGnbCellCfgMode***	The default value is recommended. If this parameter is set to **AUTO**, this parameter must be set to **AUTO** for the NR TDD cell.
SUL Algorithm Switch	**NRDUCellCollabCoop**.***SulAlgoSwitch***	The **SCHEDULE_FIRST_HALF_SLOT_SW** option must be selected to enable symbol-level scheduling on the SUL carrier. The **SCHEDULE_FIRST_HALF_SLOT_SW** option of the **NRDUCellCollabServ**.***SulAlgoSwitch*** parameter must also be selected for the corresponding NR TDD cell. Otherwise, UL and DL Decoupling cannot take effect for the corresponding NR TDD and NR FDD cells. Select the **SUL_WITH_MULTI_SLOT_CFG_NUL_SW** option when an SUL carrier needs to be bound to NUL carriers with different slot configurations in scenarios where UL and DL Decoupling and super uplink are both enabled.

Parameter Name	Parameter ID	Setting Notes
CoMP Configuration Policy	**NRDUCellComp**.***CompConfigPol***	Select the **UL_2CELL_COMP_BEAR_POL_SW** option to enable SUL CoMP to take effect for UEs with specific QCIs.
Service Differentiation Algorithm Switch	**gNBQciBearer**.***ServiceDiffAlgoSwitch***	Select the **COMP_SW** option for UEs requiring SUL CoMP for a specific QCI.

5.4.1.2 Using MML Commands

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

An NR TDD cell needs to be configured before running the following commands. For details about how to configure the NR TDD cell, see *Cell Management*. In addition, refer to **5.3 Requirements** and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

The following is an example of activation commands for intra-gNodeB UL and DL Decoupling. Assume that:

- The NR TDD cell ID is 100, the MCC is 460, the MNC is 08, and the gNodeB ID is 750074.
- The NR FDD cell ID is 120, the MCC is 460, the MNC is 08, and the gNodeB ID is 750074.
- The NR TDD frequency band is n78, the SUL frequency band is n84, and the NR FDD frequency band is n1.

```
//NR TDD cell configuration:
//Adding a neighbor relationship between NR TDD and NR FDD cells
ADD NRCELLRELATION: NrCellId=100, Mcc="460", Mnc="08", gNBId=750074, CellId=120;
//Adding the frequency of the NR FDD cell that serves as the neighboring cell of the NR TDD cell
ADD NRCELLFREQRELATION: NrCellId=100, SsbFreqPos=5309, FrequencyBand=N1,
SubcarrierSpacing=15KHZ;
//Adding an SUL frequency for the NR TDD cell
ADD NRCELLSULFREQUENCY: NrCellId=100, SulFreqIndex=0, SsbFreqPos=5309,
SulInterFreqA5RsrpThld2=-110;
//Setting the A5 measurement waiting timer
MOD GNODEBALGO: MultiCarrierMeasWaitTimer=30;
//Adding the association between the NR TDD cell and the SUL frequency band
ADD NRDUCELLSULRELATION: NrDuCellId=100, SulIndex=0, SulType=DYN_SUL, FddFrequencyBand=N1,
FddUlNrarfcn=386000, FddUlBandwidth=FDD_UL_BW_20M, FddUlBwp0ConfigMode=AUTO,
FddSubcarrierSpacing=15KHZ, FddCyclicPrefixLength=NCP;
//(Optional) Configuring the TA offset for the NR TDD cell if the TA offset configuration does not meet the
requirements
```



```
MOD NRDUCELL: NrDuCellId=100, FrequencyBand=N78, TaOffset=25600TC;
//(Optional) Configuring the frame offset for the NR TDD cell if the frame offset configuration does not
meet the requirements
MOD GNBFREQBANDCONFIG: FrequencyBand=N78, FrameOffset=0;
//Setting the thresholds for uplink carrier selection
MOD NRDUCELLCOLLABSERV: NrDuCellId=100, Nul1TxToSul1TxSwitchTime=MICROSECOND0,
RsrpThld=-110, SrsRsrpThld=-111;
//Modifying the RSRP threshold
MOD NRDUCELLCOLLABSERV: NrDuCellId=100, RsrpThld=-105;
//Setting the SUL coverage-based handover RSRP threshold offset
MOD NRCELLMEASCONFIG: NrCellId=100, SulCovHoRsrpThldOffset=-5;
//Setting the threshold for triggering an SUL carrier change
MOD NRDUCELLSULRELATION: NrDuCellId=100, SulIndex=0, SulType=DYN_SUL, SulCarrChangeSinrThld=-2;
//Binding one NR TDD cell to one FDD cell co-deployed with the SUL cell
MOD NRDUCELLCOLLABSERV: NrDuCellId=100, NulSulMaxInterGnbDelay= DELAY_0DOT5MS,
NulToSulMappingType=1TO1, SulCellID=120, SulMcc="460", SulMnc="08", SulGnodebId=750074;
//Enabling UL and DL Decoupling for the NR TDD cell
MOD NRDUCELLALGOSWITCH: NrDuCellId=100, UlDlDecouplingSwitch=ON;
//Turning on the UE cooperation configuration switch for the NR TDD cell
MOD NRDUCELLUECOOP: NrDuCellId=100, SpecUeCooperationSwitch=SPEC_UE_IDENTIFY_SW-1;
//(Required only for LampSite base stations) Modifying the SUL SSB SINR threshold and offset for the NR
TDD cell
MOD NRCELLSULFREQUENCY: NrCellId=100, SulFreqIndex=0, SulSsbSinrThld=DB10, SulSsbSinrOffset=20;
//(Optional) Setting the SUL CCE adjustment policy
MOD NRDUCELLPDCCH: NrDuCellId=100, PdcchAlgoSwitch=UL_DL_CCE_SHARING_SW-1;
MOD NRDUCELLPDCCH: NrDuCellId=100, SulCceAdjPolicy=FIX50;
//(Optional) Setting the level of flow control against multi-carrier operations at the gNodeB
MOD GNODEBALGO: MultiCarrierFlowCtrlLevel=MEDIUM;
//Enabling symbol-level scheduling
MOD NRDUCELLCOLLABSERV: NrDuCellId=100, SulAlgoSwitch=SCHEDULE_FIRST_HALF_SLOT_SW-1;
//(Optional) Enabling standard interface capability selection for UL and DL Decoupling to ensure
compatibility with some UEs that report the SUL capability but do not support UL and DL Decoupling in
terms of some capabilities
MOD NRCELLFEATURESWITCH: NrCellId=0, MultiFreqAlgoSwitch=UL_DL_DECOUPLE_STD_CAP_SEL_SW-1;
//(Optional) Enabling the function of disabling R16-compliant 140 µs-based UL and DL Decoupling to
ensure compatibility with some UEs that report the SUL capability but do not support UL and DL
Decoupling in terms of some capabilities
MOD GNBUECOMPATSW: UeInfoIndex=1, ParamCtrlType=ALL,
BlacklistCtrlExtSwitch=R16_140_UL_DL_DECOUPLE_SW_OFF-1;
//(Optional) Enabling UE random access control based on cell radius
MOD NRDUCELLPRACH: NrDuCellId=0, PrachAlgoExtSwitch=ACCESS_BYND_CELL_RAD_PROHIB_SW-1;

//NR FDD cell configuration:
//Adding a neighbor relationship between NR FDD and NR TDD cells
ADD NRCELLRELATION: NrCellId=120, Mcc="460", Mnc="08", gNBId=750074, CellId=100;
//(Optional) Configuring the TA offset for the NR FDD cell if the TA offset configuration does not meet the
requirements
MOD NRDUCELL: NrDuCellId=120, FrequencyBand=N1, TaOffset=25600TC;
//(Optional) Configuring the frame offset for the NR FDD cell if the frame offset configuration does not
meet the requirements
MOD GNBFREQBANDCONFIG: FrequencyBand=N1, FrameOffset=0;
//Binding one NR FDD cell to one TDD cell
MOD NRDUCELLCOLLABCOOP: NrDuCellId=120, NulSulMaxInterGnbDelay= DELAY_0DOT5MS,
SulToNulMappingType=1TO1, NulCellID=100, NulMcc="460", NulMnc="08", NulGnodebId=750074;
//(Optional) Setting the multiple frequency uplink PRB usage threshold
MOD NRDUCELLSULRELATION: NrDuCellId=120, SulIndex=0, SulType=DYN_SUL,
MultiFreqUlPrbUsageThld=50;
//(Optional) Selecting the SUL_WITH_MULTI_SLOT_CFG_NUL_SW option when an SUL carrier needs to be
bound to NUL carriers with different slot configurations in scenarios where UL and DL Decoupling and
super uplink are both enabled
MOD NRDUCELLCOLLABCOOP: NrDuCellId=120, SulAlgoSwitch=SUL_WITH_MULTI_SLOT_CFG_NUL_SW-1;
//Enabling UL and DL Decoupling for the NR FDD cell
MOD NRDUCELLALGOSWITCH: NrDuCellId=120, UlDlDecouplingSwitch=ON;
//Enabling symbol-level scheduling
MOD NRDUCELLCOLLABCOOP: NrDuCellId=120, SulAlgoSwitch=SCHEDULE_FIRST_HALF_SLOT_SW-1;
//Setting the CoMP configuration policy
MOD NRDUCELLCOMP: NrDuCellId=120, CompConfigPol=UL_2CELL_COMP_BEAR_POL_SW-1;
//Selecting the COMP_SW option of the Service Differentiation Algorithm Switch parameter for QCI 9
MOD GNBQCIBEARER: Qci=9, ServiceDiffAlgoSwitch=COMP_SW-1;
```


The following is an example of activation commands for inter-gNodeB UL and DL Decoupling. Assume that:

- The NR TDD cell ID is 100, the gNodeB ID is 750074, the MCC is 460, and the MNC is 08.
- The NR FDD cell ID is 220, the gNodeB ID is 750382, the MCC is 460, and the MNC is 08.
- The NR TDD frequency band is n41, the SUL frequency band is n83, and the NR FDD frequency band is n28.

An NR TDD cell must have already been configured before running the following commands. For details, see *Cell Management*.

```
//NR TDD cell configuration:
//Adding an external neighboring NR FDD cell
ADD NREXTERNALNCELL: Mcc="460", Mnc="08", gNBId=750382, CellId=220, PhysicalCellId=220, Tac=7,
SsbFreqPos=1955, NrNetworkingOption=SA;
//Specifying a neighbor relationship between NR TDD and NR FDD cells and setting the Inter-gNodeB SUL
Flag to TRUE. For this command, the Inter-gNodeB SUL Flag is manually configured.
MOD NRCELLRELATION: NrCellId=100, Mcc="460", Mnc="08", gNBId=750382, CellId=220,
InterGnodebSulFlag=TRUE;
//(Optional) Running the following command when the Inter-gNodeB SUL Flag is automatically configured:
MOD NRCELLANR: NrCellId=100, InterGnbCellCfgMode=AUTO;
//Adding an NR TDD cell frequency relationship
ADD NRCELLFREQRELATION: NrCellId=100, SsbFreqPos=1955, FrequencyBand=N28,
SubcarrierSpacing=15KHZ;
//Add an SUL frequency for the NR TDD cell
ADD NRCELLSULFREQUENCY: NrCellId=100, SulFreqIndex=0, SsbFreqPos=1955;
//Setting the A5 measurement waiting timer
MOD GNODEBALGO: MultiCarrierMeasWaitTimer=30;
//Adding the association between the NR TDD cell and the SUL frequency band
ADD NRDUCELLSULRELATION: NrDuCellId=100, SulIndex=0, SulType=DYN_SUL, FddFrequencyBand=N28,
FddUlNrarfcn=143000, FddUlBandwidth=FDD_UL_BW_20M, FddUlBwp0ConfigMode=AUTO,
FddSubcarrierSpacing=15KHZ, FddCyclicPrefixLength=NCP;
//Modifying the inter-gNodeB delay of the NR TDD and NR FDD cells
MOD NRDUCELLCOLLABSERV: NrDuCellId=100, NulSulMaxInterGnbDelay= DELAY_0DOT5MS;
//(Optional) Configuring the TA offset for the NR TDD cell if the TA offset configuration does not meet the
requirements
MOD NRDUCELL: NrDuCellId=100, FrequencyBand=N41, TaOffset=25600TC;
//(Optional) Configuring the frame offset for the NR TDD cell if the frame offset configuration does not
meet the requirements
MOD GNBFREQBANDCONFIG: FrequencyBand=N41, FrameOffset=0;
//Setting the thresholds for uplink carrier selection
MOD NRDUCELLCOLLABSERV: NrDuCellId=100, Nul1TxToSul1TxSwitchTime=MICROSECOND0,
RsrpThld=-110, SrsRsrpThld=-111;
//Modifying the RSRP threshold
MOD NRDUCELLCOLLABSERV: NrDuCellId=100, RsrpThld=-105;
//Setting the SUL coverage-based handover RSRP threshold offset
MOD NRCELLMEASCONFIG: NrCellId=100, SulCovHoRsrpThldOffset=-5;
//Binding one NR TDD cell to one FDD cell co-deployed with the SUL cell
MOD NRDUCELLCOLLABSERV: NrDuCellId=100, NulToSulMappingType=1TO1, SulCellID=220,
SulMcc="460", SulMnc="08", SulGnodebId=750382;
//Enabling UL and DL Decoupling for the NR TDD cell
MOD NRDUCELLALGOSWITCH: NrDuCellId=100, UlDlDecouplingSwitch=ON;
//(Optional) Setting the SUL CCE adjustment policy
MOD NRDUCELLPDCCH: NrDuCellId=100, PdcchAlgoSwitch=UL_DL_CCE_SHARING_SW-1;
MOD NRDUCELLPDCCH: NrDuCellId=100, SulCceAdjPolicy=FIX50;
//(Optional) Setting the level of flow control against multi-carrier operations at the gNodeB
MOD GNODEBALGO: MultiCarrierFlowCtrlLevel=MEDIUM;
//Enabling symbol-level scheduling
MOD NRDUCELLCOLLABSERV: NrDuCellId=100, SulAlgoSwitch=SCHEDULE_FIRST_HALF_SLOT_SW-1;
//(Optional) Enabling standard interface capability selection for UL and DL Decoupling to ensure
compatibility with some UEs that report the SUL capability but do not support UL and DL Decoupling in
terms of some capabilities
MOD NRCELLFEATURESWITCH: NrCellId=0, MultiFreqAlgoSwitch=UL_DL_DECOUPLE_STD_CAP_SEL_SW-1;
//(Optional) Enabling the function of disabling R16-compliant 140 µs-based UL and DL Decoupling to
ensure compatibility with some UEs that report the SUL capability but do not support UL and DL
```

```
Decoupling in terms of some capabilities
MOD GNBUECOMPATSW: UeInfoIndex=1, ParamCtrlType=ALL,
BlacklistCtrlExtSwitch=R16_140_UL_DL_DECOUPLE_SW_OFF-1;
//(Optional) Enabling UE random access control based on cell radius
MOD NRDUCELLPRACH: NrDuCellId=0, PrachAlgoExtSwitch=ACCESS_BYND_CELL_RAD_PROHIB_SW-1;

//NR FDD cell configuration:
//Adding an external neighboring NR TDD cell
ADD NREXTERNALNCELL: Mcc="460", Mnc="08", gNBId=750074, CellId=100, PhysicalCellId=100, Tac=7,
SsbFreqPos=6312, NrNetworkingOption=SA;
//Adding a neighbor relationship between NR FDD and NR TDD cells and setting the Inter-gNodeB SUL
Flag to TRUE. For this command, the Inter-gNodeB SUL Flag is manually configured.
ADD NRCELLRELATION: NrCellId=220, Mcc="460", Mnc="08", gNBId=750074, CellId=100,
InterGnodebSulFlag=TRUE;
//(Optional) Running the following command when the Inter-gNodeB SUL Flag is automatically configured:
MOD NRCELLANR: NrCellId=220, InterGnbCellCfgMode=AUTO;
//Binding one NR FDD cell to one TDD cell
MOD NRDUCELLCOLLABCOOP: NrDuCellId=220, NulSulMaxInterGnbDelay= DELAY_0DOT5MS,
SulToNulMappingType=1TO1, NulCellID=100, NulMcc="460", NulMnc="08", NulGnodebId=750074;
//(Optional) Setting the multiple frequency uplink PRB usage threshold
MOD NRDUCELLSULRELATION: NrDuCellId=220, SulIndex=0, SulType=DYN_SUL,
MultiFreqUlPrbUsageThld=50;
//(Optional) Selecting the SUL_WITH_MULTI_SLOT_CFG_NUL_SW option when an SUL carrier needs to be
bound to NUL carriers with different slot configurations in scenarios where UL and DL Decoupling and
super uplink are both enabled
MOD NRDUCELLCOLLABCOOP: NrDuCellId=220, SulAlgoSwitch=SUL_WITH_MULTI_SLOT_CFG_NUL_SW-1;
//Enabling UL and DL Decoupling for the NR FDD cell
MOD NRDUCELLALGOSWITCH: NrDuCellId=220, UlDlDecouplingSwitch=ON;
//(Optional) Configuring the TA offset for the NR FDD cell if the TA offset configuration does not meet the
requirements
MOD NRDUCELL: NrDuCellId=220, FrequencyBand=N28, TaOffset=25600TC;
//(Optional) Configuring the frame offset for the NR FDD cell if the frame offset configuration does not
meet the requirements
MOD GNBFREQBANDCONFIG: FrequencyBand=N28, FrameOffset=0;
//Enabling symbol-level scheduling
MOD NRDUCELLCOLLABCOOP: NrDuCellId=220, SulAlgoSwitch=SCHEDULE_FIRST_HALF_SLOT_SW-1;
//Setting the CoMP configuration policy
MOD NRDUCELLCOMP: NrDuCellId=220, CompConfigPol=UL_2CELL_COMP_BEAR_POL_SW-1;
//Selecting the COMP_SW option of the Service Differentiation Algorithm Switch parameter for QCI 9
MOD GNBQCIBEARER: Qci=9, ServiceDiffAlgoSwitch=COMP_SW-1;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

```
//On the gNodeB side
//(Optional) Disabling standard interface capability selection for UL and DL Decoupling
MOD NRCELLFEATURESWITCH: NrCellId=0, MultiFreqAlgoSwitch=UL_DL_DECOUPLE_STD_CAP_SEL_SW-0;
//(Optional) Disabling the function of disabling R16-compliant 140 µs-based UL and DL Decoupling
MOD GNBUECOMPATSW: UeInfoIndex=1, ParamCtrlType=ALL,
BlacklistCtrlExtSwitch=R16_140_UL_DL_DECOUPLE_SW_OFF-0;
//Disabling UL and DL Decoupling
MOD NRDUCELLALGOSWITCH: NrDuCellId=0, UlDlDecouplingSwitch=OFF;
//Deselecting the COMP_SW option of the Service Differentiation Algorithm Switch parameter for QCI 9
MOD GNBQCIBEARER: Qci=9, ServiceDiffAlgoSwitch=COMP_SW-0;
//Restoring the CoMP configuration policy
MOD NRDUCELLCOMP: NrDuCellId=0, CompConfigPol=UL_2CELL_COMP_BEAR_POL_SW-0;
```

5.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

5.4.2 Activation Verification

Basic Functions of UL and DL Decoupling

Use either of the following methods to check whether the basic functions of UL and DL Decoupling have taken effect:

- Using MML commands

  Run the **DSP NRDUCELL** command on the gNodeB for the SUL cell. If the value of **NR DU Cell State** is **Normal**, the basic functions of UL and DL Decoupling have taken effect.
- Observing counters

  When the basic functions of UL and DL Decoupling are enabled and used by UEs, observe the values of the counters listed in the table below. If any of them has a non-zero value, the basic functions of UL and DL Decoupling have taken effect. If the counter values are always zero, the basic functions of UL and DL Decoupling have not taken effect.

Counter ID	Counter Name
1911816547	**N.User.Decouple.Avg**
1911816832	**N.Decoupling.PUCCH.IntraCell.NonSultoSul.Att**
1911816833	**N.Decoupling.PUCCH.IntraCell.NonSultoSul.Succ**
1911816834	**N.Decoupling.PUCCH.IntraCell.SultoNonSul.Att**
1911816835	**N.Decoupling.PUCCH.IntraCell.SultoNonSul.Succ**

5.4.3 Network Monitoring

Basic Functions of UL and DL Decoupling

The basic functions of UL and DL Decoupling improve the average uplink UE throughput, and decrease the proportion of UEs whose rates fall within a low-rate range in a cell.

- The average uplink throughput of UEs in a cell is measured by **User Uplink Average Throughput (DU)**.
- The proportion of UEs whose rates fall within a low-rate range can be obtained by dividing the number of samples in the low-rate range by the total number of samples. A low-rate range is relative. For example, range 0 is a low-rate range compared with range 1. The table below lists the counters used to observe the number of samples in each rate range.

Counter ID	Counter Name
1911816708	**N.Thp.UL.Samp.Index0**
1911816709	**N.Thp.UL.Samp.Index1**
1911816710	**N.Thp.UL.Samp.Index2**
1911816711	**N.Thp.UL.Samp.Index3**
1911816712	**N.Thp.UL.Samp.Index4**
1911816713	**N.Thp.UL.Samp.Index5**
1911816714	**N.Thp.UL.Samp.Index6**
1911816715	**N.Thp.UL.Samp.Index7**
1911816716	**N.Thp.UL.Samp.Index8**
1911816717	**N.Thp.UL.Samp.Index9**
1911816937	**N.Thp.UL.Samp.Index10**
1911816941	**N.Thp.UL.Samp.Index11**
1911816938	**N.Thp.UL.Samp.Index12**
1911816939	**N.Thp.UL.Samp.Index13**
1911816940	**N.Thp.UL.Samp.Index14**

SUL CoMP

To compare the throughput before and after SUL CoMP is enabled, you are advised to select the **INTRA_GNB_OVERLAPPING_MEAS_SW** option of the **NRDUCellAlgoSwitch.*CompSwitch*** parameter for the NR TDD cell one week before SUL CoMP is enabled so that statistics of throughput-related counters for UEs in overlapping areas can be collected in advance.

SUL CoMP increases the uplink throughput of UEs in overlapping areas excluding small-packet throughput without decreasing the average uplink UE throughput and the uplink UE throughput excluding small-packet throughput. You can evaluate the gains of SUL CoMP by comparing the values of the following performance counters before and after this function is enabled. The counter values depend on parameter settings.

To observe the gains of SUL CoMP, you are advised to set the **NRCellComp.*OverLappingRsrpOffset*** (for the NR FDD cell) and **NRCellComp**.***UlCompRsrpOffset*** (for the NR TDD cell where super uplink UEs are located) parameters to the same value. The reason is as follows: The offset for event A3 in the overlapping area of the NR FDD cells is determined by the **NRCellComp.*OverLappingRsrpOffset*** parameter. The RSRP difference for event A5 indicating whether SUL CoMP takes effect for UEs for which UL and DL Decoupling has taken effect is determined by the **NRCellComp**.***UlCompRsrpOffset*** parameter.

- Uplink throughput of UEs in overlapping areas excluding small-packet throughput = (**N.ThpVol.UL.CellOverlap – N.ThpVol.UL.SmallPkt.CellOverlap**)/ **N.ThpTime.UL.RmvSmallPkt.CellOverlap**
- Uplink UE throughput excluding small-packet throughput = (**N.ThpVol.UL – N.ThpVol.UE.UL.SmallPkt**)/**N.ThpTime.UE.UL.RmvSmallPkt**
- Average uplink UE throughput = **N.ThpVol.UL**/**N.ThpTime.UL**

6 Parameters

The following hyperlinked EXCEL files of parameter reference match the software version with which this document is released.

- Node Parameter Reference: contains device and transport parameters.
- Node Used Reserved Parameter List: contains the reserved parameters that are in use and those that have been disused.
- gNodeBFunction Parameter Reference: contains all parameters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- 3900 & 5900 Series Base Station gNodeBFunction Used Reserved Parameter List: contains the reserved parameters that are in use and the reserved parameters that fall into disuse in the current R version.

NOTE

You can find the EXCEL files of parameter reference and used reserved parameter list for the software version used on the live network from the "Man-Machine Interface Reference" node in the *3900 & 5900 Series Base Station Product Documentation* delivered with that version.

FAQ 1: How do I find the parameters related to a certain feature from parameter reference?

Step 1 Open the EXCEL file of parameter reference.

Step 2 On the **Parameter List** sheet, filter the **Feature ID** column. Click **Text Filters** and choose **Contains**. Enter the feature ID, for example, FBFD-010011.

Step 3 Click **OK**. All parameters related to the feature are displayed.

----End

FAQ 2: How do I find the information about a certain reserved parameter from the used reserved parameter list?

Step 1 Open the EXCEL file of the used reserved parameter list.

Step 2 On the **Used Reserved Parameter List** sheet, use the **MO**, **Parameter ID**, and **BIT** columns to locate the reserved parameter. View its information, including the meaning, values, and impacts.

----End

7 Counters

The following hyperlinked EXCEL files of performance counter reference match the software version with which this document is released.

- Node Performance Counter Summary: contains device and transport counters.
- gNodeBFunction Performance Counter Summary: contains all counters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- gNodeBFunction Used Reserved Counter List: contains the reserved counters that are in use and those that have been disused.

NOTE

You can find the EXCEL files of performance counter reference and used reserved counter list for the software version used on the live network from the "Man-Machine Interface Reference" node in the *3900 & 5900 Series Base Station Product Documentation* delivered with that version.

FAQ: How do I find the counters related to a certain feature from performance counter reference?

Step 1 Open the EXCEL file of performance counter reference.

Step 2 On the **Counter Summary(En)** sheet, filter the **Feature ID** column. Click **Text Filters** and choose **Contains**. Enter the feature ID, for example, FBFD-010011.

Step 3 Click **OK**. All counters related to the feature are displayed.

----End

8 Glossary

For the acronyms, abbreviations, terms, and definitions, see *Glossary*.

9 Reference Documents

- *LTE FDD and NR SUL Dynamic Spectrum Sharing*
- Documents in *5G RAN Feature Documentation*:
  - *Power Control*
  - *EPC-based NSA Basics*
  - *EPC-based NSA Performance Enhancement*
  - *Cell Management*
  - *5G Networking and Signaling*
  - *Carrier Aggregation*
  - *High Speed Mobility*
  - *Hyper Cell*
  - *Cell Combination*
  - *URLLC*
  - *Super Uplink*
  - *UE Power Saving*
  - *CoMP*
  - *SA Mobility Management in Connected Mode*
  - *Network Slicing*
  - *Mobility Load Balancing in Connected Mode*
  - *Interoperability Between E-UTRAN and NG-RAN in Connected Mode*
  - *Channel Management*
  - *Resource Control in NSA Networking*
  - *Multi-Frequency Smart Selection*
  - *Remote Interference Management (Low-Frequency TDD)*
  - *Fusion Cell (Low-Frequency TDD)*
  - *Energy Conservation and Emission Reduction*
  - *CA Experience Improvement*
  - *Downlink Scheduling*
  - *VoNR*

  - *QCI-based Resource Control*
  - *eXn Self-Management*
  - *LTE FDD and NR Dynamic Spectrum Sharing*
  - *LTE FDD and NR Hybrid Dynamic Spectrum Sharing*
  - *Distributed Antenna Solutions (Low-Frequency TDD)*
  - *Random Access Control*
  - *RedCap*
- *Carrier Aggregation* in *eRAN Feature Documentation*
- 3GPP TS 36.213 "Evolved Universal Terrestrial Radio Access (E-UTRA); Physical layer procedures"
- 3GPP TS 36.331 "Evolved Universal Terrestrial Radio Access (E-UTRA); Radio Resource Control (RRC); Protocol specification"
- 3GPP TS 36.423 "Evolved Universal Terrestrial Radio Access Network(E-UTRAN); X2 application protocol (X2AP)"
- 3GPP TS 38.104 "NR; Base Station (BS) radio transmission and reception"
- 3GPP TS 38.300 "NR and NG-RAN Overall Description; Stage 2"
- 3GPP TS 38.304 "NR; User Equipment (UE) procedures in Idle mode and RRC"
- 3GPP TS 38.331 "NR; Radio Resource Control (RRC) protocol specification"